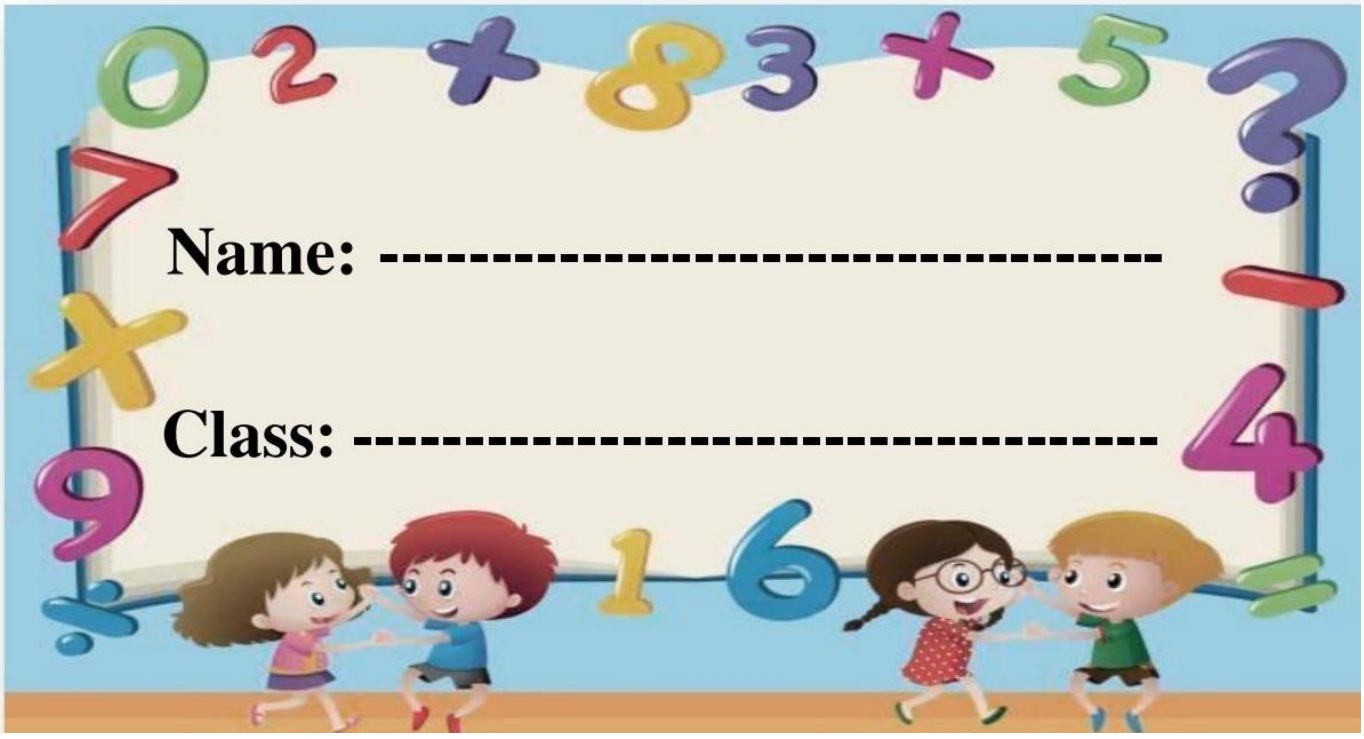
مدارس مصر للغات
MISR LANGUAGE SCHOOLS

Primary
6

Mathematics
First Term



Theme 1: Number Sense and Operations:

Expressions and Equations

Unit 1: Division, Factors, and Multiples

Concept 1: Revision, Division Algorithm, GCF and LCM

Lesson 1: Divisibility

Lesson 2: Factorizing a Number to its Prime Factors

Lesson 3: Writing Expressions Using the G.C.

Lesson 4: Analyzing Least Common Multiples

Date……

Unit (1), Lesson (1)

Divisibility

Any number is divisible by another if the remainder of the division operation is zero.

Ex: $12 \div 3 = 4\ R\ 0$ so, 12 is divisible by 3

$12 \div 5 = 2\ R\ 2$ so, 12 is not divisible by 5

(A) Divisibility rules:

Divisibility rule of factor 2	**All even numbers are divisible be 2** **Even numbers have "0,2,4,6,8" in the ones place** **For example: 530, 794 are even numbers (divisible by 2)** **But 821, 427 are odd numbers (not divisible by 2)**
Divisibility rule of factor 3	**number is divisible by 3 if the sum of its digits is a multiple of 3** **For example:** **423 is divisible by 3 because 4+2+3 = 9 and 9 is a multiple of 3** **83 is not divisible of 3 because 8+3=11 and 11 isn't multiple**
Divisibility rule of factor 6	**Numbers which are divisible by 2 and 3 in the same time are divisible by 6** **For example:** **132 is divisible by 6 because** **➞one's digit is 2 (even so divisible by 2)** **➞1+3+2=6 (divisible by 3)**
Divisibility rule of factor 5	**Numbers which have the digit 0 or 5 in the ones place are divisible by 5** **For example:** **230,625, 100, 95 are divisible by 5**
Divisibility rule of factor 10	**Numbers which have the digit 0 in the ones place are divisible by 10** **For example:** **140, 7450, 90, 300 are divisible by 10**
Divisibility rule of factor 4	**Numbers which have the last 2 digits (ones and tens places) 00 or forming a multiple of 4 are divisible by 4** **For example:** **524 is divisible by 4 because if you check ones and tens (24 is a multiple of 4)** **600 is divisible by 4 because if you check ones and tens (00)** **319 is not divisible by 4 because if you check ones and tens digits (19 isn't a multiple of 4)**

Notes: -

1. 25 is divisible by 5 because 25 is a multiple of 5. So, divisible by = multiple of

2. 1 is factor for all numbers. So, each number is divisible by 1

3. Each number is divisible by itself except zero.

Q1. Complete:

1) The number is divisible by 2 if its one's digit is.....

2) 2 x 3 = 6 then is…… divisible by…… and…….

3) 2 x 3 = 6 then ……..is multiple of……. and……..

4) Each number is divisible by.....

5) The number is divisible by 10 if its one's digit is……

6) The number 423 is ……..(divisible / not divisible) by 6

7) The number 125 is..... (divisible / not divisible) by 10

8) The number 2036 is ……(divisible / not divisible) by 4

9) The number 70 is ……(divisible / not divisible) by 5

10) Each number except zero is divisible by.....

11) The number 1,537 is…… (divisible / not divisible) by 3

12) The smallest 3-digit number divisible by 2, 5 and 10 is.....

13) The smallest 3-digit number divisible be 2, 3 and 5 is…….

14) (41 +....) is divisible by 10

15) (715 −.....) is divisible by 4

Q2. Choose:

1) The number is divisible by both 4 and 5

a)200 b)315 c)210 d)745

2) The number……. is divisible by 2, 3, 4, 5, 6 and 10

a)20 b)30 c)40 d)60

3) The number 3032 is divisible by.....

a)4 b)5 c)6 d)10

4) The number..... is not divisible by 10

a)450 b)900 c)215 d)0

5) The number is divisible by 3 if the sum of its digits is divisible by.....

a)6 b)2 c)3 d)5

6) The number…… is divisible by 2

a)113 b)32 c)55 d)247

7) If a number is divisible by 2 and 3, it is divisible by……

a)6 b)10 c)4 d)5

8) 7203 is a multiple of……..

a)6 b)2 c)3 d)4

9) The number..... is divisible by 4

a)214 **b)709** **c)550** **d)500**

10)…… is factor for all numbers

a)0 **b)5** **c)2** **d)1**

Q3. A charity wants to distribute 8,730 pounds among 6 poor families; can it divide them all equally? explain your answer

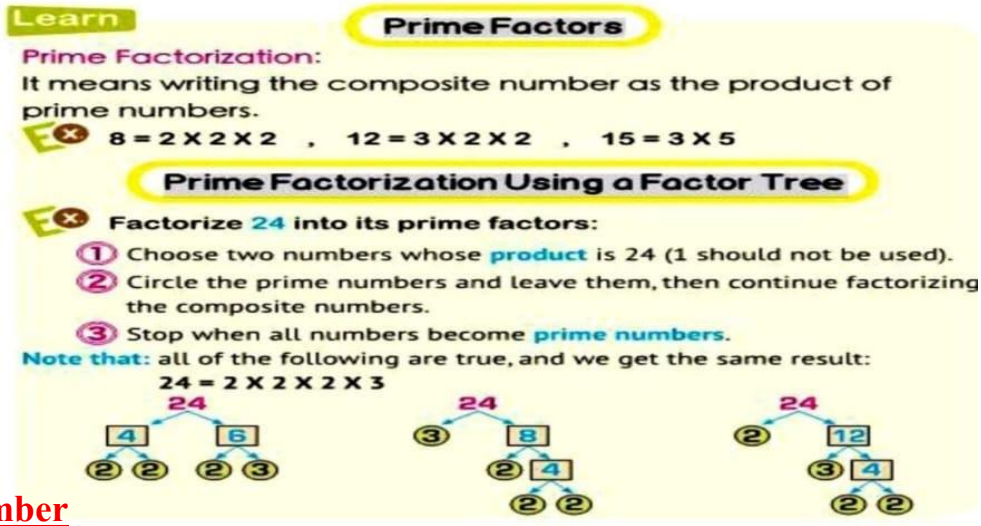
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Lesson 2

Factorizing a number to its prime Factors

G.C.F and L.C.M



Remember

A prime number:

It is a number that has only 2 factors 1 and the number itself as
$(2, 3, 5, 7, 11, 13, 17, 19, 23, - - -)$

A composite number:

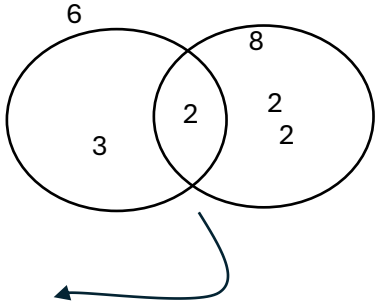
It is a number that has more than 2 factors as $(4, 6, 8, 9, 10, 12, 14, 15, 16, - - -)$

*** The common factor of all number is 1**
***The greatest common factor (G.C.F) of any two prime numbers is 1**
***The G.C.F of the two numbers which haven't any common prime factors is 1**
***The common multiple of all numbers is 0**
***The (L.C.M) of any two prime numbers is their product**

Example 1:

Find the GCF and LCM of 6 and 8 using Venn diagram:

6 = 2 x 3
8 = 2 x 2 x 2
GCF = 2
LCM = 2 x 3x 2 x 2 = 24



(The intersection is the GCF)

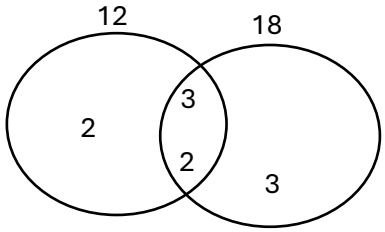
(The product of all prime factors in the Venn diagram is the LCM)

Example 2:

Find the GCF and LCM of 12 and 18 using Venn diagram

$12 = 2 \times 2 \times 3$

$18 = 2 \times 3 \times 3$



$GCF = 2 \times 3 = 6$

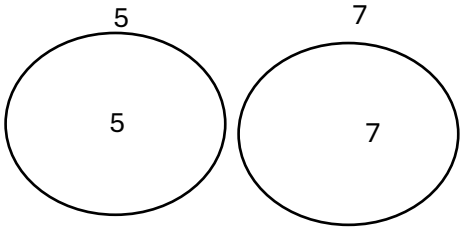
$LCM = 2 \times 3 \times 2 \times 3 = 36$

Example 3:

Find the GCF and LCM of 5 and 7 using Venn diagram:

$5 = 5 \times 1$

$7 = 7 \times 1$



$GCF = 1$

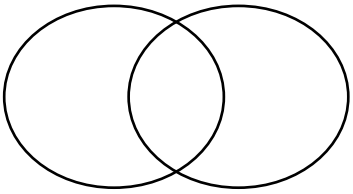
$LCM = 5 \times 7 = 35$

Let's Try:

1) Find the GCF and LCM of 20 and 30 using Venn diagram:

20 =……...

30 =……...



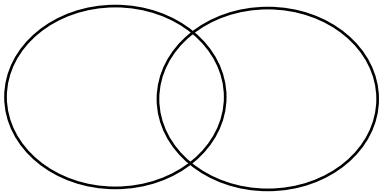
GCF =……...

LCM =……...

2) Find the GCF and LCM of 10 and 35 using Venn diagram:

10 =……...

35 =……...



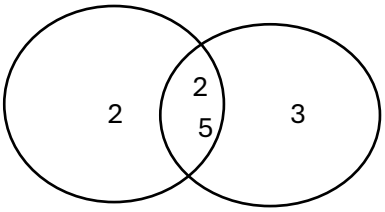
GCF =……...

LCM=……...

3) From the Venn diagram find:

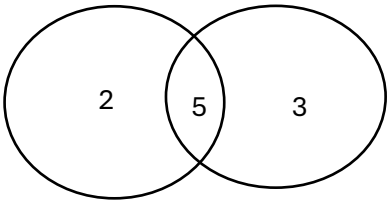
GCF =……..

LCM =……..



4) The two numbers represented by the

Venn diagram are …… and……



Evaluation:

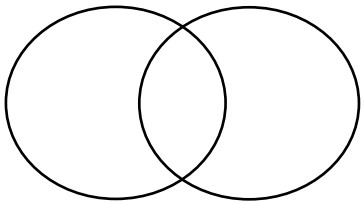
1) Find the GCF and LCM of 15 and 6 using Venn diagram:

15 =……..

6 =……..

GCF =……..

LCM =……..



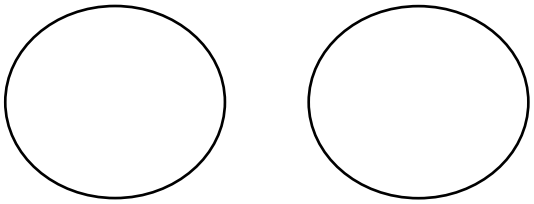
2) Find the GCF and LCM of 4 and 11 using Venn diagram:

4 =……..

11 =……..

GCF =……..

LCM =……..



3) Choose the correct answer:

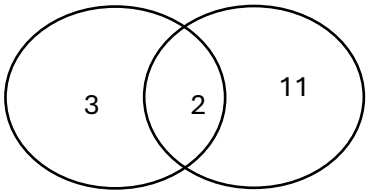
1) The common multiple of all numbers is …….

a) 0 **b) 1** **c) 2** **d) 3**

2) The greatest common factor of any two prime numbers is …….

a) 0 **b) 1** **c) 2** **d) 3**

3) From the opposite Venn diagram, the LCM is …….



a) 6 **c) 22**

b) 2 **d) 66**

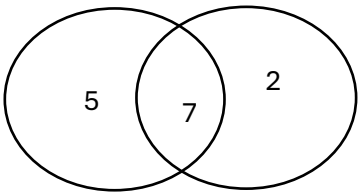
4) The greatest common factor of 6 and 8 is …….

a) 1 **b) 2** **c) 3** **d) 4**

5) The greatest common factor of 2 and 3 is ……..

a) 1 **b) 2** **c) 3** **d) 6**

6) From the opposite Venn diagram, the GCF is ……



a) 5 **c) 14**

b) 35 **d) 7**

7) The LCM of any two prime numbers is their ……..

a) sum **b)differenc e** **c) product** **d)otherwise**

8) All prime numbers are odd except

a) 0 **b) 2** **c) 4** **d) 3**

9) …….. is neither a prime nor a composite number.

a) 1 **b) 3** **c) 2** **d) 0**

10) The smallest odd prime number is ……

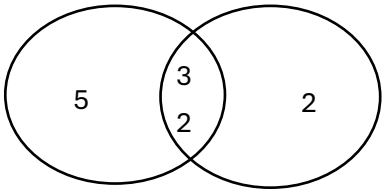
a) 5 **b) 3** **c) 1** **d) 0**

Q:4) Using the opposite Venn diagram, Complete:

a) The two numbers represented in the Venn diagram are…… and……

b) The GCF for the two numbers are……

c) The LCM for the two numbers are……



Date……

Lesson 3

Writing Expressions Using the G.C.F

Remember

- The Distributive Property states that multiplying a number by the sum of two addends is the same as multiplying that number by each addend individually and then adding those products.

Ex. $7 \times (3 + 9) = (7 \times 3) + (7 \times 9)$

Writing Numerical Expressions Using the Greatest Common Factor

- The greatest common factor is used to solve real-word problems that usually involve

Dividing — Breaking

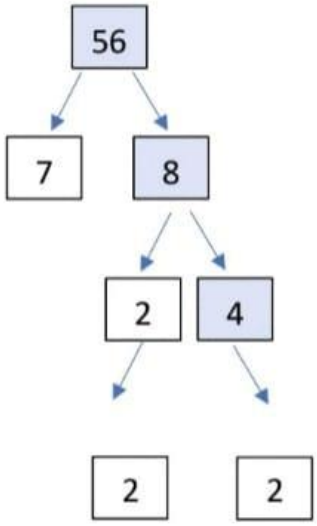
- Cutting things into pieces
- Separating things into groups
- Distributing Equally

Example 1:

Use the distributive property and G.C.F to write the expression of $7 + 56$

Solution

7



$7 = 7$

$56 = 7 \times 2 \times 2 \times 2$

G.C.F = 7

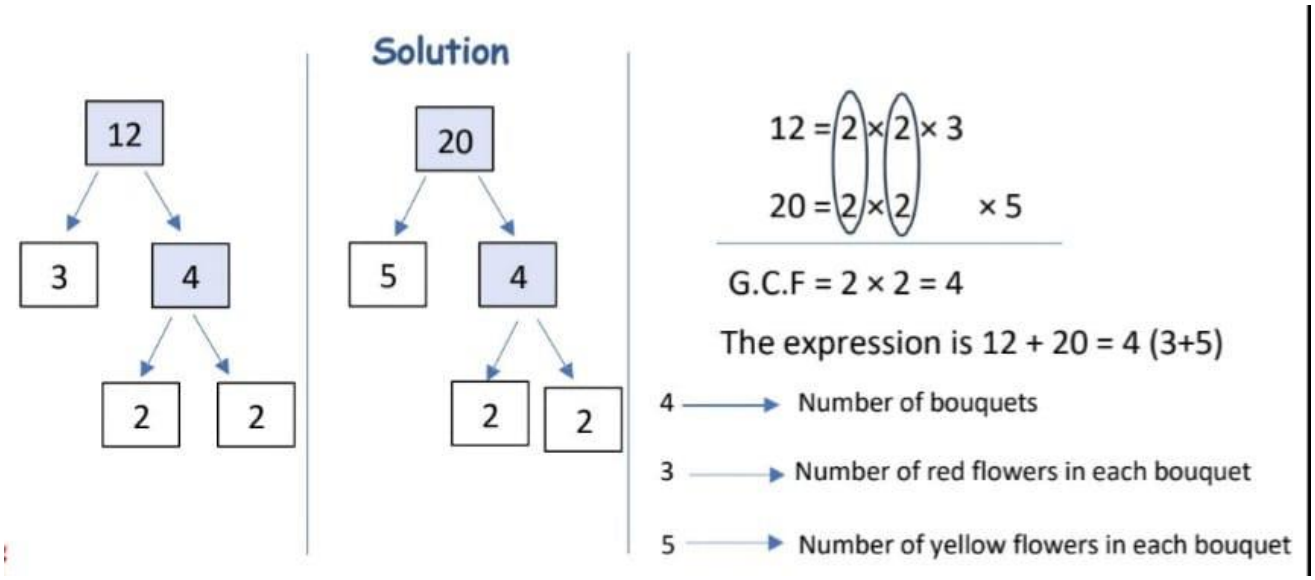
The expression is: $7 + 56 = 7\,(1+8)$

Example 2:

Sara has 12 red flowers and 20 yellow flowers. She wants to make bouquets with the same number of each colour flower in each bouquet.

What is the greatest number of bouquets she can make?

Write the expression that represent this situation.



Let's Try

1) Complete:

a) $18 + 6 = 6(---+---)$

b) $5 \times (---+---) = (45 + 10)$

c) $14 + 21 = 7 \times (---+---)$

d) $--- \times (---+---) = 5 \times 7 + 5 \times 4$

e) $3(5 + 4) = --- \times --- + --- \times ---$

f) $30 + 50 = ---(---+---)$

g) $9(1 + 2) = 9 + ---$

h) $8 \times (---+---) = (--- \times 9) + (--- \times 2)$

2) Use the distributive property and G.C.F to write the expression of the following addition problems:

a) $15 + 45$

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b) 20 + 60

3) Karma has 12 red crayons and 15 blue crayons. What is the greatest number of groups can Karma divide the crayons into so that all groups contain the same number of both colors?

4) Rana divided 12 oranges and 8 candies into bags so that the bags contained the same number of oranges and the same number of candies. Write a numerical expression for this situation.

5) Sam wanted to divide the 24 pens and 36 notebooks into groups, so that each group contained the same number of tools.

What is the largest number of groups that can be configured for each type of supply to have for each group?

How many pens are in each group?

What is the number of notebooks in each group?

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6) Bassant has 25 blue roses and 15 red roses that she wants to distribute in bouquets, so that each bouquet contains the same number of roses of each colour.

Write numerical expressions using the greatest common factor

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Evaluation

1) Complete the following:

a) $5 \times (7 + 2) = (5 \times ---) + (--- \times 2)$

b) $2 \times (--- + ---) = (--- \times 4) + (--- \times 6)$

c) The G. C. F for 9 & 6 is $---$

d) The G. C. F of all numbers is $---$

e) $--- \times (--- + ---) = (7 \times 2) + (7 \times 4)$

f) $--- \times (4 + 6) = (9 \times ---) + (9 \times ---)$

g) $5 \times (3 + 6) = (- - - \times - - -) + (- - - \times - - -)$

h) $8 \times (- - - + - - -) = (- - - \times 9) + (9 \times - - -)$

2) Use the distributive property and G.C.F to write the expression of the following addition problems:

a) 35 + 40

b) 70 + 80

c) 18 + 9

3) Yasmeen collected 12 bags of legumes and 8 boxes of cheese to prepare cartons of donations for the poor. Write a numerical expression that represents the largest number of cartons possible so that all cartons include the same number of the two types of food.

4) A hospital staffed by 12 doctors and 28 nurses, find the greatest number of equal groups that can be made of doctors and nurses together. How many doctors are there in each group? How many nurses are there in each group?

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5) A merchant has 18 kg of oranges and 27 kg of apples, so if the merchant wants to divide the oranges and apples in bags of the same mass, what is the largest number of bags that can be made for each type of fruit to be:

Does each bag have the same mass?

How many kilograms of oranges will each bag contain?

How many kilograms of apples will each bag contain?

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6) Dina prepared bags contain snacks. She has 6 bananas and 12 pieces of fruit. Dina wants the snacks in the bags to be distributed evenly without any food left.

How many bags of snacks can Dina prepare?

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Lesson 4

Analyzing Least Common Multiples

Remember

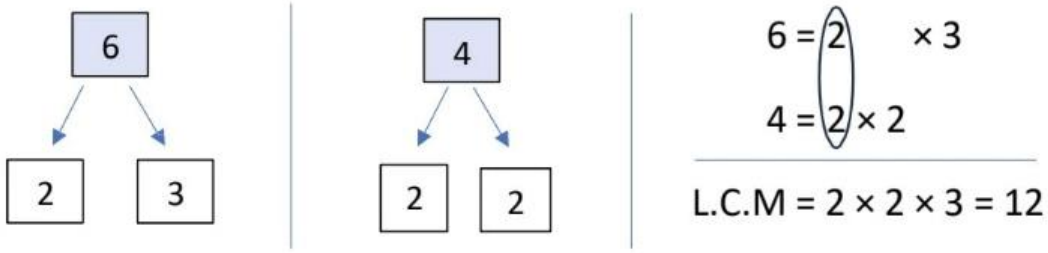
Example: Adding and Subtracting Fractions with Unlike Denominators Using the LCM

Find the result of the following in the simplest form:

Solution steps:

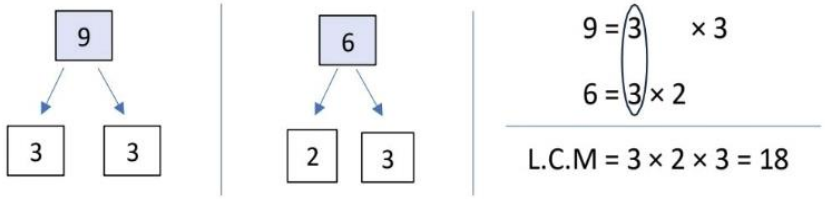
1. Find the LCM for the denominators.
2. Replace these fractions with equivalent fractions with a like denominator.
3. Add or subtract, putting the answer in its simplest form if possible.

a) $\frac{1}{6}+\frac{1}{4}=$ ----------



$$\frac{1}{6}+\frac{1}{4}=\frac{1\times 2}{6\times 2}-\frac{1\times 3}{4\times 3}=\frac{2}{12}-\frac{3}{12}=\frac{5}{12}$$

b) $\frac{7}{9}-\frac{1}{6}=$----------



$$\frac{7}{9}-\frac{1}{6}=\frac{7\times 2}{9\times 2}-\frac{1\times 3}{6\times 3}=\frac{14}{18}-\frac{3}{18}=\frac{11}{18}$$

Let's Try

1) Find the result of the following in the simplest form:

a) $\frac{4}{5} + \frac{2}{3} = - - -$

b) $\frac{7}{8} - \frac{1}{6} = - - -$

c) $1\frac{2}{3} + 1\frac{15}{24} = - - -$

d) $9\frac{1}{10} - 5\frac{7}{12} = - - -$

e) $\frac{2}{5} + \frac{3}{8} + 1 = - - -$

f) $\mathbf{1 - \frac{1}{4} - \frac{1}{5} = - - -}$

2) Alaa made 4 trays of basbousa of the same size and cut each tray of basbousa in a different way. After the end of the party, she noticed that there was left over basbousa in each tray, as follows: $\frac{1}{3}$ **of the first tray,** $\frac{1}{6}$ **of the second tray,** $\frac{5}{12}$ **of the third tray,** $\frac{1}{4}$ **of the last tray. What is the total amount of basbousa left?**

3) Yomna had 12 meters of fabric, of which she used $4\frac{3}{4}$ **meters to make a dress. How many meters of fabric does she have left?**

Evaluation

1) Find the result of the following in the simplest form:

a) $5\frac{1}{6} + 3\frac{1}{3} = - - -$

b) $7\frac{3}{4} - 3\frac{2}{5} = - - -$

c) $9\frac{1}{2} + 2\frac{1}{4} = - - -$

d) $7\frac{1}{3} - 2\frac{1}{7} = - - -$

e) $\frac{1}{2} + \frac{3}{4} + 1 = - - -$

f) $3 - \frac{2}{5} - \frac{1}{6} = - - -$

2) Ali has $15\frac{1}{2}$ **pounds, he bought a ruler for** $4\frac{1}{5}$ **pounds and a pen for** $6\frac{1}{9}$ **pounds. How much money is left with Ali?**

3) Mother ate $\frac{1}{3}$ **of the pizza and** $\frac{3}{8}$ **father ate of it. How much of the pizza has been eaten and how much is left?**

4) Bassant has found 5 packages of 4 apples each. She opened each package to eat some from each, trying to find the ones that had the best taste.

The packages had the remaining portions left

$$: \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{4}{4}, \frac{5}{4}$$

If you were to repackage the remaining pieces of fruit to create full packages of apples.

a) How many packages of apples would remain?

b) How many packages did Bassant eat?

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Date……

Unit One, Concept One Assessment

1) Choose:

1) The common multiple of all numbers is……..

a) 0 **b) 1** **c) 2** **d) 3**

2) The greatest common factor of any two prime numbers is……..

a) 0 **b) 1** **c) 2** **d) 3**

3) The LCM of 5 and 15 is……..

a) 3 **b) 15** **c) 5** **d) 75**

4) The greatest common factor of 9 and 27 is……..

a) 3 **b) 9** **c) 18** **d) 27**

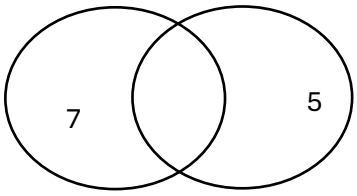
5) Mona saves L.E 480 monthly, how much does she save daily?

a) 16 **b) 20** **c) 7** **d) 8**

6) From the opposite Venn diagram, the LCM is……..

a) 0

c) 35



b) 7

d) 1

7) The Common Factor of all numbers is……..

α) 0 **b) 2** **c) 3** **d) 1**

8) The LCM of 12 and 18 is……..

a) 6 **b) 36** **c) 9** **d) 18**

9) ---------is a multiple of any number

a) 0 **b) 1** **c) 2** **d) 3**

10) The G.C.F of 15 & 30 is……..

a) 15 **b) 30** **c) 5** **d) 3**

11) $\frac{2}{7}+\frac{1}{7}+\frac{3}{7}+\frac{4}{7}=$ **……..**

a) $\frac{7}{9}$ **b)** $1\frac{2}{7}$ **c)** $1\frac{3}{7}$ **d)** $8\frac{1}{4}$

12) Which of the following are relatively prime numbers?

a) 4 & 6 **b) 8 & 15** **c) 8 & 18** **d) 8 & 24**

13) $1\frac{1}{6}-\frac{2}{3}=$**……..**

a) $1\frac{1}{3}$ **b)** $1\frac{2}{3}$ **c)** $\frac{1}{2}$ **d)** $\frac{1}{4}$

14) $45+27=9$ **[……..]**

a) 9 + 3 **b) 5 + 3** **c) 9-3** **d) 4 + 2**

15) L.C.M of 7 & 14 is……..

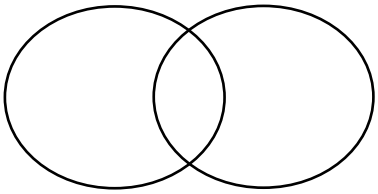
a) 0 **b) 1** **c) 7** **d) 14**

16) $\frac{1}{4}+\frac{1}{5}=$**……..**

a) $\frac{2}{9}$ **b)** $\frac{9}{5}$ **c)** $\frac{9}{20}$ **d)** $\frac{1}{20}$

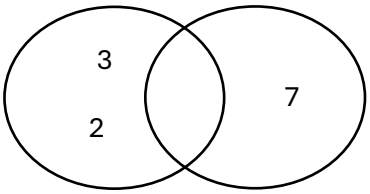
2) Answer the following:

a) Find the GCF and LCM of 10 and 30 using Venn diagram



……
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……
……
……
……
……

b) In the opposite Venn diagram:



i) The GCF is……..

ii) The LCM is……..

iii) The two numbers are…….. and……..

c) Write two numbers whose GCF is 1, ……..and……..

d) 24 volunteers at the food bank donated a total of 3936 hours for the year. Each volunteer worked the same number of hours. How many hours did each volunteer donate to the food bank?

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e) Find L.C.M and G.C.F of two numbers 24 and 30 by using Venn diagram.

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f) A merchant paid 5,287 L.E. to buy 17 boxes of apples. Find the price of each box.

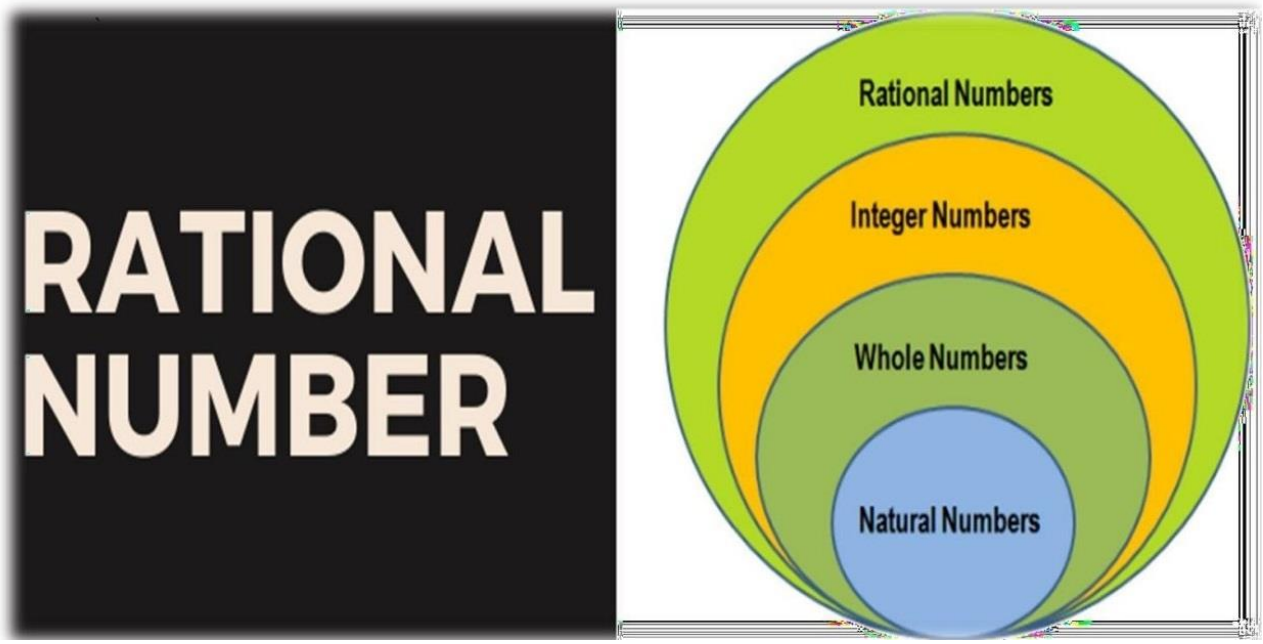
g) If Sally has 28 pieces of chocolate and 35 pieces of marshmallow, what is the greatest number of bags he can prepare with no pieces left over?

Write the expression which represents the total number of pieces.

h) Malek found the packages of bananas. He opened 4 packages each package contains 8 and use some to make some bananas pudding. This what remains of banana packages. $\frac{3}{8}, \frac{5}{8}, \frac{2}{8}, \frac{6}{8}$

1) If you were to regroup the bananas into full packages, how many full packages could be made from the remaining bananas?

2) How much in terms of whole packages did Malek use?



Unit 2 Rational Numbers

Concept 1: Explore the Number Line

Lesson 1: Using a Number Line to Describe Data

Lesson 2: Using a Number Line and Symbols to Compare numbers

Concept 2: Investigate Rational Numbers

Lesson 3: Analyzing Rational Numbers by Using Models

Lesson 4: Comparing and Ordering Rational Numbers

Concept 3: Interpret and Use Absolute Value

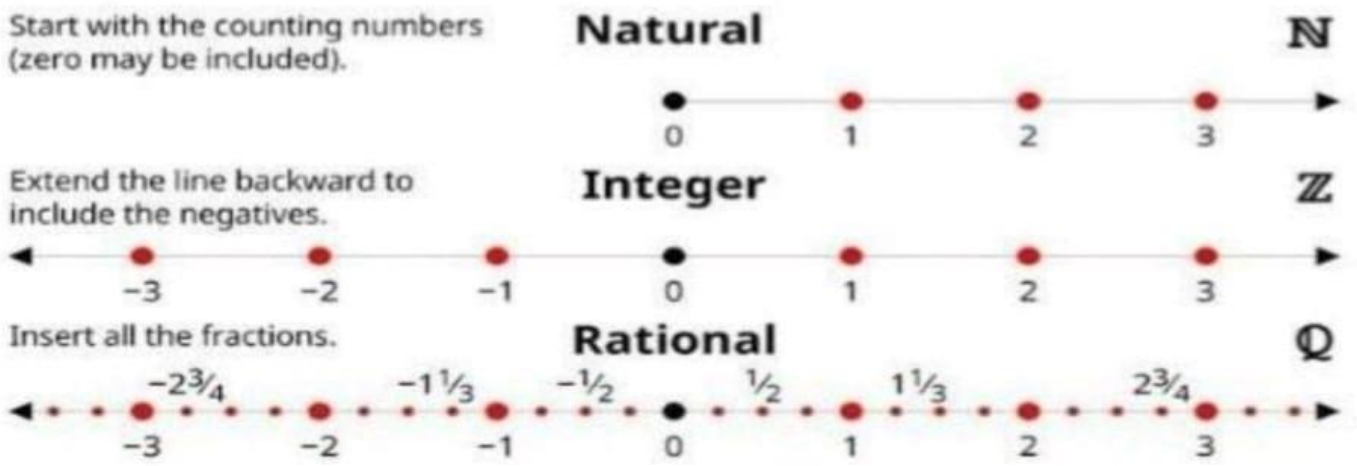
Lesson 5: Exploring Absolute Value

Lesson 6: Comparing Absolute Values

Date……

Lessons 1& 2

Using a Number Line to Describe Data Using a Number Line and Symbols to Compare Numbers

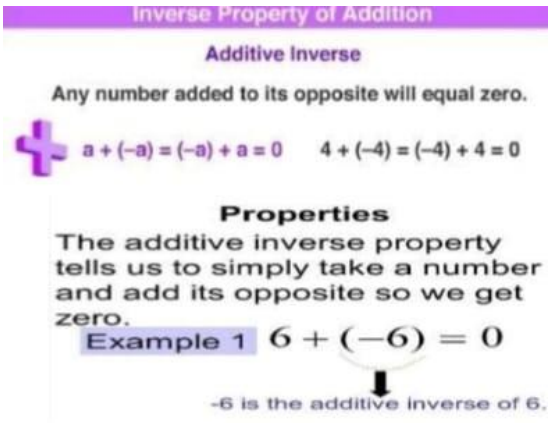


***The set of counting numbers C ={1, 2, 3, 4, 5, 6 ,.....}**

***The set of Natural numbers N ={ 0, 1, 2, 3, 4, 5, 6 ,.....}**
***The set of Integer numbers Z ={ - 6, - 5, - 4, - 3, - 2, - 1, 0, 1, 2, 3, 4, 5, 6 ,....}**

***The set of Rational numbers Q can be made by dividing two integers**

- It can be a whole number, fraction, decimal or a mixed number as: $(7, \frac{8}{10} = 0.8, \frac{0}{5} = 0, \frac{9}{3} = 3, 5\frac{1}{5})$



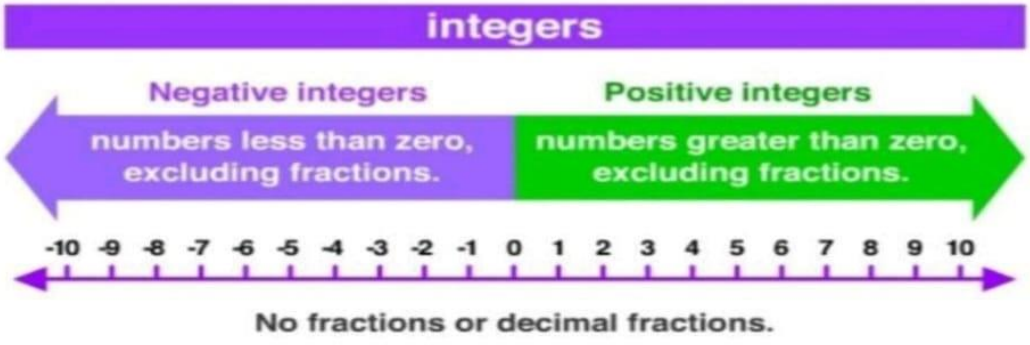
The integer numbers are infinite

The smallest positive integer is 1

The greatest negative integer is -1

Zero is neither negative nor positive

The opposite number (Additive inverse)



Let's Try

1) Write an integer that represents the following:

a) The temperature is 5 below zero……

b) The value of profit is L.E 35……

c) 5 meters below the sea level……

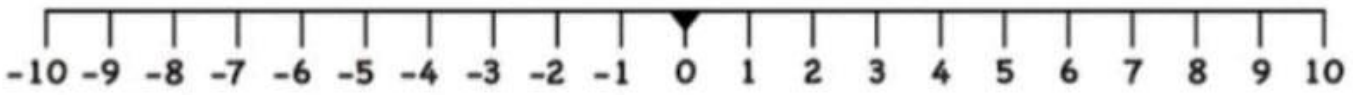
d) Move 4 steps forward……

e) Deposit L.E 850……

f) A loss of L.E 20……

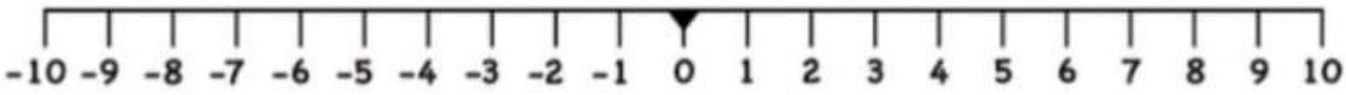
2) Represent the following numbers on the number line:

a) 5, -6, 2, -4, 0, 6, -1



3) Represent each number and its inverse on the number line

a) 4
b) -5
c) 0



1) Put<,> or =:

a) 7--------6

b) 3--------4

c) 1--------0

d) -5--------9

e) -100--------3

f) -30--------8

g) -4--------(-4)

h) –(-3)--------(-7)

i) 2--------2

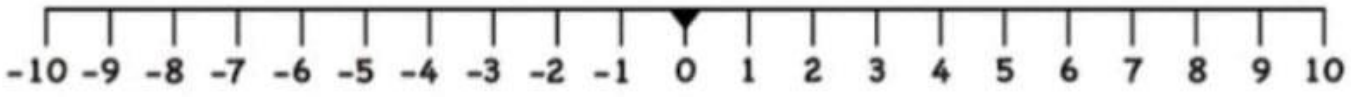
2) Arrange the following in descending order:

5, -6, 2, -4, 0, 7, 8 and -1

--------,--------,--------,--------,--------,--------,--------,--------

3) Represent the following on the number line

7, -5, 4, -3, 0, -7



4) Represent each number and its inverse on the number line

a) **4**
b) **-5**
c) **0**

5) Complete:

a) The greatest negative number is ------- while the smallest positive number is

b) The smallest natural number is--------- while the smallest counting number is ---------

c) The number -------- is neither positive nor negative.

d) The integers between -4 and 2 are ---------

e) The integer just after -1 is ---------

f) The integer just before -13 is ---------

g) The distance between -2 and 3 on the number line equals ---------

6) Write an integer that represents the following:

a) The temperature is 5 below zero ---------

b) The value of profit is L.E 35 ---------

c) 5 meters below the sea level----------

d) A gain of 5 kg = ---------

e) A decrease of L.E 100 = ---------

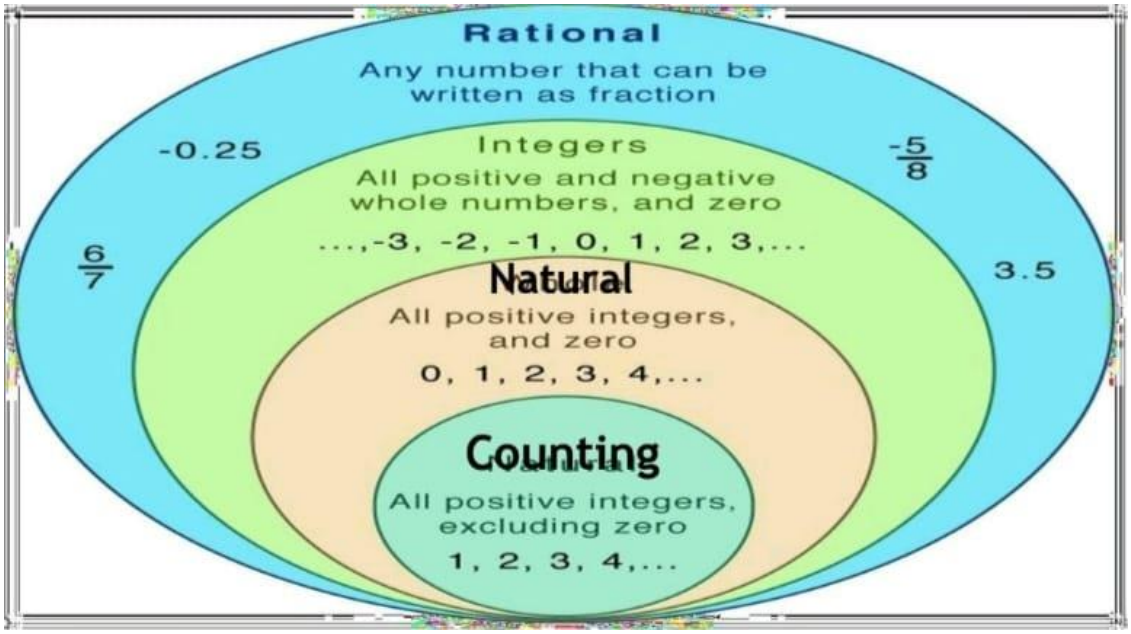
7) Write the additive inverse of the following:

a) 8 ---------
b) -9 ---------
c) 0 ---------
d) 0.7 ---------
e) $-\frac{6}{11}$ ---------
f) $2\frac{4}{5}$ ---------

Date……

Lesson 3

Analyzing Rational numbers by using models



The set of Rational numbers Q

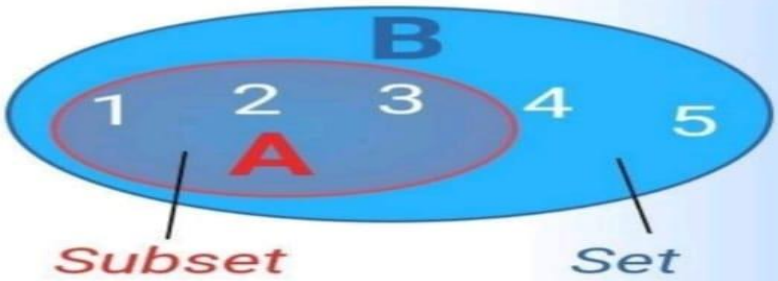
It can be made by dividing two integers, it can be written in the form of $\frac{a}{b}$, b≠ 0 it can be a whole number, fraction, decimal or a mixed number

All Counting numbers, Natural numbers and Integers are rational numbers

Remember $\frac{1}{2}$ = 0.5, $\frac{1}{4}$ = 0.25, $\frac{3}{4}$ = 0.75, $\frac{1}{8}$ = 0.125, $\frac{1}{5}$ = 0.2 , 0.1=$\frac{1}{10}$, 3=$\frac{3}{1}$, 0=$\frac{0}{1}$

Part of another set.

A is a subset of B when every member of A is a member of B.

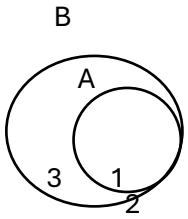


Example: B = {1,2,3,4,5}
Then A = {1,2,3} is a subset of B
Other subsets of B include {2,3} or {1,4,5} or {4} etc…
But {1,2,6} is NOT a subset of B as it has 6 (which is not in B)

Ex.1

$A = (1, 2), B = (1, 2, 3)$

$A \subset B$



A subset of B

Symbol	Meaning
$\in$	'is a member of' or 'is an element of'
$\notin$	'is not a member of' or 'is not an element of'

$A \subset B$, $B \not\subset A$

$2 \in A$, $3 \notin A$ (2 is an element of set A , 3 is not an element of set A)

$\{1, 2\} \subset B$

Let's Try

1) **Write the following Rational numbers in the form $\frac{a}{b}$:**

a) -6 =--------- b) -0.24 =-------- c) $4\frac{2}{5}$ =---------

d) 8 = --------- e) $4\frac{2}{5}$ =--------- f) 7.3 = ---------

2) Match the numbers to the best subset:

(Rational number- Integer - Counting numbers - Natural numbers)

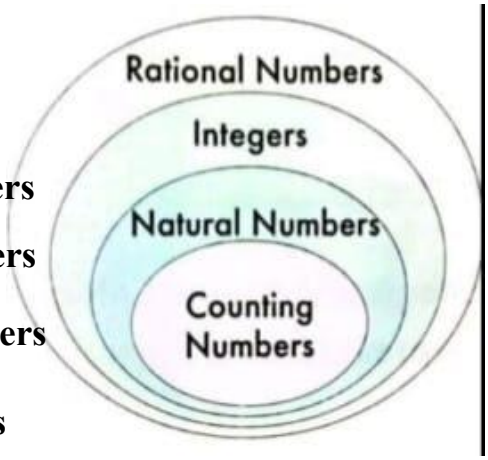
a) 0.674 =---------- b) -3 =---------- c) 0 =----------

d) $\frac{3}{8}$ =---------- e) 6 =---------- f) 5.9 =----------

g) 18.62 =---------- h) -38 =---------- i) 2541 =----------

j) $5\frac{1}{2}$ =---------- k) 11 =---------- l) -6.7=----------

3) Write belongs or doesn't belong:



a) $-\frac{3}{8}$**----------the set of Integers**

b) 0---------- the set of Rational numbers

c) -6----------the set of counting numbers

d) $\frac{8}{4}$**----------the set of Counting numbers**

e) $\frac{0}{9}$**----------the set of Natural numbers**

4) Write subset or not subset:

a) The set of Counting numbers---------- the set of Integers

b) The set of Rational numbers numbers---------- the set of Natural

c) The set of Natural numbers numbers---------- the set of Counting

d) The set of Integers---------- the set of Rational numbers

Evaluation

1) Choose:

1) The additive inverse of -7.3 is----------

a) 73 **b) 10** **c) 3.7** **d) -3.7**

2) The best subset for the number -5 is a/an ----------number

a) Counting **b) Natural** **c) Integer** **d) Rational**

3) The greatest negative integer is----------

a) -99 **b) 0** **c) -1** **d) 1**

4) $\left|\frac{-5}{11}\right|$ **belongs to the set of ----------numbers**

a) Counting **b) Natural** **c) Integer** **d) Rational**

5) Which of the following is the smallest number?

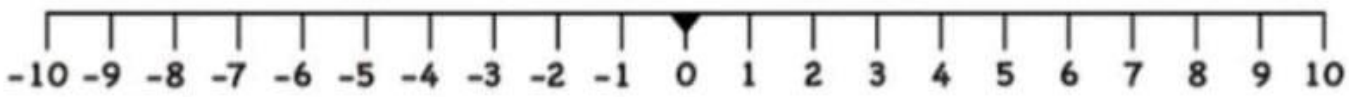
a)-3.8 **b) -8.3** **c) -2.3** **d) -3.2**

6) The best subset of the number 1 is---------- number

a) Counting **b) Natural** **c) Integer** **d) Rational**

2) Represent the following on the number line:

$$4\frac{1}{2}, 2.5, -3\frac{1}{3}, -6.25, 0, -8$$



Date……

Lesson 4

Comparing and Ordering Rational Numbers

Ex.1

Order fractions with the same denominator

To order fractions of the same denominator we need to look at the numerator.

$\frac{6}{8}, \frac{1}{8}, \frac{4}{8}, \frac{3}{8}, \frac{7}{8}$

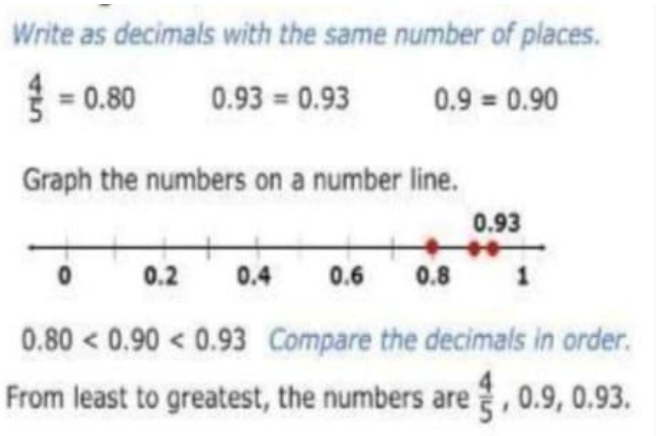
Smallest to largest

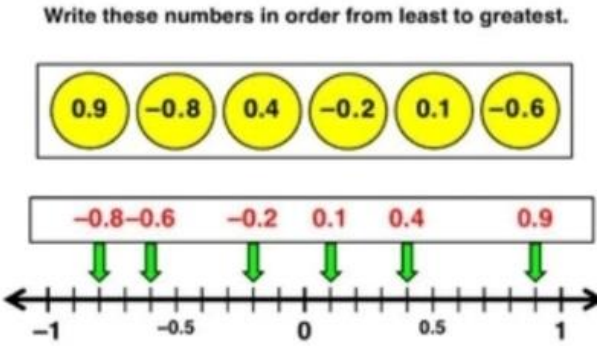
$\frac{1}{8}, \frac{3}{8}, \frac{4}{8}, \frac{6}{8}, \frac{7}{8}$

Ex.2

Order fractions and decimals

Order, 0.93, and 0.9 from least to greatest.





Let's Try

1) Arrange the following from the least to the greatest:

$0.26, 0.74, -0.5, -0.45, 0.1$

--------,--------,--------,--------,--------

2) Arrange the following from the greatest to the least:

$3.5, 3\frac{3}{5}, -3, 3\frac{1}{2}, -11$

--------,--------,--------,--------,--------

3) Write a rational number lying between the following:

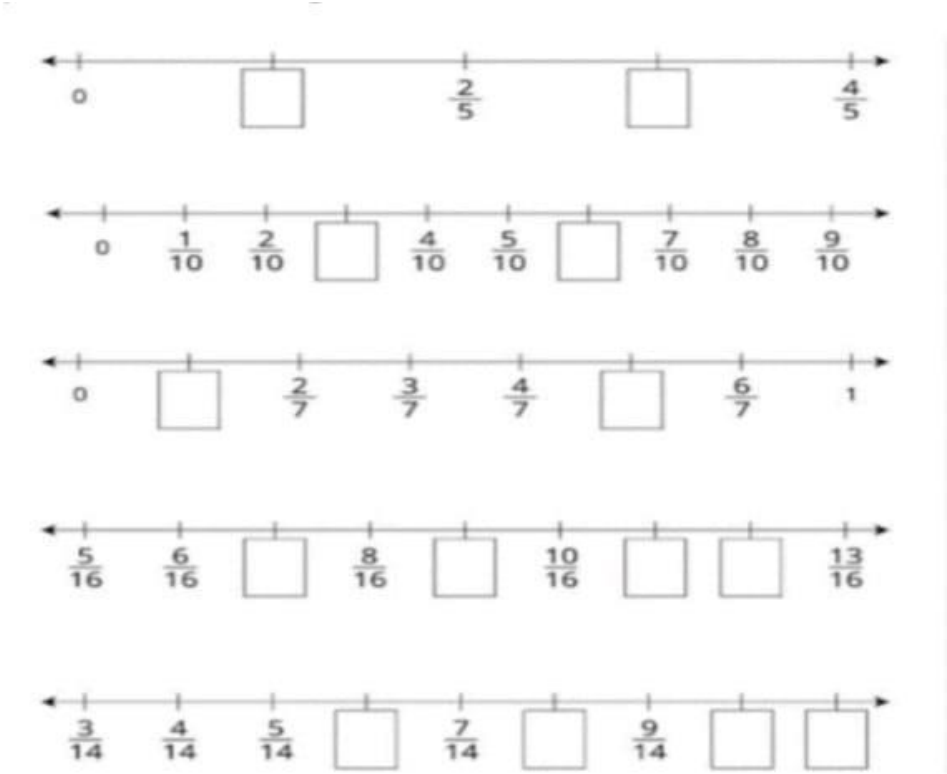
a) $\frac{1}{8}$ -------- $\frac{3}{8}$

b) $\frac{1}{3}$ -------- $\frac{5}{6}$

c) -8.2 ------- -8

d) $\frac{1}{5}$--------$\frac{1}{2}$

4) Complete the missing number on the number line:



Evaluation

1) Choose:

1) The rational number between -4.5 and -4.4 is ……

a) -4.63 b) -4.43 c) -4.2 d) -4.3

2) The number $-2\frac{1}{2}$ in the form $\frac{a}{b}$ is ……

a) $-\frac{5}{4}$ b) $-\frac{9}{4}$ c) $-\frac{5}{2}$ d) $-3/2$

3) The opposite of 0 is ……

a) -1 **b) 0** **c) 1** **d) otherwise**

4) $-2.25 <$ ……

a) -5.3 **a) $3\frac{1}{2}$** **c) 0** **d) -4**

5) The smallest number from the following is ……

a) $\frac{1}{2}$ **b) 1** **c) 0.12** **d) 0.21**

6) $-1.62 <$ ……

a) -1.6 **b) -1.63** **c) -2.6** **d) -3**

<u>2) Draw a number line and write a number located between each pair of numbers:</u>

a) 7.85 and 7.86

……

b) $-\frac{3}{4}$ and $-\frac{1}{2}$

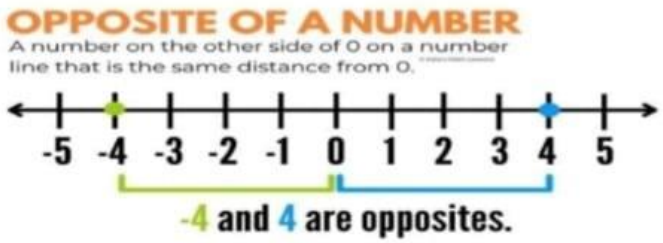
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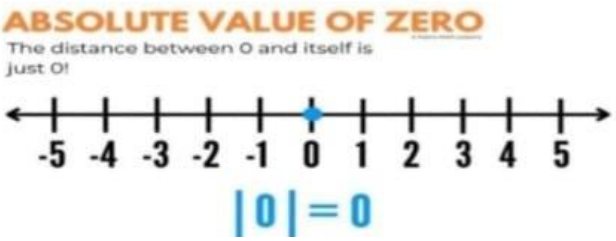
Date……

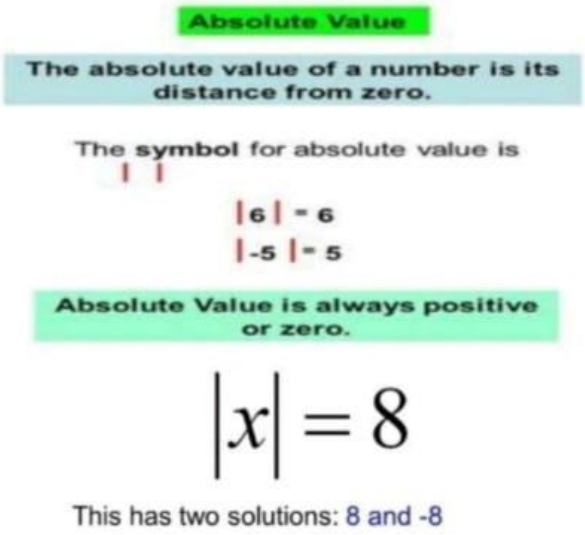
Lessons 5 & 6

Exploring Absolute Value

Comparing Absolute Values







Let's Try

1) Find the value of the following:

a) $|-5| =$…… **b)** $-|7| =$…… **c)** $\left|\frac{3}{4}\right| =$……

d) $|0| =$ …… **e)** $\left|-\frac{6}{11}\right| =$…… **f)** $|-6| =$ ……

g) $|-3| - |-3| =$ …… **h)** $|-2| \times |-7| =$ …… **i)** $|-24| + |6| =$……

2) Find the value of x:

a) $|x| = 6$ **b)** $|x| = 0$ **c)** $|-1| = x$

$x =$…… $x =$…… $x =$……

Evaluation

1) Put<, >or =:

a) $|-5| = \ldots\ldots\ldots\ldots |-3|$

b) $|0| \ldots\ldots\ldots\ldots |-6|$

c) $|2.5| \ldots\ldots\ldots\ldots \left|-2\frac{1}{2}\right|$

d) $|-9.3| \ldots\ldots\ldots\ldots |-10|$

e) $|-7.54| \ldots\ldots\ldots\ldots |7.5|$

f) $4\frac{1}{3} \ldots\ldots\ldots\ldots \left|-\frac{12}{3}\right|$

2) Find the value of the following:

a) $|0| + |-5| = \ldots\ldots\ldots\ldots$

b) $|-9| + |3| = \ldots\ldots\ldots\ldots$

c) $|-90| - |-40| = \ldots\ldots\ldots\ldots$

d) $|14| - |-14| = \ldots\ldots\ldots\ldots$

e) $|-40| \div |8| = \ldots\ldots\ldots\ldots$

f) $|-36| \div |-6| = \ldots\ldots\ldots\ldots$

g) $|-9| \times |-4| = \ldots\ldots\ldots\ldots$

h) $|0| \times |-7| = \ldots\ldots\ldots\ldots$

i) $|-24| \div |6| = \ldots\ldots\ldots\ldots$

j) $|-28| \div |4| = \ldots\ldots\ldots\ldots$

3) Find the value of x:

a) $|x| = 9$

$x = \ldots\ldots\ldots\ldots$

b) $|x| = 15$

$x = \ldots\ldots\ldots\ldots$

c) $|x| = 0$

$x = \ldots\ldots\ldots\ldots$

d) $|5| = x$

$x = \ldots\ldots\ldots\ldots$

e) $|4| = x$

$x = \ldots\ldots\ldots\ldots$

f) $|-87| = x$

$x = \ldots\ldots\ldots\ldots$

Date……

Unit Two Assessment

1) Choose:

a) $|2| \times |-6|$=……

a) 0 **b) 12** **c) -12** **d) -4**

b) $|-8| <$ ……

a) $|-7|$ **b) $|-6|$** **c) $|-9|$** **d) $|-8|$**

c) $\left|-3\frac{1}{4}\right| \div \left|\frac{1}{2}\right| =$ ……

a) $-6\frac{1}{2}$ **b) $6\frac{1}{2}$** **c) $6\frac{1}{4}$** **d) $\frac{1}{8}$**

d) The distance between 5 and | -5 | on the number line is ……units.

a) 10 **b) 0** **c) -5** **d) 5**

e) $|8| \times -|-8| =$ ……

a) -64 **b) 16** **c) 0** **d) 64**

f) $|-3.27| <$ ……

a) 3 **b) 3.26** **c) 3.28** **d) 2**

g) The absolute value of $-3\frac{1}{4}$ = ……

a) $3\frac{1}{4}$ **b) 0** **c) $-3\frac{1}{4}$** **d) $\frac{13}{4}$**

h) The opposite of $-\frac{4}{9}$ is ……

a) $-\frac{4}{9}$ **b) $\frac{4}{9}$** **c) $-\frac{|4|}{9}$** **d) $\frac{9}{4}$**

i) All integers are also …… numbers

a) Counting **b) Natural** **c) Rational** **d) Even**

j) The rational number between -4.1 and -4.17 is……

a) -4.2 **b) 4.15** **c) -4.18** **d) -4.3**

2) Answer the following:

a. Arrange in descending order

$$-8\ ,\ 3\frac{3}{5}\ ,\ |-8|\ ,\ 2\ ,\ \left|-5\frac{3}{5}\right|$$

--------,--------,--------,--------,--------

b. Arrange in ascending order

$$\left|-\frac{12}{4}\right|\ ,\ 2\ \frac{1}{5}\ ,\ -|-4|\ ,\ 0\ ,\ \left|-1\frac{1}{2}\right|$$

--------,--------,--------,--------,--------

c. Find the rational number lying between 7.5 and 7.6 --------

d. The opposite of 0 is--------

e. The number $3\frac{1}{5}$ in the form of is $\frac{a}{b}$ is --------

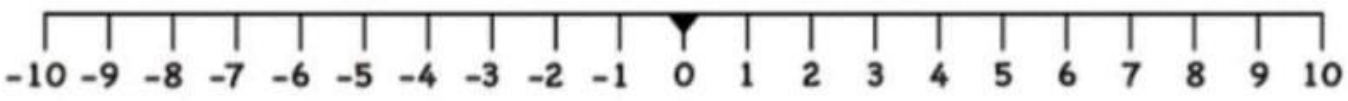
f. The opposite of $|-7|$ is -------

g. $|-45| = x$, then x =--------

h. $|x| = 4$ then x =--------

3) Order the numbers on the number line

$$4.5\ ,\ |-3\frac{1}{4}|\ ,\ -|-7|\ ,\ |-8.5|\ ,\ -2$$





Unit 3: Algebraic Expressions

Concept 1: Use and Analyze Expressions

Lesson 1: Creating Mathematical Expressions

Lesson 2: Analyze Mathematical Expressions

Lesson 3: Writing Algebraic Expressions

Date:……..

Lesson 1 & 2

Creating Mathematical Expressions

Analyzing Mathematical Expressions

Using Letters or Symbols to Represent Unknown Values in Mathematical Expressions

Example:

On a sale, a clothes store sells any piece of clothes for half its original price.

1) Complete the table

Original Price	Price after dicount
120LE	--------
150LE	--------
--------	80LE
--------	50LE

$120 \div 2 = 60$

$150 \div 2 = 75$

$80 \times 2 = 160$

$50 \times 2 = 100$

2) Write mathematical expression to represent the price after discount of a piece of clothes whose original price is KL.E.

(K÷2) L.E.

Let's define some important concepts:

1) Variable (unknown): It's a letter or a symbol that represents the unknown value.

Such as: x, y, z,--------

2) Constant (absolute term): It's a number in which its value is the same (can't be changed).

Such as: 1, 7, 9, --------



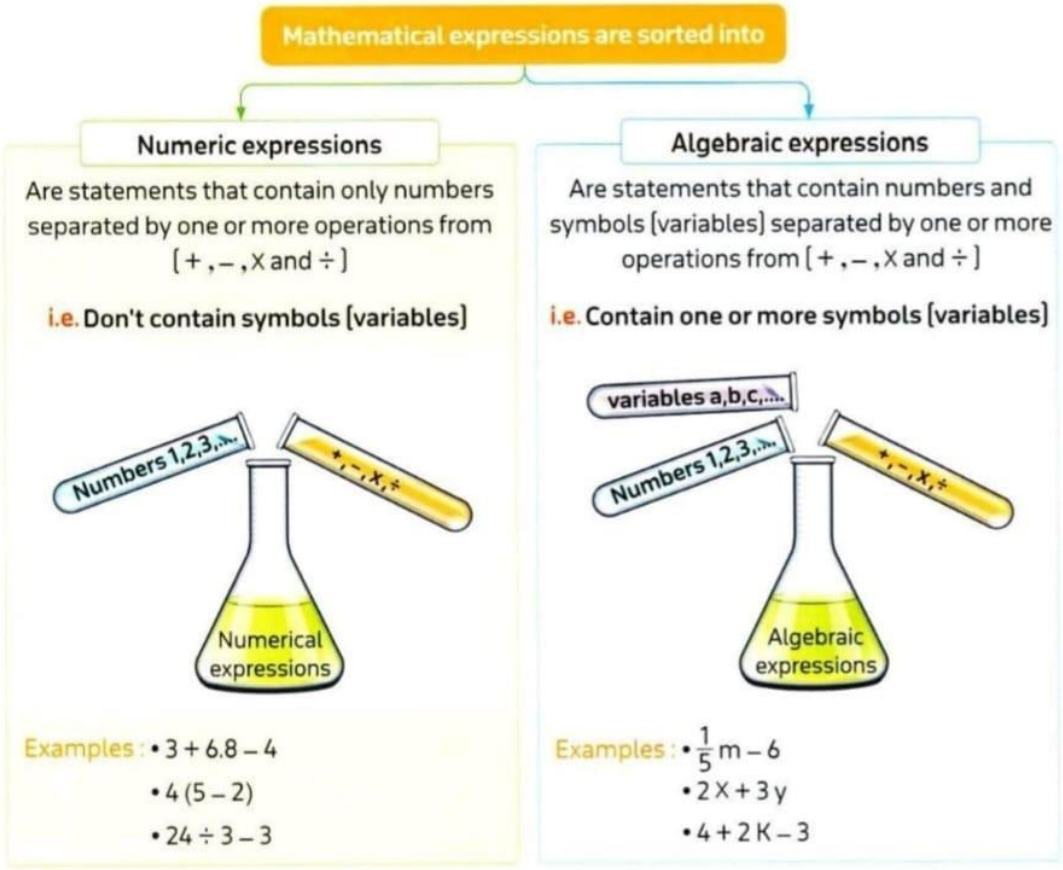
3) Mathematical Expression: Is a statement contains numbers or numbers and symbols separated by one or more operations as: [+, -, x and÷] and doesn't contain the equal sign " = ".

Such as: $7.4 + 2.5 - 1.5$

10×35

$2.5 + m$

$49 - x - 24.5$





4) Term:

Each content (number of variable or product of both) of a mathematical expression that separated by the sign + or -

5) Equation:

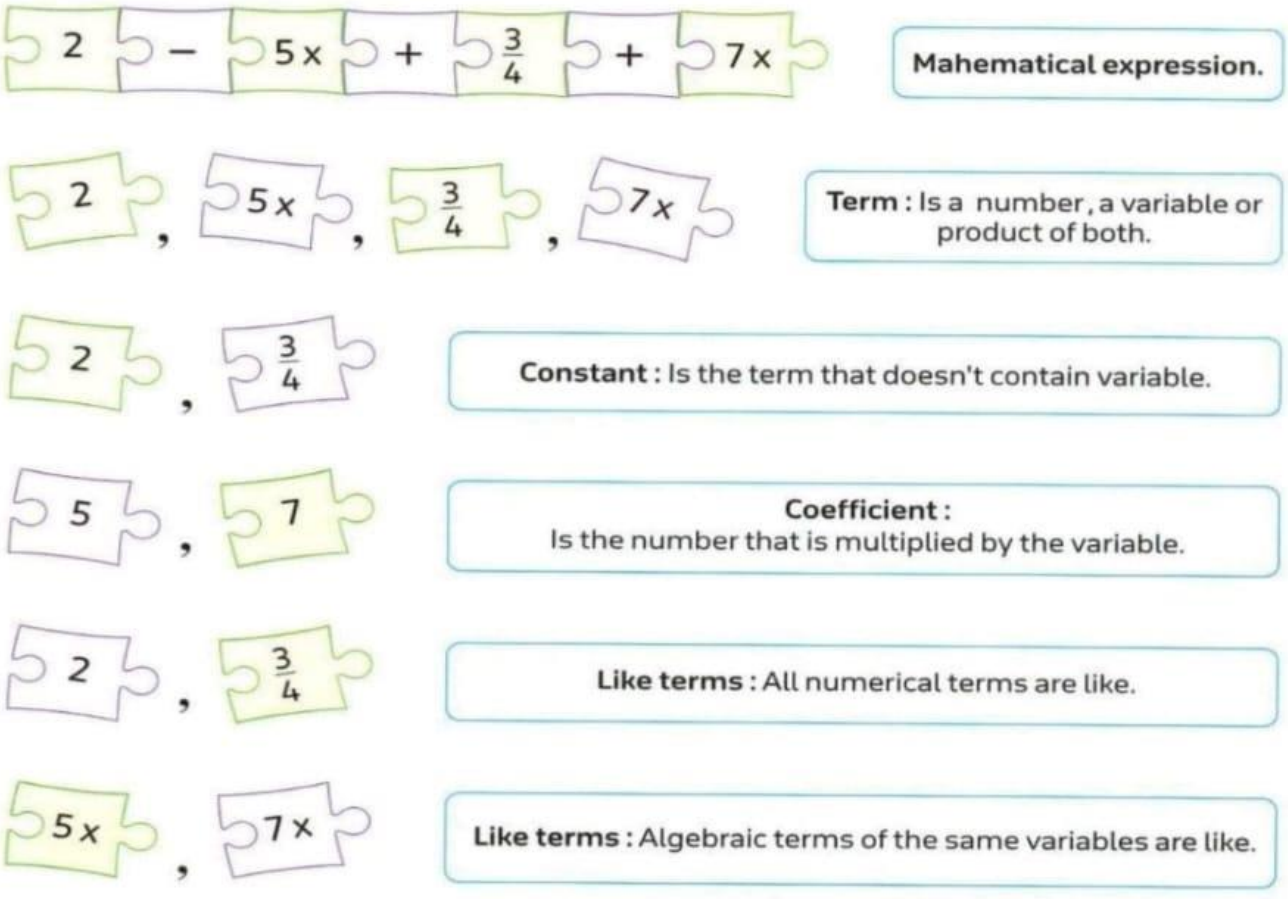
Is a mathematical expression containing the equal sign " = ".

Such as: $24.8 - x = 17.5$

$36.5 + 14.1 = k$

$4.2 + 1.5 = 8.9 - 3.2$

$7.36 + 1.036 + 2.5 = b$



Let's Try

1) Put a tick (✓) to classify the following mathematical sentences into "Equation" or "Mathematical Expression" or "other":

	Equation	Mathematical Expression	Other
1) $15 \times 7 = x$			
2) The sum of the two numbers is 13.8.			
3) $25.06 + 6.2 + 5$			
4) Aya ran 8 km last week			
5) $56 - x = 47.5$			
6) $30 \times m = 300$			
7) The sum of two numbers is 13.8			
8) $37.125 - 13.7$			
9) $3.5 + 2.456 = 2.5 + 3.456$			
10) Sara bought a shirt for 145.75 L.E. and a skirt for 189.5 L.E.			
11) $4.7 + 3.6 = m$			

2) Complete the following tables:

a)

Algebraic Term	Number of Factors	Coefficient	Algebraic Factors
$9xy$			
$3m$			
$-5y$			
$\frac{1}{3}ab$			
$-\frac{3}{7}n$			
$6xyz$			

b)

Expression	Number of terms	Like terms	Constants	Coefficients
$5a - 3b + 4$				
$k + k + 6k$				
$3 - b + 5$				
4				
$\frac{1}{2}n - 2n + k$				
$0.5a + 0.1a + 3a$				
$7 + 3 + 2$				
Z				

c)

Algebraic Term	Number of Factors	Coefficient	Algebraic Factors
$-3ab$			
$5x$			
$-3y$			
$\frac{1}{5}xy$			
$-\frac{2}{8}m$			
$8abc$			
$5xyz$			
$7r$			
$6n$			
$\frac{3}{7}km$			
$23ab$			
$\frac{1}{6}yz$			

3) If the ticket of the amusement park is 50 L.E. and fees of 20 L.E. for each game you play inside.

a) Write an algebraic expression to represent the relation between the number of games you played and the total cost.

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b) How much money you will pay in total if you want to play 5 games?

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Evaluation

1) Complete the following tables:

a)

Mathematical Expression	Variable	Absolute Term	Coefficients
$5a + 8$			
$9x - 5$			
$3a + 4b + 7$			
$6x + 15 + 2.5$			
$5y + 63$			
$8r + 1.3 + 7$			
$0.2m + 12 - 0.3h - 5$			
$3p + 15c + 4 + 2$			
$6w + 2 + 3 + 0.2$			
$2.5\,q + 7 + 3 + 2.4$			

b)

Mathematical Expression	Number of Terms	Like Terms
9		
3.9		
$5a$		
$7m$		
$3x + 7x$		
$8a + 5a$		
$2y + 7$		
$15 + 3 + x + y$		
$36 + \frac{1}{2}a\,a + 12 + a$		
$3b + 5b + 5 + 2b$		

c)

Expression	Number of terms	Like terms	Constants	Coefficients
$2m + 5$				
$k + 3 + 4k$				
7				
$\frac{4}{5}b + 5q - 7h$				
$8a$				
$3 + 5$				
2×4				

2) In the midtown car park, the charge is 10 L.E. per hour. Think about the relation between the parking time (in hour) and the charge (in L.E.), then answer the following:

a) Complete the following table:

Number of hours	How to calculate	Charge
3		
5		
7		

b) What values are always the same in this situation and what values vary?

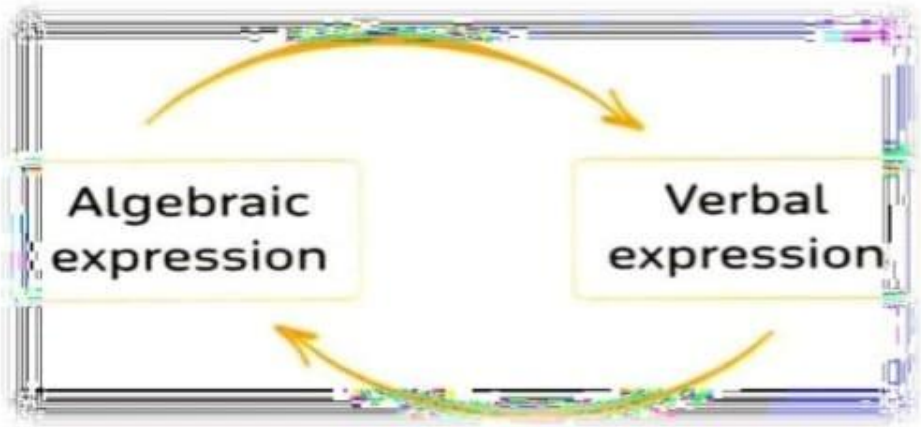
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c) What algebraic expression would represent the charge in L.E. where the time is (t) hour?

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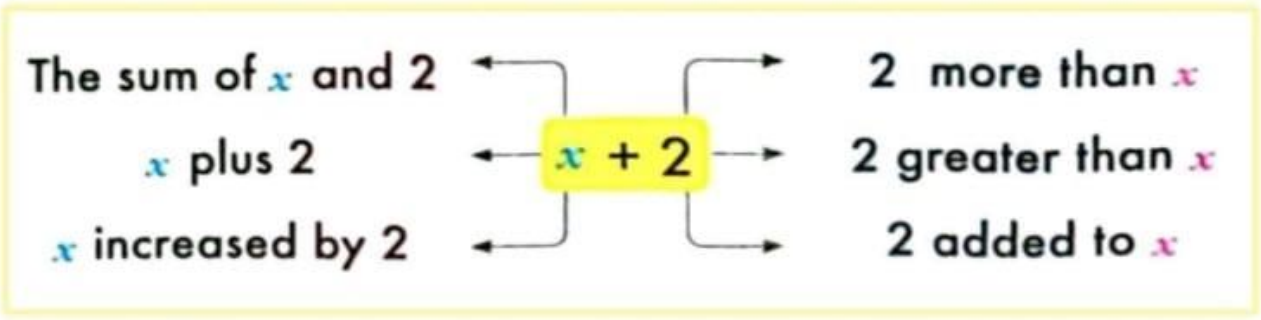
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Lesson 3
Writing Algebraic Expressions



Addition	Subtraction	Multiplication	Division
add sum increased by more than exceeds total plus in all gain deposit	subtract difference decreased by fewer less than diminished by minus take away withdraw reduced by magnitude of increase	multiplied by Product of times double triple twice	per divide quotient divided equally split into fraction ratio of half third

The following algebraic expression can be expressed using the verbal form in more than one way:



Note:

- **x plus 8 is $x + 8$**
- **x minus 3 is $x - 3$**
- **3 times x is $3x$**
- **Half of x is $\frac{1}{2}x = \frac{x}{2}$**
- **The sum of x and 2 is $x + 2$**
- **7 minus X is $7 - x$**
- **4 times x is $4x$**
- **Third of x is $\frac{1}{3}x = \frac{x}{3}$**
- **Twice x is $2x$**
- **5 times x is $5x$**
- **Sum of the double of x and 5 is $2x + 5$**
- **Double of the sum of x and 5 is $2(x + 5)$**
- **The difference between 7 times x and 3 is $7x - 3$**
- **7 times the difference between x and 3 is $7(x - 3)$**

Let's Try

1) Complete the following table:

Verbal expression	Algebraic expression
Add 3 to the double of the number x	
Subtract 5 from the double pf the number y	
Add 7 to three times of the number z	
Subtract 3 from the half of the number x	
Add 6 to one third of a number	
Four times the sum of a number and seven	
The sum of four times a number and seven	
The quotient of a number by 8 is increased by 12	
Half the product of a number by itself	
8 decreased by three times a number	
Twice the sum of a number and three	
17 less the product of a number by itself	

2) Write verbal expression for each of the following algebraic expressions:

a) $\frac{1}{2}b + 15$

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b) $8 - 3n$

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c) $2m + 7$

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d) $3(m + 7)$

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e) $x - 6$

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f) $2n - 8$

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g) $5k - 2$

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h) $2y$

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Evaluation

1) Write an algebraic expression for each of the following verbal expressions:

a) Add 5 to x

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b) Three fifths a number n

……

c) Take away a number k from 18

……

d) Quotient of a number h by 9

……

e) A number x multiplied by itself.

……

f) Seven increased by a number s

……

g) Nine more than a number x

……

h) Divide the number m by 3

……

i) A number w decreased by 5

……

j) A number less than ten.

……

k) Three times the sum of a number and 4

……

l) Roudy deposits 5,000 pounds in her bank account.

……

m) An amount of money is divided among 5 persons, then each of them spends 27 pounds.

……

n) Moustafa bought 7 boxes, each x pens.

……

o) The perimeter of rectangle whose length 10 cm and width w cm.

……

p) Subtract 3 from, 5 times a number.

……

q) Distribute (x + 5) pieces of candy among 8 boys.

……

r) 8 less than twice a number n

……

s) The product of a number and itself.

……

t) The sum of 2 and 5 multiplied by x .

……

u) The sum of 2 times x and 5 .

……

v) Twelve less than three groups of y. .

……

w) The product of 2 x and 5 .

……

2) Write a verbal statement or real-world situation that could be represented by an algebraic expression that uses division. What is the algebraic expression that models your verbal statement?

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3) A square of side length I, write an algebraic expression for the perimeter of this square.

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4) Sara has 150 pounds, she saves x pound daily, after a week she paid third of the money as a donation.

Write an algebraic expression for the remainder money with Sara.

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5) Laila ate x sandwiches, and her father Ahmed ate 3 sandwiches, if this price for each sandwich is 12 pounds.

Write two algebraic expressions to find the cost of the meal.

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6) Write an algebraic expression to compute the number of liters of gas Dina needs to buy to make the round trip to visit her grandma. Her car gets 15 kilometers per liter.

Use the variable d to represent the distance in kilometers to her grandma's house.

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Date: ……

Unit Three, concept One Assessment

1) Complete the following :

a) The verbal form for the algebraic expression 6 m is ……

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b) Nada is now "Y" years old. How old was she 3 years ago?

…..

c) A square of side length "S" cm has a perimeter of …… cm.

d) The price of a kilogram of meat increased by 120 pounds. If its price becomes "x" after the increase, then its price before the increment is ……

…..

e) The algebraic expression that represents (three times the sum of the number "m") and 12 is ……

f) If we subtract 5 from number "X" , the result is ……

g) The constant in the algebraic expression 5b + 3.2 is ……

h) The number of terms of the algebraic expressions

2.5x + 2xy - 4 is

……

i) Ahmed and Nader have 60 pounds. If what Ahmed has is x pounds, then what Nader has is …… pounds.

j) The value that expresses the verbal form "three times b" is:

…..……

k) The verbal form for the algebraic expression $\frac{a}{5}$ + 3 is:

……

l) If the height of the school building is "m" meters and the height of the tree adjacent to this building is 10 meters less than it, then the height of the tree is …… meters.

2) Complete using the mathematical expression "5x + 2y + 6x + 3":

a) The number of terms of the mathematical expression is: ……

b) Like terms are: ……

c) Coefficients are: ……

d) Constants are: …….

3) Write each of these algebraic expressions using the verbal form:

a) 7 − 2x

……

……

……

b) 7.5 − f

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c) 5(c − 3)

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4) Bassem runs one kilometer in 15 minutes. Write a mathematical expression that expresses the number of kilometers that Bassem runs in "t" minutes.

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5) Write a mathematical expression that represents each of the following situations:

a) The mass of the astronaut on the moon is 1/6 of his mass on earth. If the mass of an astronaut on earth is m kg, then his mass on the moon is:

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b) If Ahmed sleeps for 7 hours a day, then the number of hours he sleeps in n days is:

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c) Kareem bought 4 boxes, each containing "c" cookies.

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d) Half of x:

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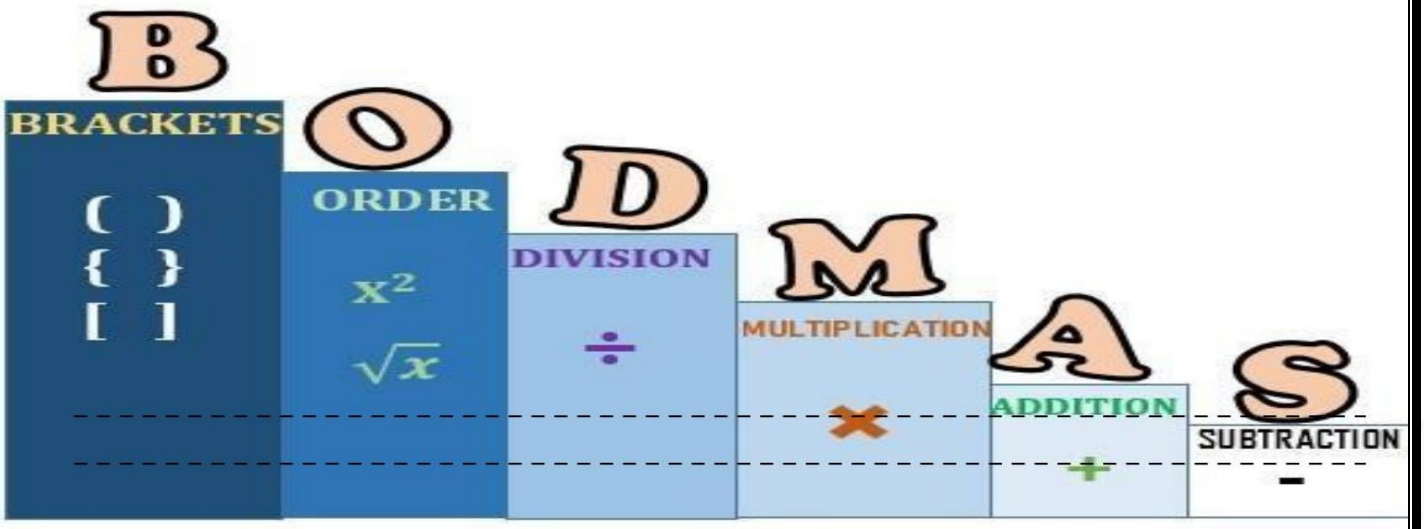
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e) 5 more than k:

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Unit 3, Concept 2

Algebraic Expressions and Exponents

Lesson 4: Ordering of Operations and Exponents

Lesson 5: Evaluating Algebraic Expressions

Lesson 6: Applications on Algebraic Expressions

Lesson 7: Determining Equivalent Algebraic Expressions

Date: .....

Lessons 4, 5 & 6

Ordering of Operations and Exponents

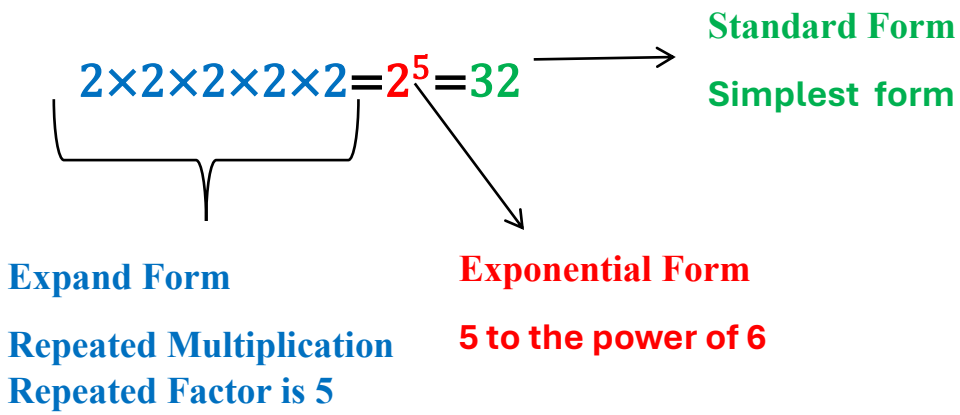
Evaluating Algebraic Expressions

Applications on Algebraic Expressions

Exponent: is a mathematical form for the repeated multiplication focused on a repeated factor which is called "base" and the number of times the base is used as a factor is called "exponent", "power" or "index".

product of repeated multiplication can be written in 3 forms:

For Example:



Any number to the first power is the number itself.

For example:

$5^1=5$ $7^1=7$ $9^1=9$

The number one to the power of any number is the number one itself.

For example

$1^3=1\times1\times1$ $1^5=1\times1\times1\times1\times1$

The second power of a number is called the square of this number.

7^2 is read as "the square of 7"

The third power of a number is called the cube of this number.

7^3 is read as "the cube of 7"

1) Preform operations within parentheses → ( ).

1) Multiply or divide from left to right.
2) Add or subtract from left to right.

1) Preform operations within brackets → [ ].

1) Multiply or divide from left to right.

2) Add or subtract from left to right.

3) Preform operations outside of parentheses or brackets.

1) Multiply or divide from left to right.

2) Add or subtract from left to right.

EX:

operations within parentheses

Adding → $8 \times [24 \div (4 + 2) - 1] + 5$

operations within brackets

1) Dividing →

2) Subtracting → $= 8 \times [24 \div 6 - 1] + 5$

1) →

2) Adding → $= 8 \times [4 - 1] + 5$

$= 8 \times 3 + 5$

1) Preform operations within parentheses ( ). →

- **1) Exponents**
- **2) Multiply or divide from left to right.**
- **3) Add or subtract from left to right.**

2) Preform operations within brackets [ ]. →

- **1) Exponents**
- **2) Multiply or divide from left to right.**
- **3) Add or subtract from left to right.**

4) Preform operations outside of parentheses or brackets.

- **1) Exponents**
- **2) Multiply or divide from left to right.**
- **3) Add or subtract from left to right.**

EX. Use the order of operations to evaluate the expression:

a)

$(15-9)+3\times 4^2\div 2$

$=\ 6\ +3\times 4^2\div 2$

$=\ 6\ +3\times 16\div 2$

$=\ 6\ +\ 48\ \div 2$

$=\ 6\ +\ 24$

$=\ 30$

b)

$6^2\div(17-8)\times(3+4)$

$=\ 6^2\div\ 9\ \times\ 7$

$=\ 36\div\ 9\ \times\ 7$

$=\ 4\ \times\ 7$

$=\ 28$

Let's Try:

1) Complete the following tables:

a)

Exponential form	Base	Exponent	Expanded form	Simplest form
2^3	___	___	___	___
___	7	2	___	___
___	___	___	$3\times 3\times 3\times 3$	___

b)

Base	Exponent	Exponential form	Expanded form	Simplest form
5	2	___	___	___
___	___	3^3	___	___
___	___	___	$2 \times 2 \times 2 \times 2 \times 2$	___
4	___	___	___ × ___ × ___	___
___	___	___	___	9

2) <u>Complete:</u>

a) Three cubed = ……

b) 9 to the power of

c) 4 = …….

d) 3^3 = ……..

e) 5 squared = ……

f) $K \times K \times K \times K \times K$ = …….

g) $Y \times Y$ = ……

h) $6 \times 6 \times 6 \times 6 =$ **……**

i) $7 \times 7 \times 7 =$ **……**

3) Use order of operations and exponents to simplify each of the following:

a) $8 + 4^2 - 5 + 6(60 \div 20)^2$

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b) $(8 - 6 \div 2)^2 + 3 \times 4$

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c) $(13 - 5) \times 3 + 7 + 16 - 8$

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d) $12 - 8 \div 4 + [(6 + 1) - 2]^2 \times 3$

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4) Evaluate each algebraic expression for the given value of the variable.

a) $7k - 15 \ \ at \ k = 6$

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b) $6 \div (8x - 3) \ \ when \ x = 0.5$

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c) $\mathbf{5}\boldsymbol{h}^{\mathbf{2}}\mathbf{(6 - 4)}\ \ \boldsymbol{for}\ \boldsymbol{h} = \mathbf{3}$

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d) $\mathbf{9 + (}\boldsymbol{p}^{\mathbf{2}}\mathbf{- 3) \div 2}\ \ \boldsymbol{for}\ \boldsymbol{p} = \mathbf{5}$

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5) You want to buy some T-shirts. Each T-shirt is 100 L.E. and you have a coupon for 40 L.E. off your entire purchase.

a) What algebraic expression can you write to represent the situation?

Explain what each symbol in your expression represents.

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b) What is the cost of 4 T-shirts? Explain how you found that cost.

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6) The fare of a taxicab is 15 L.E. as a base fare and 8 L.E. for each Km.

a) Write an algebraic expression to represent the total fare of the taxicab.

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b) What is the fare for traveling 7 Km?

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Evaluation:

1) Use order of operations and exponents to simplify each of the following:

a) $7^2 - 3(15 - 10) \div 3$

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b) $(17 - 11) + 3 \times 4^2 + 8$

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c) $13 - 7 + 8 \times 9 + 63 \div 7$

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d) $\mathbf{4(3 + 2 \times 3) - 5^2}$

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2) Evaluate each algebraic expression for the given value of the variable.

a) $\mathbf{4(h + 8)}$ $\boldsymbol{at\ h = 1}$

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b) $34 + b(4^2 \div 2)$ when b = 6

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c) $9 + 3(m^2 + 2)$ for m = 5

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d) $n^2 \div 2 + 5$ for n = 4

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3) Nada has 25 L.E. in her money box; she will save 20 L.E. daily.

a) What algebraic expression represents this situation?

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b) How much money is in the money box after 3 days?

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c) How much money is in the money box after 6 days?

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Date: .....

Lessons 7

Determining Equivalent Algebraic Expressions

Algebraic expressions are equivalent if they represent the same value for every value of the variable.

For Example:

- **The two algebraic expressions 2x + 10 and 2(x + 1) + 8 are equivalent.**

Substitution value for the variable	2x + 10	2(x + 1) + 8	Result
For x = 3	2×3+10 = 6+10 = 16	2(3+1)+8 = 2×4+8 = 8+8 = 16	Same
For x = 5	2×5+10 = 10+10 = 20	2(5+1)+8 = 2×6+8 = 12+8 = 20	Same
For x = 1	2×1+10 = 2+10 = 12	2(1+1)+8 = 2×2+8 = 4+8 = 12	Same

- **The two algebraic expressions 3x + 1 and 2x + 5 are NOT equivalent**

Substitution value for the variable	3x + 1	2x + 5	Result
For x = 3	3×3+1 = 9+1 = 10	2×3+5 = 6+5 = 11	Different
For x = 4	3×4+1 = 12+1 = 13	2×4+5 = 8+5 = 13	Same
For x = 5	3×5+1 = 15+1 = 16	2×5+5 = 10+5 = 15	Different

~~Notice:~~

When the algebraic expressions give different results for the first trial, it is sufficient to conclude they are NOT equivalent. No need to try another value.

Let's Try:

1) If the two algebraic expressions $(2x + 1)$ and $[x + 5 + 3(2x + 1) - 5x - 7]$ are equivalent, find the value of the second expression for $x = 10$ in the least number of steps.

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2) Are they equivalent? Examine these two expressions and determine whether they are equal. If so, consider whether they are always equal. Complete each task. (Use positive integer values for x)

$$3(x + 1) \quad , \quad 2x + x + 3$$

a) Try to find a value for x that will make these expressions equal.

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……

……

b) Try to find a value for x that will make the expressions NOT equal.

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c) Decide if these two expressions are always equal and if they should be considered equivalent expressions.

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3) Evaluate each of these algebraic expressions using two positive integers of your choice. Determine if they are equivalent or not.

a) $\mathbf{3 + 3x \ \ and \ \ 3(x + 1)}$

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b) $\mathbf{5x + 5 \ \ and \ \ 5(x + 1)}$

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c) $2x + 3x + 10$ *and* $5(x + 2)$

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Evaluation:

Evaluate each of these algebraic expressions using two positive integers of your choice. Determine if they are equivalent or not.

a) $8x - 4$ *and* $4x + 3 + 4x + 1$

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b) $9(x + 6)$ *and* $9x + 6$

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c) $3(2x + 1)$ ***and*** $6x + 3$

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d) $4x + 6$ ***and*** $3(x + 2)$

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e) $2(x + 1)$ ***and*** $2x + x$

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f) Five times x added to 3 and 3 times x increased by 5

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Date: ……

Unit Three, Concept Two Assessment

1) Follow the order of performing operations, then find the value of each of the following:

a) $6 \times 6 + 4$

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b) $5 \times 2 + 3 \times 4$

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c) $7 \times 7 - 24$

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d) $[3 \times (9 - 4)] - 10$

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e) $5 \times [12 \div (4 + 2)]$

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f) $(3 + 6) \times 2$

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g) $6^2 \div 9 + 5$

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……

h) $4^3 + 2 \times 5$

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……

i) $(2^5 + 3) \div (2^3 - 1)$

……

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……

j) $[5^2 \times (6^2 \div 9)] - 24$

……

……

……

k) $4^2 + (15 - 7) \times 2$

……

……

……

2) Find the value of the algebraic expression in each of the following:

a) $6x + 15 \ \ (If\ x = 3)$

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b) $9z - 15 \ \ (If\ z = 2.1)$

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c) $18 \div (9 - 2c) \ \ (If\ c = 1.5)$

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d) $g^2 - 32 \div 8 \ \ (If\ g = 5)$

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3) Shereen studies for "k" hours a day for 5 days, then studies for 6 hours on the sixth day.

a) Write an algebraic expression for the number of hours she studies in the 6 days.

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b) If the number of hours she studies in each of the five days is 4 hours, how many hours did she study in the 6 days?

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4) A worker in a factory receives a daily wage of 100 pounds for working for a specified number of hours, in addition to an amount of 30 pounds for every additional hour of work.

a) Write an algebraic expression that expresses the wage that the worker receives on a day on which he worked y overtime hours.

…….

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…….

b) If the number of overtime hours is 3 hours, what is the amount that he gets for that day?

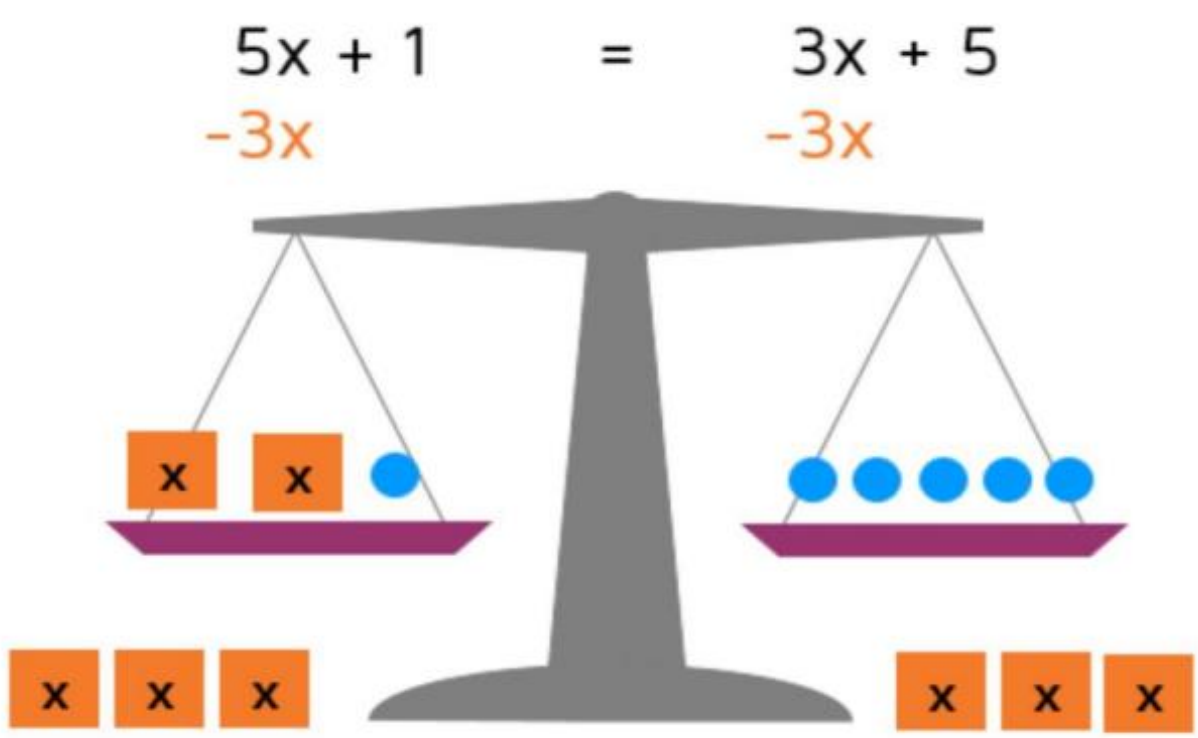
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Unit 4, Concept 1

Concept 1: Write and Solve Equations and Inequalities

Lesson 1: Solving Algebraic Equations

Lesson 2: Exploring Inequalities

Lesson 3: Solving Inequalities

Date: ……

Lesson 1

Solving algebraic Equations

Equation: is a mathematical sentence that includes an equality relation between its two sides.

Examples:

$$x + 5 = 9$$

$$24 - a = 17$$

$$3k = 15$$

$$\frac{b}{5} = 5$$

Solving equations: means finding the value of the unknown which satisfies the equation. In this term we will solve one-step equations only.

You can solve an equation in many ways:

1)Mental math

Example: 16 + x = 20

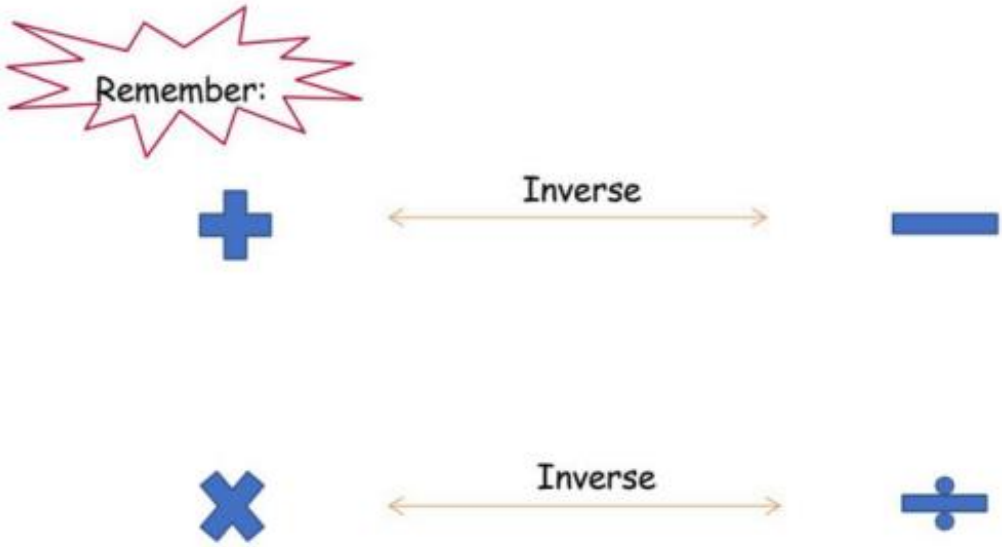
What number plus 16 equals 20?

The answer is 4, so x = 4.

2)Inverse operation

Example: y - 3 = 8

Then, y = 8 + 3 = 11



Let's Try:

Solve the following equations:

1) $3 + k = 13$

……

……

……

2) $4n = 28$

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3) $z - 6 = 6$

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.....

.....

4) $v \div 5 = 7$

.....

.....

.....

5) $\frac{a}{8} = 3$

.....

.....

.....

6) $3 + 9 = 5 + h$

……

……

……

7) $36c = 36$

……

……

……

8) $3p - 5 = 7$

……

……

……

Evaluation:

1) $x + 4 = 9$

.....

.....

.....

2) $3z = 9$

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.....

3) $13 - b = 4$

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4) $2 + 8 = 5 + y$

.....

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.....

5) $\frac{x}{7} = 6$

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Date:…….

Lesson (2) & (3)

Exploring Inequalities

- **Inequality:** **Is a mathematical sentence that includes sign of inequality $(\leq , \geq , < , >)$ between the two sides.**

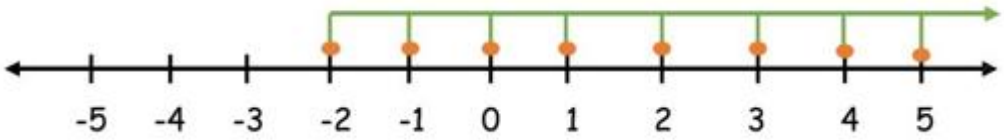
- **EX:** $a + 4 \geq 6$

- **Solution of the inequality:** **Is any number that makes an inequality true**

- **You can represent the solution of the inequality $x > -3$ on the number line in the set of integers**

- **The integers which are greater than -3 are: $-2 , -1 , 0 , 1 , 2 ,$ … … ….**

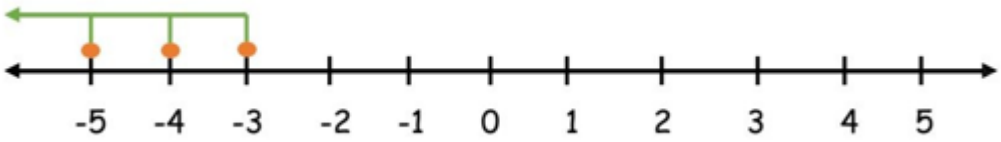
  Represent these numbers on the number line as follows



- **To represent the solution of the inequality $x \leq -3$ on the number line in the set of integers**

- **The integers which are less than or equal -3 are: $-3 , -4 , -5 ,$ … ….**

  Represent these numbers on the number line as follows



Inequality

Choose the correct answer:

1) Mona reads for at least 30 minutes everyday
Which of the following values may be her reading time today?
a) 20 min. b) 40 min. c) 15 min. d) 28 min

2) Which of the following values is a solution of $x > 5$?
a) -5 b) 5 c) 4 d) 8

3) Which of the following values is a solution of $x \leq -2.5$?
a) 2 b) -1 c) -3 d) -2

4) Which of the following values is a solution of $x > -1$?
a) -1 b) 0 c) -5 d) -25

5) If the weather temperature in summer is more than $35^{\circ}C$
Which of the following degrees may be in 25th of June ?
a) $25^{\circ}C$ b) $45^{\circ}C$ c) $21^{\circ}C$ d) $30^{\circ}C$

6) Which of the following values is a solution of $x \leq 3$?
a) 5 b) 7 c) 3 d) 8

7) Which of the following values is a solution of $x < 0$?
a) 0 b) 1 c) -1 d) 0.5

8) Which of the following values is a solution of $x < -5$?
a) -5 b) -4 c) -3 d) -6

9) The inequality which represents " All values greater than 0 " is …….

a) $x > 0$ **b)** $x < 0$ **c)** $x \leq 0$ **d)** $x \geq 0$

10) Which of the following values is not a solution to the inequality $x > -1$**?**

a) 1 **b)** 0 **c)** -2 **d)** -0.5

Represent the solution on the number line:

1) $x > -1$



2) $x \geq -3$



3) $x > 3$



4) $x \geq 4$



5) $x < -2$

6) $x \leq 0$



7) $x < 4$



8) $x \leq -5$

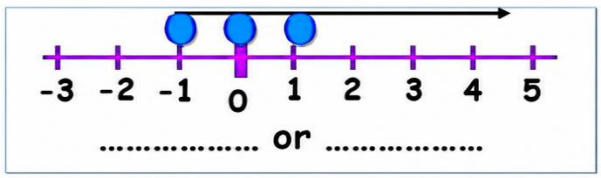


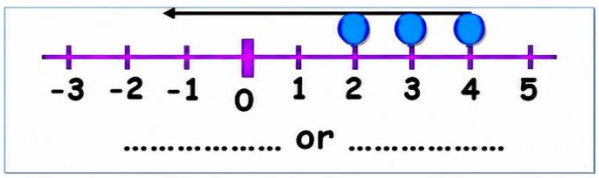
9) $x \leq 5$

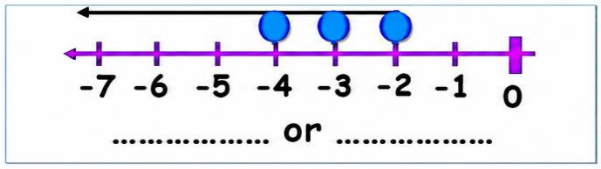


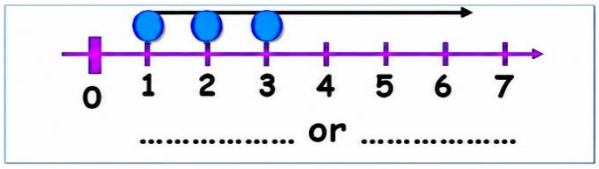
10) $x \geq 0$

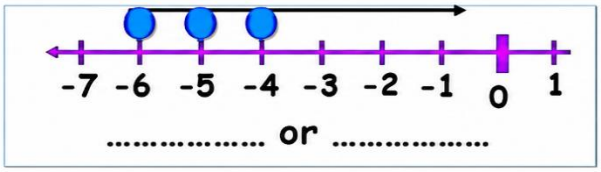
Write the inequality:

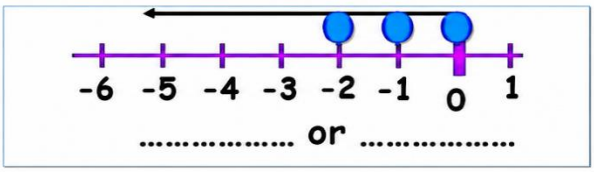












Write the algebraic expression:

1) $x + 5 = 2$ .....

2) $x > -2$ .....

3) $3x = 20$ .....

4) $x < 7$ .....

5) $x - 3 = 1$ .....

6) $x \geq 0$ .....

7) $\frac{1}{4}\, x = 16$ .....

8) $x \leq -5$ ……

Give 2 solutions in the set of integers and 2 solutions in the set of rational numbers:

1) $x > -1$



2) $x \geq -3$



3) $x > 3$



4) $x \geq 4$



5) $x < -2$



6) $x \leq 0$



7) $x < 4$

8) $x \leq -5$

9) $x \leq 5$



10) $x \geq 0$

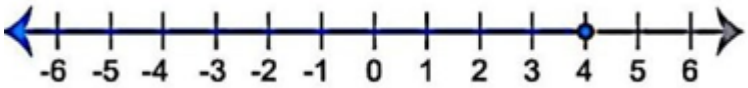


Date:…….

Lesson One: Writing Inequalities

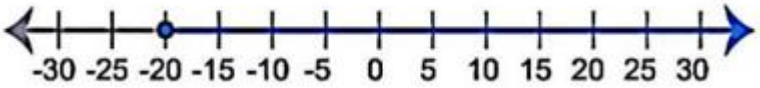
A) Write the inequality that best describes each graph

1)



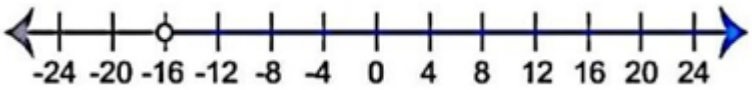
Inequality: ……..

2)



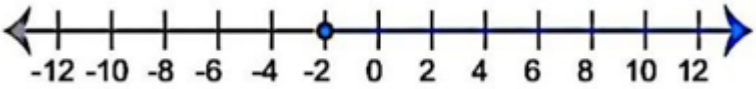
Inequality: ……..

3)

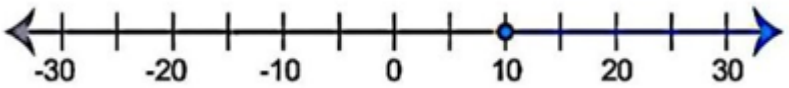


Inequality: ……..

4)

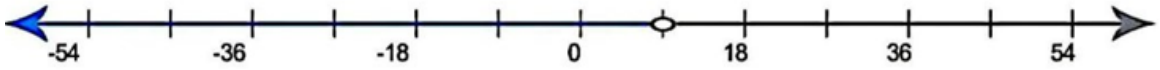


Inequality: ……..

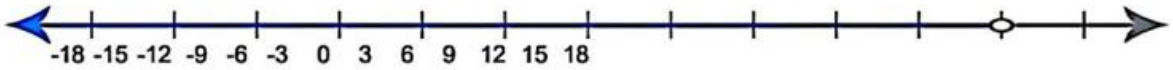
5) 

Inequality: ……..

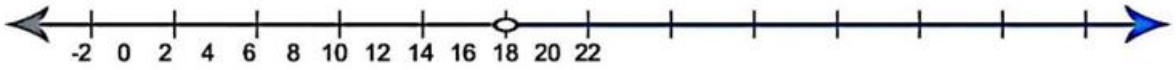
6)



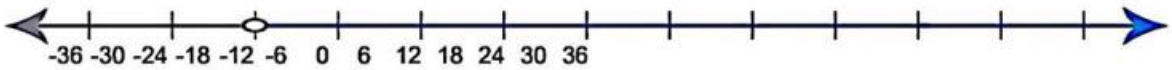
Inequality: ……..

7) 

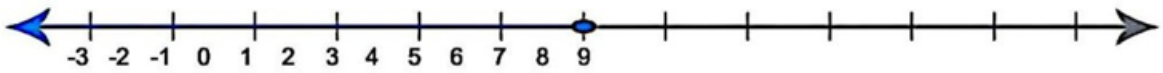
Inequality: ……..

8) 

Inequality: ……..

9) 

Inequality: ……..

10) 

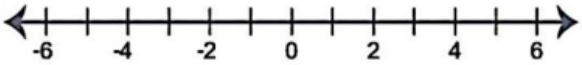
Inequality: ……..

Date:…….

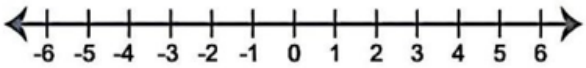
Graphing inequalities

B) Graph the inequality:

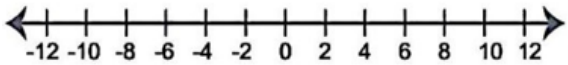
1) $x \geq -2$



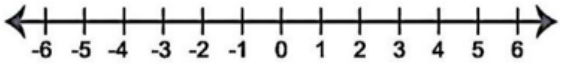
2) $x < 5$



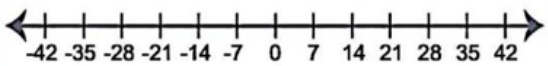
3) $x \leq 6$



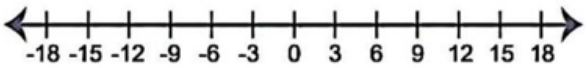
4) $4 > 1$



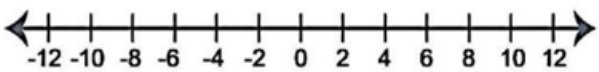
5) $x < -14$



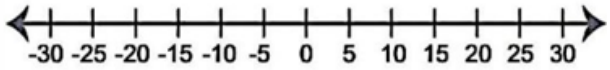
6) $x \leq 9$



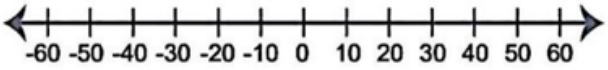
7) $x > -2$



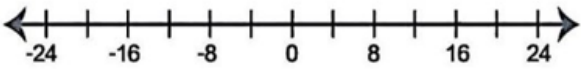
8) $x \geq -15$



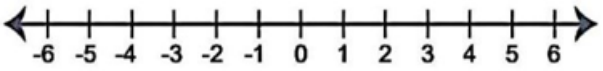
9) $x \leq -10$



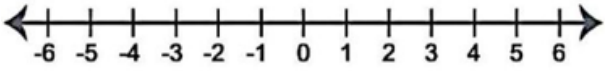
10) $x > 16$



11) $x > -1$



12) $x \geq -3$



Date:……

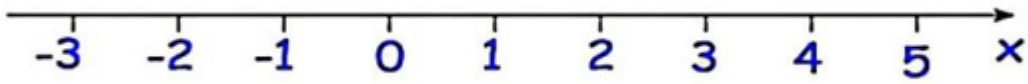
Lesson (2) & (3)

Exploring and solving inequalities

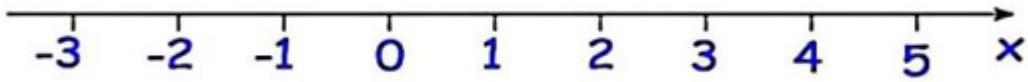
1) Match each inequality to the correct description:

1) $x > 3$	i) x is less than or equal to 3
2) $x \leq 3$	ii) x is greater than or equal to 3
3) $x \geq 3$	iii) x is less than 3
4) $x < 3$	iv) x is greater than 3

2) Represent the inequality $x > 2$ on this number line



3) Represent the inequality $x \leq 4$ on this number line



4) Find 3 solutions of x in the set of integers:

a) $x \leq 3$

b) $x > -4$

c) $x > 0$

5) Find 3 solutions of x in the set of rational numbers:

a) $x \leq -6$

b) $x < -1$

c) $x \geq 1$

Date:……

Unit Four, Concept one assessment

1) Solve the following equations:

a) $5 + x = 21$

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……

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b) $7y = 28$

……

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……

……

c) $z - 8 = 14$

……

……

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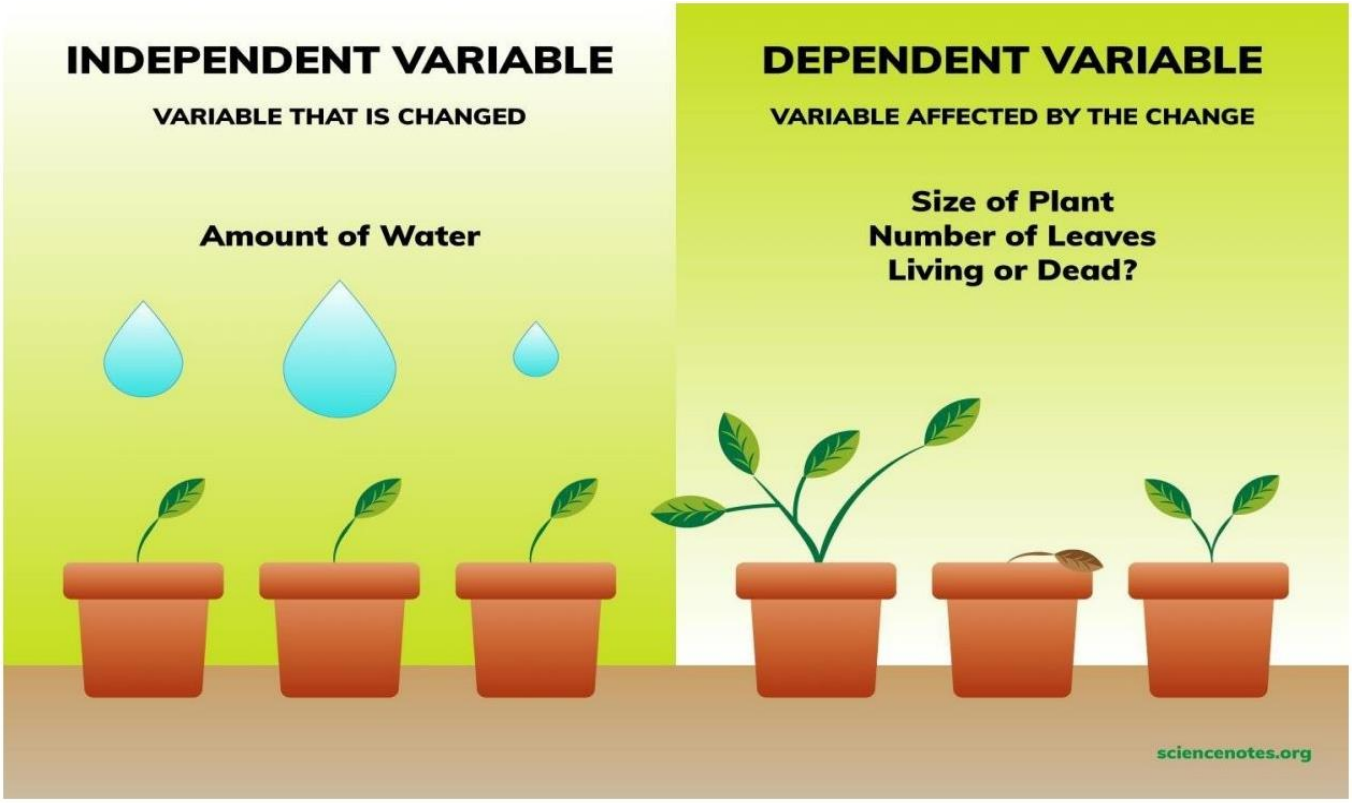
……

d) $\frac{f}{2} = 9$

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Unit 5, Concept 1

Concept 1: Dependent and Independent Variable.

Lesson 1: The Relation Between Dependent and Independent Variable.

Lesson 2: Relation Between Dependent and Independent Variables.

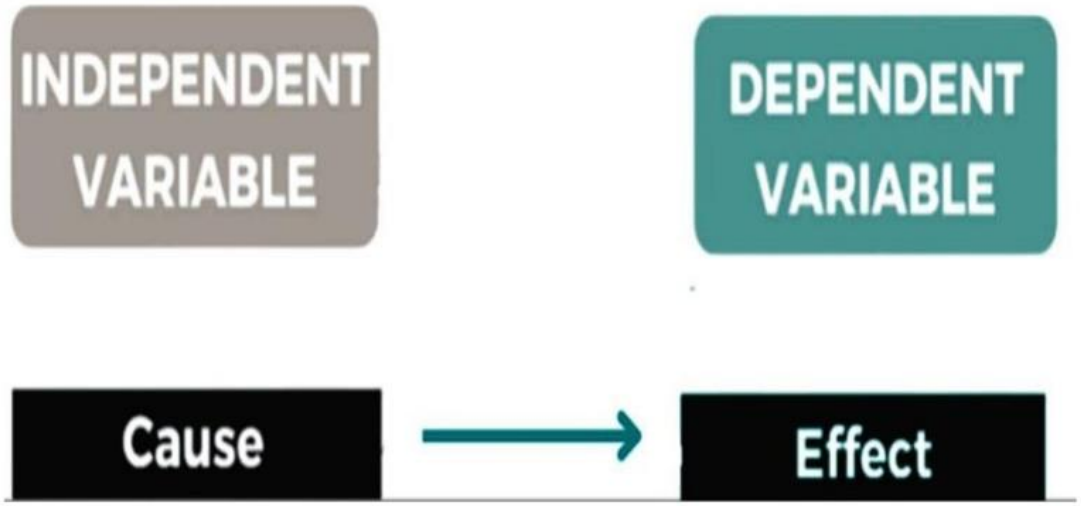
Lesson 3: Analyzing the Relationship Between Dependent and Independent Variables.

Lesson 4: Graph Representation for Dependent and Independent.

Date:…….

Lesson (1)

The relation between dependent and independent variables:



The Independent variable causes a change in the Dependent variable

OR

The dependent variable depends on or is affected by the independent variable

Example:

Time spent studying (Independent variable) causes a change in Test score (Dependent variable) and it isn't possible that Test score could cause a change in Time spent studying

OR

Your test score is affected by the time you spend studying

Let's try

Ex.1: For each of the following statements, decide which variable is dependent and which is independent:

a. How fast do you eat	**and**	**how much are you hungry**
b. What you bought	**and**	**the cost you pay**
c. The size of plant	**and**	**the amount of water**

Ex.2: Label each of the variables as independent and dependent variables

a. Dina is planning her wedding. The number of announcements she orders is determines by the number of guests she wants to invite

…….. g = The number of guests

…….. g = The number of announcements

b. Tamer goes biking every morning. The farther he bikes, the more calories he burns during the bike ride

…….. c = The number of calories Tamer burns during the bike ride

…….. d = The distance Spencer bikes

Evaluation

Ex.1: In the tables below either an independent or dependent variable is listed. Fill in the other side of the table with several variables that will make the comparison make sense

Independent variable	Dependent variable
Age of a person	----------

Independent variable	Dependent variable
----------	A student test score.

Ex.2: Label each of the variables as independent or dependent variables.

a. Tamara is a singer who will be participating in an upcoming recital for each song she plans to perform, she will require a different costume.

…….. c = The number of costumes Tamara will require

…….. s = The number of songs Tamara plans to perform

b. Sarah is organizing an open day. The amount of time that the park has been reserved for determines the number of participants will attend that open day

…….. t = The amount of time that the park has been reserved for

…….. b = The number participants will attend that open day

c. Nader is selling lemonade. The number of cups pf lemonade he sells determines how much money he earns

…….. m = The amount of money Nader earns

…….. c = The number of cups of lemonade Carson sells

d. During the winter, a trucking company ships produce from Cario to Alex. The more produce the company needs to ship, the more truckers it will hire.

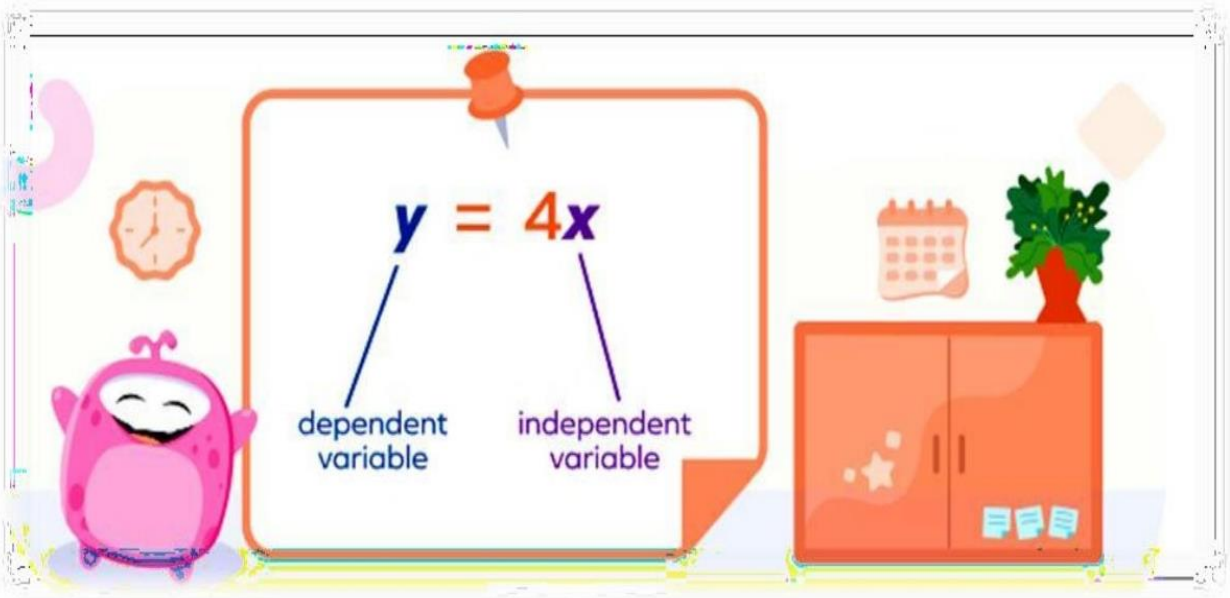
…….. t = The number of truckers the company will hire

…….. g = The amount of produce the company needs to ship

Date: .....

Lessons 2&3

Relation Between dependent and Independent Variables



Independent Variable, x (input)	Dependent Variable, $y = 4x$ (output)
0	4 x 0 = ------
1	4 x 1 = ------
2	4 x 2 = ------
3	4 x 3 = ------

Keywords of mathematical operations:

Addition	Subtraction	Multiplication	Division
• sum of • total of • plus • increased • gain • raise • more • increase of	• difference • minus • decreased • loss • less • fewer • take away	• multiply • product of • times • double • twice • triple	• quotient of • divide by • divided by • divided equally • per

Let's try:

1) Use the following equation and complete the table:

Equation	Dependent variable (output)	Independent variable (Input)
a. $m = n + 4$	.....	.....
b. $v = c - 7$	.....	.....
c. $6 - b = a$	.....	.....
d. $r = 4f - 5$	.....	.....

2) Write each phrase as an algebraic equation:

a. 4 more than x equals y

……

b. 45 more than r equal k

……

c. h increased by 13 equals e

……

d. j equals the difference between fifty and k

……

e. w equals the product of 14 and c

……

3) Write the variable phase for each of the following equations:

a. $c = f + 3$

……

b. $g = v - 12$

……

c. $5q = u$

……

Evaluation

1) Choose the correct answer:

1) In the equation: $x = 6y + 7$, the dependent variable is ……

a) x b) y c) 6 d) 7

2) In the equation: $6a + 24 = b$, the dependent variable is ……

a) a b) b c) 24 d) 6

3) 7 is more than u equals v in equation is ……

a) $u = v + 7$ b) $v = u + 7$ c) $v = 7u$ d) $v = u - 7$

4) m is equals the product of n and 9 in equation is ……

a) $n = 9m$ b) $m = 9n$ c)$m = 9 + n$ d) $m = n - 9$

---

2) Use the following equation and complete the table:

Equation	Dependent variable	Independent variable
a. $y = x - 8$	……	……
b. $h = j + 12$	……	……
c. $15b = a$	……	……

d. $w = 7f + 3$	.....	.....

3) Write the variable phrase for each of the following equations:

a. $y = x - 2$

.....

b. $t = 4 + q$

.....

c. $e = 12\ f$

.....

Date:…….

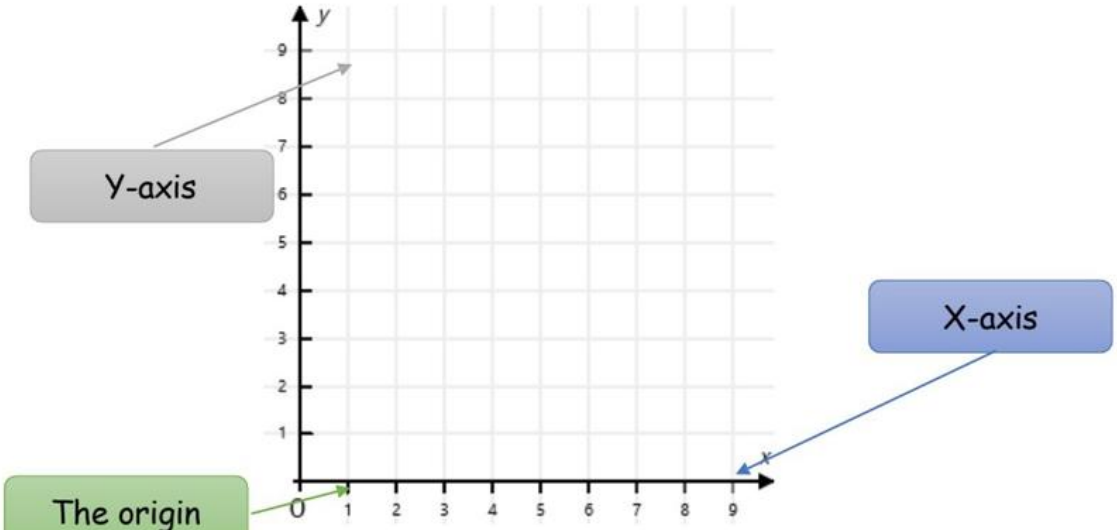
Lesson (4)

Graph representation fro dependent and independent

Remember

- **The coordinate plane:**

Is the plane determined by a horizontal line, called the x – axis and a vertical line called the y – axis , interacting at a point called the origin and its labeled as "O"



Coordinate plane

- **The ordered pair**

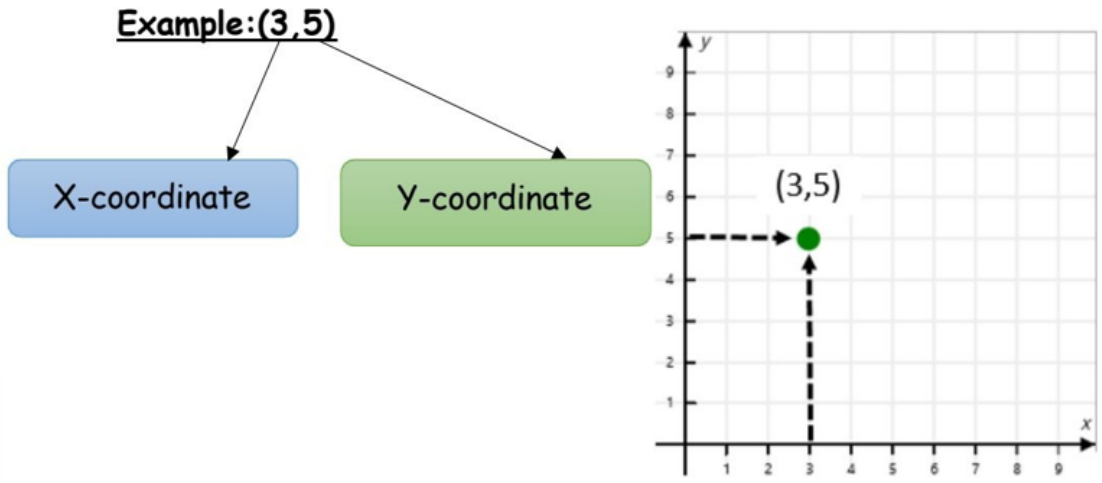
Is pair of number used to locate any point on the coordinate plane. Ordered pairs are written left to right (x , y)

- **x –coordinate**

The first number in an ordered pair, which tells how far to move left or right from the origin. It is labeled as x

- **y – coordinate**

The second number in an ordered pair, which tells how far to move up or down from the origin. It is labeled as y



X	1	2	3
y	- - - - -	----------	----------

How can we evaluate an algebraic equation?

To evaluate an algebraic equation, substitute (replace) number for the independent variable (x) and simplify.

EX.1: Fill in the table below by substituting the values of x in the equation $y = x + 3$

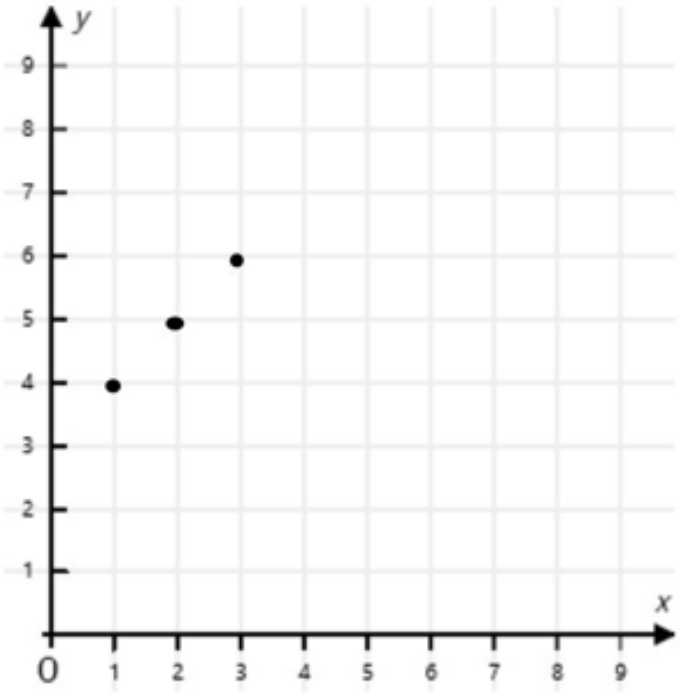
x	0	1	2
y	…….....	…….....	…….....
(x, y)	…….....	…….....	…….....

By replacing the values of x in the equation:

At $x = 1$, $y = 1 + 3 = 4$

At $x = 2$, $y = 2 + 3 = 5$

At $x = 3$, $y = 3 + 3 = 6$



Then we will have the following ordered pairs: (1 , 4) , (2 , 5) and (3 , 6)

EX.2: Fill in the table below by substituting the values of x in the equation $y = 2x + 1$

x	1	2	3
y	……...	……...	……...
(x, y)	……...	……...	……...

By replacing the values of x in the equation:

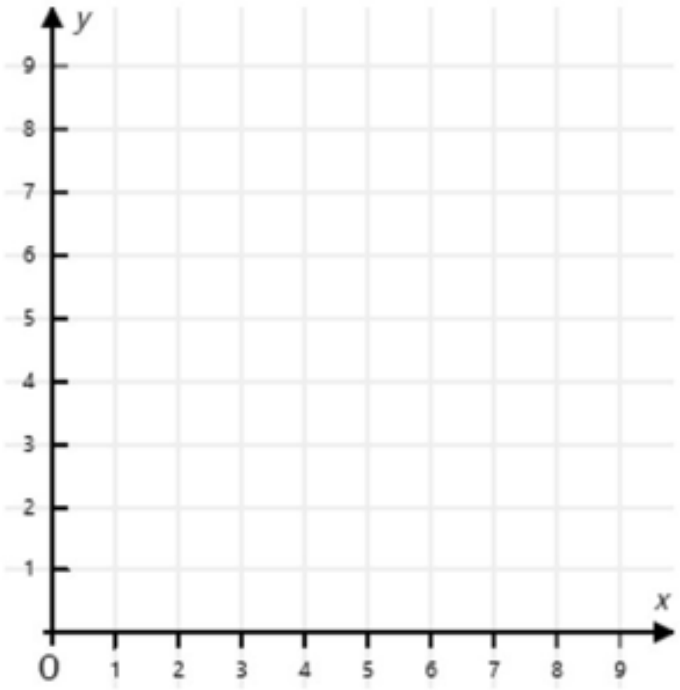
At $x = 0$, $y = 0 \times 2 + 3 =$……..

At $x = 1$, $y = 1 \times 2 + 3 =$ ……

At $x = 2$, $y = 2 \times 2 + 3 =$……..

Then we will have the following ordered pairs:

……



Let's try:

Complete the following table, and then make the graphs:

a. The equation $y = x + 2$

x	0	1	2
y	……	……	……
(x, y)	……	……	……

Show your work out here:

……

……

……

……

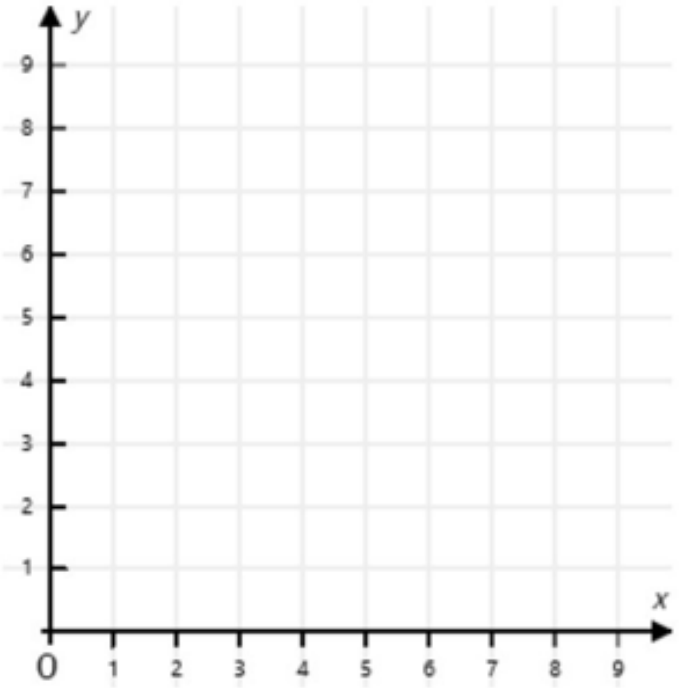
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b. The equation: $y = 2x - 1$

Evaluation

1) Complete the following:

a. In the equation: $y = 2x$, if $x = 1.5$ then y would be ……

b. $(2$, … … … $)$ satisfied the rule $y = 2x - 2$

c. In the equation: $y = x + 3$, if $x = 3.4$ then y =……

d. In the equation: $y = \frac{1}{4}\ x$, if the input is 16 , then the output is …….

e. In the equation: $y = 2x - 3$, if $x = 5$, then the value of y = …….

f. The rule is "Add 5", then the equation is …….

g. The rule is "Multiply by 4", you would write the equation as……

1) Complete the following table, and then make the graphs:

The equation: $y = 2x - 2$

x	1	2	3
y	……...	……...	……...
(x, y)	……...	……...	……...

Show your work out here:

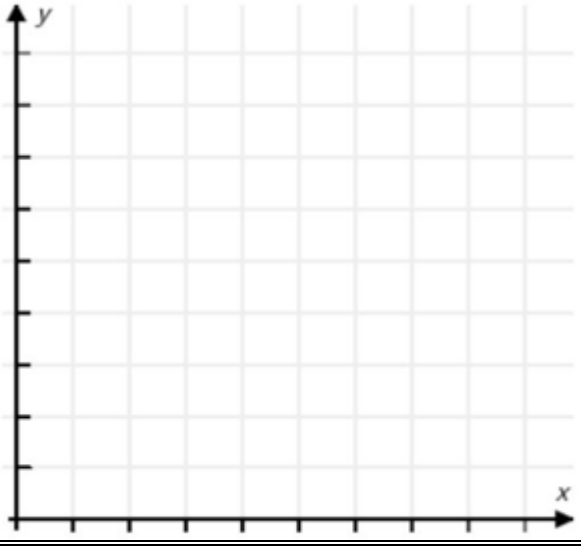
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Date:……

Unit Five, Concept One Assessment

1) Complete the following:

a. $(1, \ldots\ldots\ldots)$ **satisfied the rule** $y = 3x + 1$

b. If $x - 6 = 4$, $\frac{1}{2}\ x =$……

c. The algebraic equation of " 4 more than x **equals** y**" is …….**

d. The dependent variable in the equation: $4m - 7 = n$ **is ……..**

e. 7 more than u **equals** v **in equation is ……..**

f. In the rule $y = 4x$ **, if** $x = 1.2$ **then** y **would be ……**

g. "13 less than a number" written as ……..

h. The word phrase for the equation "$k = 3n$**" is**
……..

---

2) Choose the correct answer:

a. In the equation: $x = 3y + 4$ **, the <u>independent</u> variable is ……**

a) x **b)** y **c) 4** **d) 3**

b. In the equation: $2a + 4 = b$ **, the <u>dependent</u> variable is …….**

a) a **b)** b **c)** 4 **d)** 2

c. 7 more than u **equals** v **in equation is …….**

a) $u = v + 7$ **b)** $v = u + 7$ **c)** $v = 7u$ **d)** $v = u - 7$

d. m **equals the product of** n **and 9 in an equation is ……..**

a) $n = 9m$ **b)** $m = 9n$ **c)** $m = 9 + n$ **d)** $m = n - 9$

e. In the equation: $y = x + 1$ **, if the output is 1 , then the input is ……..**

a) 0 **b) 1** **c) 2** **d) 3**

---

3) Complete the following table, and then make the graphs:

The equation: $y = 3x + 1$

x	0	1	2	3
y	…….	…….....	…….....	…….....
(x, y)	…….	…….....	…….....	…….....

Show your work out here:

…….

…….

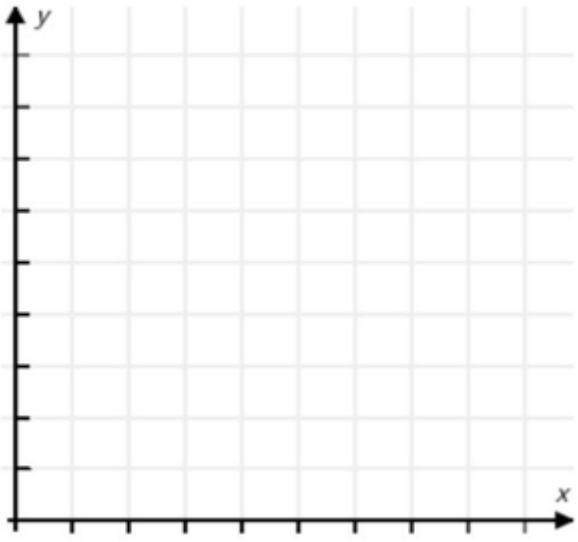
…….

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…….

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Unit 6, Concept 1

Unit 6: Data Distributions

Concept 1: Applications on Collecting and Representing Data

Lesson 1: Data and Statistical Questions

Lesson 2: Exploring the Histogram

Lesson 3: Representing Data Using Histograms

Lesson 4: Exploring Box Plot

Lesson 5: Applications on Data Representations

Date:…….

Lesson (1)

Data and statistical questions

To learn about data and how to collect it, we should first talk about what is the right question to ask to collect the needed data. That's why it is very important to know how to construct a statistical question

- **Statistical question**

Is a question that entitles different answers from which we can collect data

Example 1:

What is your favorite color? **Not a statistical question**

Because it has only one answer

What are the student's favorite colors? **A statistical question**

Because it has different answers that can be sorted and analyzed

Let's try

1) State whether each question below is a statistical question or not:

a. What is your name? **……**

b. How many times did you go to Alex throughout the last 2 years?

……

c. What is the cost of each book in your bag? ……

d. What is the age of each member in your football team?

……

e. What is the cost of this book? ……

Notice that:

- **A 'Yes' or 'No' question is NOT a statistical question**
- **A question that produces a single answer is NOT a statistical question**
- **The statistical question produces 3 or more possible answers**

Types of statistical questions:

Statistical questions are divided into categorical and numerical

- **Categorical statistical questions:**
  Are questions that provide qualitative or categorical answers

- **Numerical statistical questions:**
  Are questions that provide quantitative or numerical answers

Example 2:

a. What type of movies do you and your teammates like to watch?

Categorical statistical questions

b. What is the age of each member in your football team?

Numerical statistical questions

2) Determine the type of each of the following statistical questions:

a. What is the height of each member of your family?

……

b. In which grade is each on e of the 30 students joining the art club?

……

c. How many times did each one in your class go to the cinema last month?

……

d. What bet would each one in your team prefer to adopt?

Evaluation:

1) State whether each question below is a statistical question or not:

a. What it the number of siblings each one of your classmates have?

……

b. Are you Egyptian? **……**

c. How many books did each of one of your family read last week?

……

d. Do you like pasta? **……**

e. What is the favorite food for each student in your class?

……

f. What is your favorite food? ..……

2) Choose the correct answer:

a. What is the age of each one in your building? The statical question is ……

(numerical - categorical)

b. What is your favorite football team for each one in your family? The statistical question is ……..

(numerical - categorical)

c. What is the age of each member in grade 4? This statistical question is ….

(numerical - categorical)

d. How many letters are in the first name for each student of your class?

This statistical question is ……..

(numerical - categorical)

e. What type of books does every member in your family like to read? This statistical question is …….

(numerical - categorical)

f. What is the favorite subject for each student in grade 4? This statistical question is ……

(numerical - categorical)

g. How much money did each member of your team save last year? This statistical question is …….

(numerical - categorical)

Date:…….

Lesson (2)

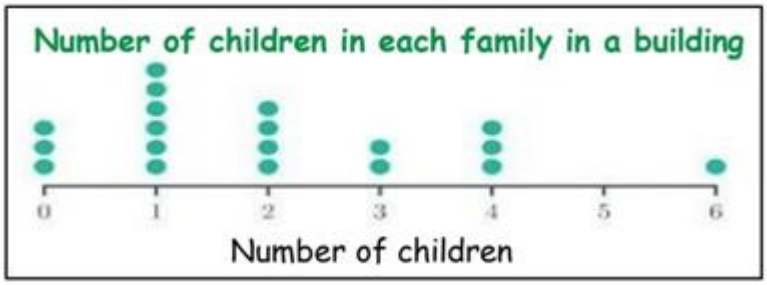
Exploring the histogram

Data can be represented using many types pf graphs. This year we'll study a new type of graph called "Histogram". Before introducing the histogram let's remember two of the previously studied graphs – namely the dot plot & the bar graph

Dot plot:

Is a graph that can be used to display data by plotting dots for each observation

Example 1:

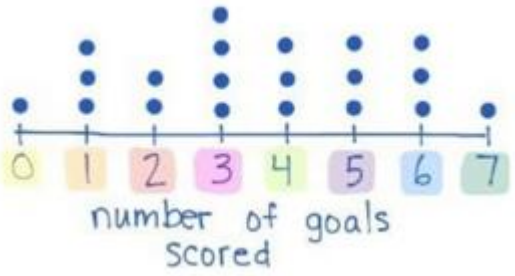


From this dot plot, we notice that:

- **The statistical question "what is the number of children in each family in your building?"**
- **The data represented in graph in numerical data**
- **The total number of families that were surveyed is 19 families (total number of observation)**
- **No family has 5 children**
- **Most families have 1 child**

Many other information can be noticed and analyzed using this dot plot.

Let's try:



a. A suitable statistical question for this dot plot is:

……

b. What is the type of data represented in this graph?

……

c. What title can we write for this dot plot?

……

d. How many people were surveyed?

……

e. State two more information you can deduce from this graph:

1) ……

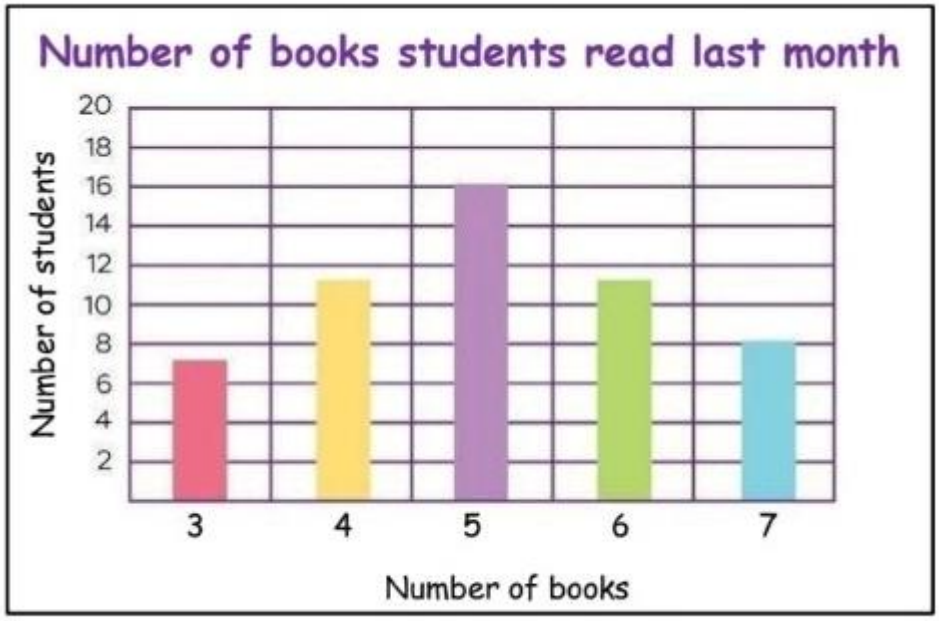
……

2)……

……

Bar graph: is a graph used to represent numerical or categorial data using a rectangular bar for each observation. Bars can be plotted vertically or horizontally.

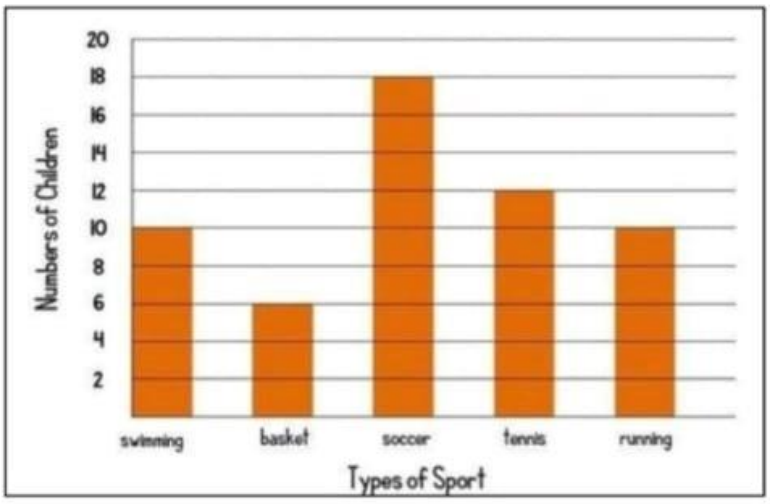
Example 2:



From this bar graph, we notice that:

the statistical question is "how many books did each student read last month?"
the data represented in this graph is numerical data.
the total number of students is (7+11 +16 +11+8) 53 students (total number of observations).
the fewest number of students read only 3 books (only 7 students).
the majority of student read 5 books (16 students).
Many other information can be noticed and analyzed using this bar graph.

Let's Try



a) A suitable statistical question for this bar graph is……..

……

b) What is the type of data represented in this graph?.....

……

c) What title can we write for this bar graph?.....

……

d)How many people were surveyed?.....

……

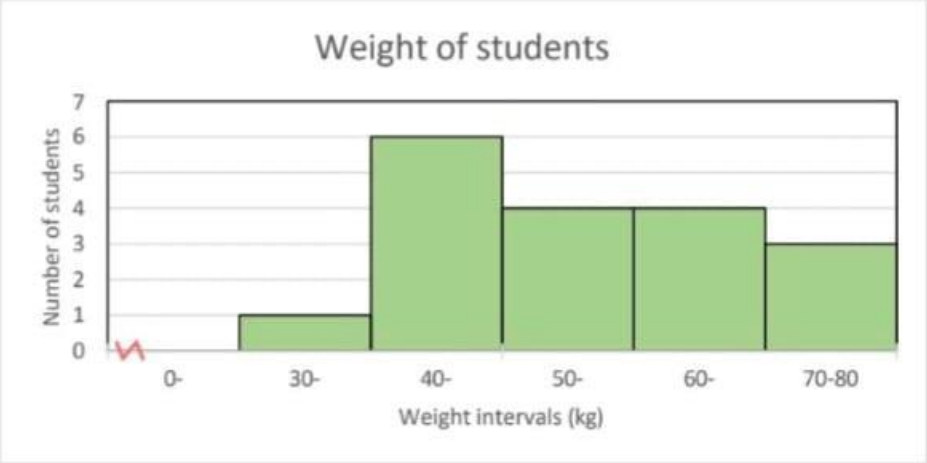
e) State two more information you can deduct from this graph:

a) ……

b)……

Histogram: is a to represent numerical data by displaying data intervals on the x-axis & using bars to represent the frequency of values (number of observations) for each interval.

Example 3:

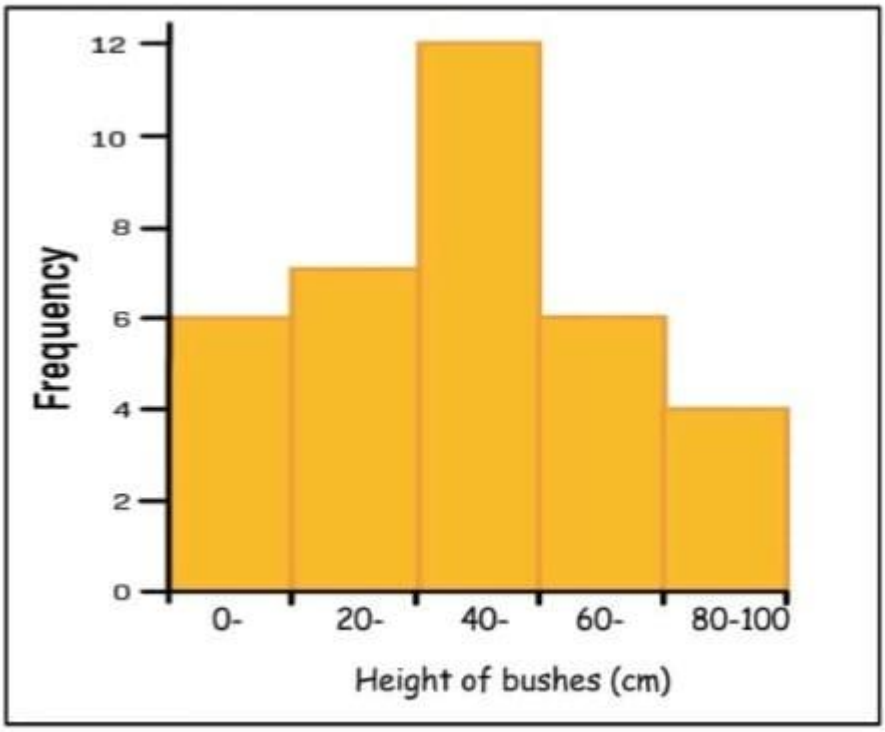


From this bar graph, we notice that:
the statistical question is "what is the weight of each student in your class?"

the data represented in this graph is numerical data.
we started the x-axis with 0 & put this sign w because distance between 0 & 30 distance between 30 & 40. Means I cut part of the x-axis then wrote my equal intervals.
the weight interval is 10 kg.
the total number of students is (1+6+4+4+3) 18 students (total number of observations).
there were no students with weight below 30 kg.
only 1 student recorded weight between 30 & 40 kg.
most students' weight is between 40 & 50 kg (6 students).
the number of students who recorded weight between 50 & 60 kg is exactly equal to the number of students who recorded weight between 60 & 70 kg.
the highest weight of students is 80 kg.

Many other information can be noticed and analyzed using this bar graph.

Let's Try



1) A suitable statistical question for this histogram is……
……

2) What is the type of data represented in this graph?.....
……

3) What title can we write for this histogram?.....
……

4) How many bushes were measured?.....
……

5) How many bushes are shorter than 40 cm?.....
……

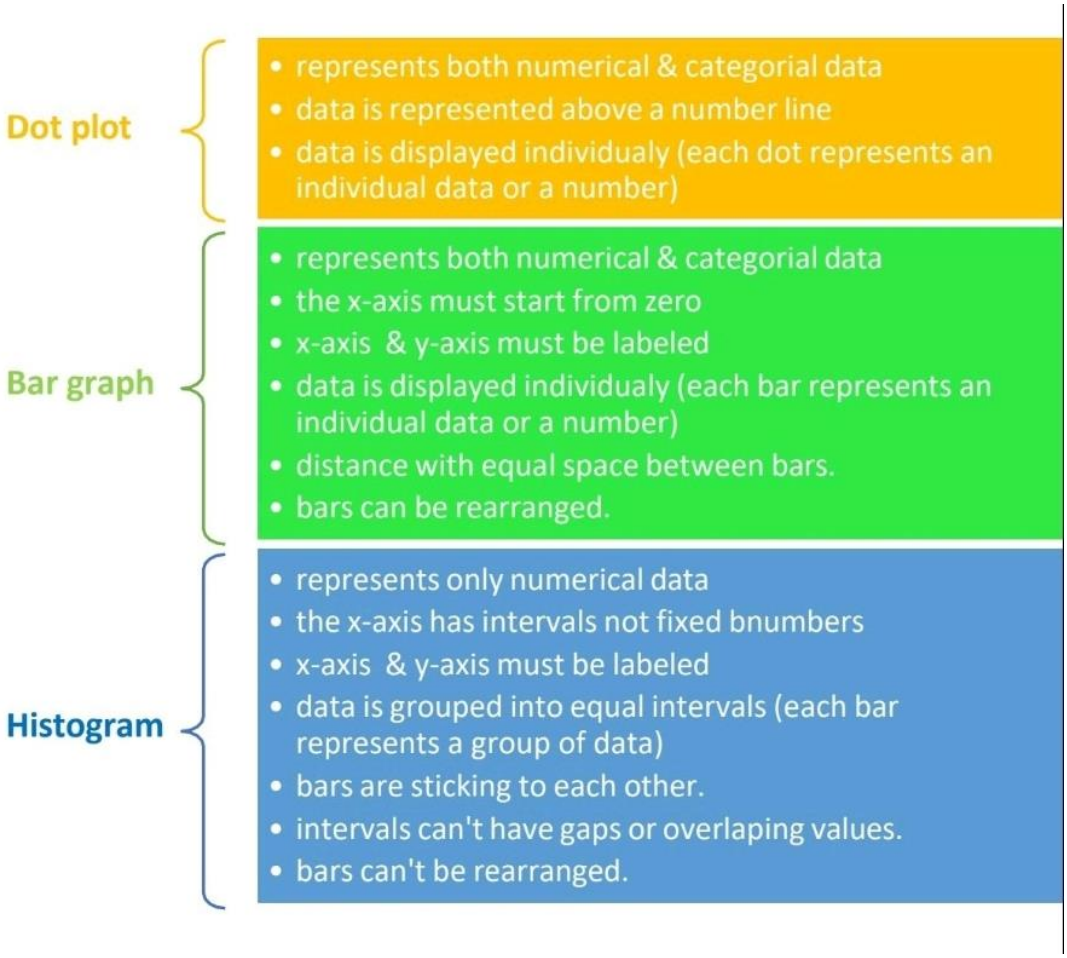
6) How many bushes are between 40 & 80 cm?.....
……

7) State two more information you can deduct from this graph:

a) ……

b) ……

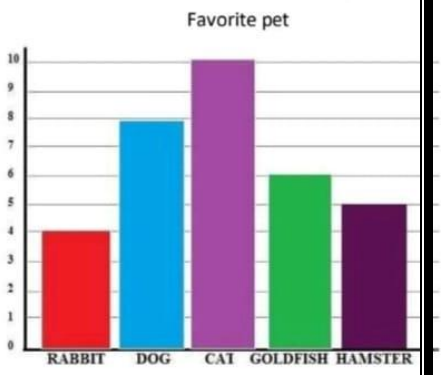
Comparing dot plot, bar graph & histogram



Each of these types of graphs must have a title & show the unit of measurement for the numerical data.

Evaluation

1) **Look at the following graphs then answer the questions below each graph**:



A) According to this bar graph:

i) What is the type of data represented in this graph?

……
……

ii) A suitable statistical question for this bar graph is
……
……..

iii) How many people were surveyed?
……

iv) What can be the label for the x-axis?
……

v) What can be the label for the y-axis?
……

vi) The most favored pet is. pet is while the least favored
……

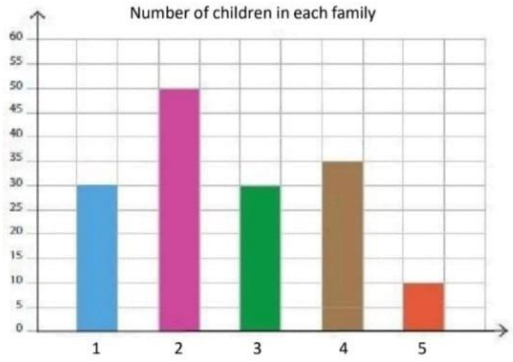
vii) The number of people who prefer goldfish is
……

viii) The number of people who prefer cats is more than who prefer rabbits by how many?

……

……

B) According to this bar graph:



i) What is the type of data represented in this graph?

……

……

ii) A suitable statistical question for this bar graph is

……

……

iii) How many families were surveyed?

……

……

iv) What can be the label for the x-axis?

……

……

v) What can be the label for the y-axis?

……

……

vi) How many families have only 1 child?

……

……

vii) How many children do most of the families have?

……

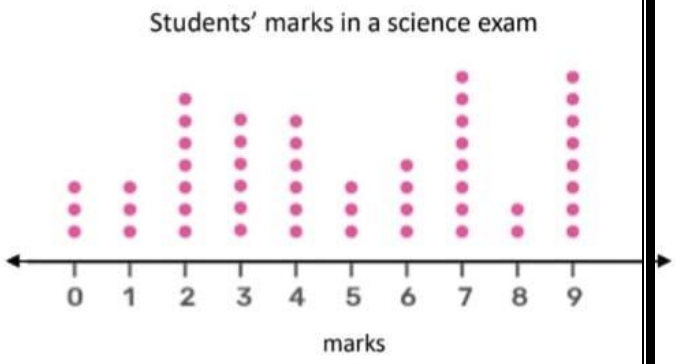
……

viii) The highest number of children recorded is

……

……

C) According to this dot plot:



i) What is the type of data represented in this graph?

……

……

ii) A suitable statistical question for this dot plot is

……

……

iii) How many people were surveyed?

……

……

iv) What is the highest recorded mark?

……

……

v) What the mark for most of the students?

……

……

vi) The least number of students recoded a mark =

……

……

vii) How many students recorded a mark less than 5?

……

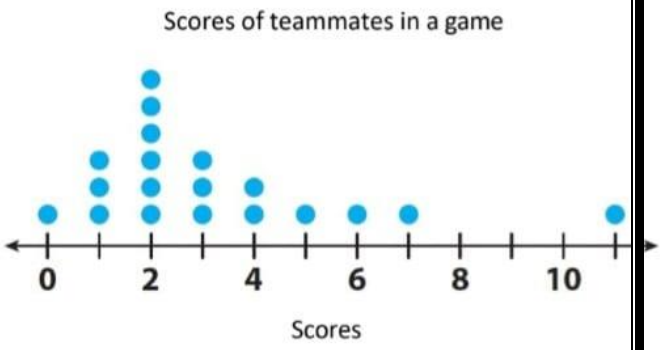
……

viii) How many students recorded a mark higher than 7?

……

……

D) According to this dot plot:



i) What is the type of data represented in this graph?

……

……

ii) A suitable statistical question for this dot plot is

……

……

iii) How many people were surveyed?

……

……

iv) What is the highest recorded score?

……

……

v) How many team members did not record any score?

……

……

vi) The majority of the team members recorded scores.

……

……

vii) How many team members recorded 8 scores?

……

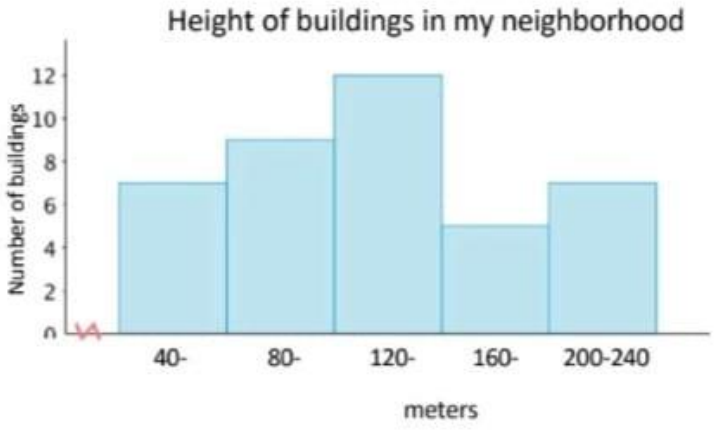
……

viii) How many team members recorded more than 5 scores?

……

……

E) According to this histogram:



i) What is the type of data represented in this graph?

……

……

ii) A suitable statistical question for this bar graph is

……

……

iii) How many buildings were surveyed?

……

……

iv) What is the height of most buildings?

……

……

v) What is the number of buildings with height more than 120 meters?

……

……

vi) What is the number of buildings with height less than 40 meters?

……

……

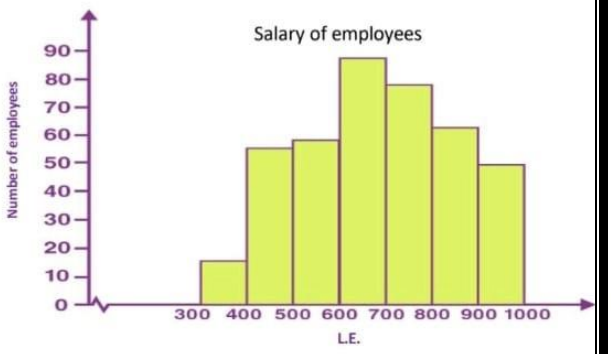
vii) What is the number of buildings with height less than 80 meters?

……

……

viii) The highest buildings recorded a height between……&……

F) According to this histogram:



i) What is the type of data represented in this graph?

……

……

ii) A suitable statistical question for this bar graph is

……

……

iii) How many employees were surveyed?

……

……

iv) What is the salary interval?

……

……

v) How many get a salary between 700 & 900 LE.?

……

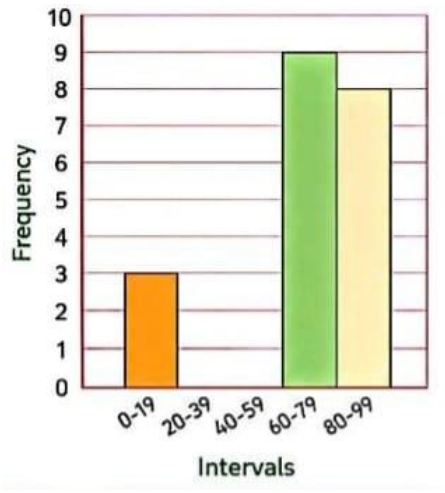
……

1. Using the opposite histogram:

a. Complete the table.

b. Complete the graph.

Interval	frequency
0 - 19	3
20 - 39	…….
40 - 59	7
60 - 79	…….
80 - 99	8



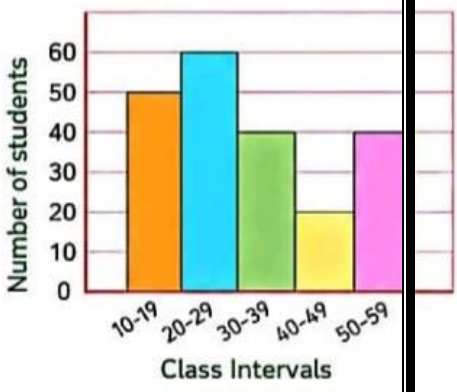
2. In the opposite histogram:

a. How many students got more than 39 marks?

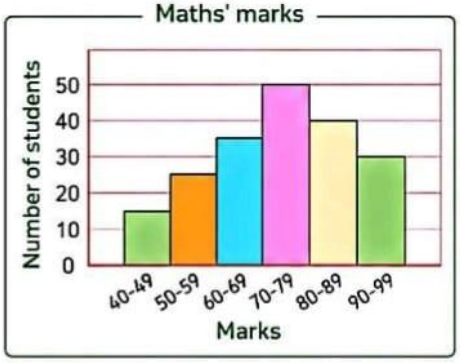
……

b. Which interval has the minimum number of students?

……



3. From the opposite histogram answer the following questions:



a. How many students were surveyed?

b. How many students whose marks are there in the interval [80-89]?

c. Which marks interval has maximum number of students?

d. Which marks interval has minimum number of students?

e. How many students whose marks are less than 60?

f. How many students whose marks are 70 or more?

g. How many students whose marks are 50 or greater than 50 and less than 80?

Date:…….

Lesson 3

Representing Data Using Histograms

Example 1:

The table below shows the weight (in kg) of students in your class. Use this table to construct a frequency table & draw a histogram.

40.7	40.5	62.4	69.7	30	51.4
42	72.4	56.3	54.9	54	76
41.7	42.3	71	42.6	63.8	65.7

It's obvious that the collected data show different answers & not all observed answers are repeated. To represent such data, we need to make intervals (groups) & a frequency table following these steps:

Notice that the minimum wight observed is 30 kg while the maximum weight observed is 76.

This means the range = maximum number - minimum number 76-30 = 46

I can round the range, so 46 = 50.

Range

Notice that finding the range here is optional because I know the x-axis on the graph must start from O so I need an interval from O till the maximum number or more = 50 in this case
I can mentally figure out that from 0 till 50 can be divided into 5 intervals (length of each interval = 10) or it can be divided into 10 intervals (length of each interval = 5) I'll choose 5 intervals, each interval size = 10,

The 1st interval is students with weight from 30 to 40 kg, the 2nd interval is students with weight from 40 to 50 kg, ...etc.

Weight Interval	Frequency
30 -	1
40 -	6
50 -	4
60 -	4
70 - 80	3
Total	18

The total number of observed data (total number of students inside the class)

To draw the graph, I'll do the following steps:

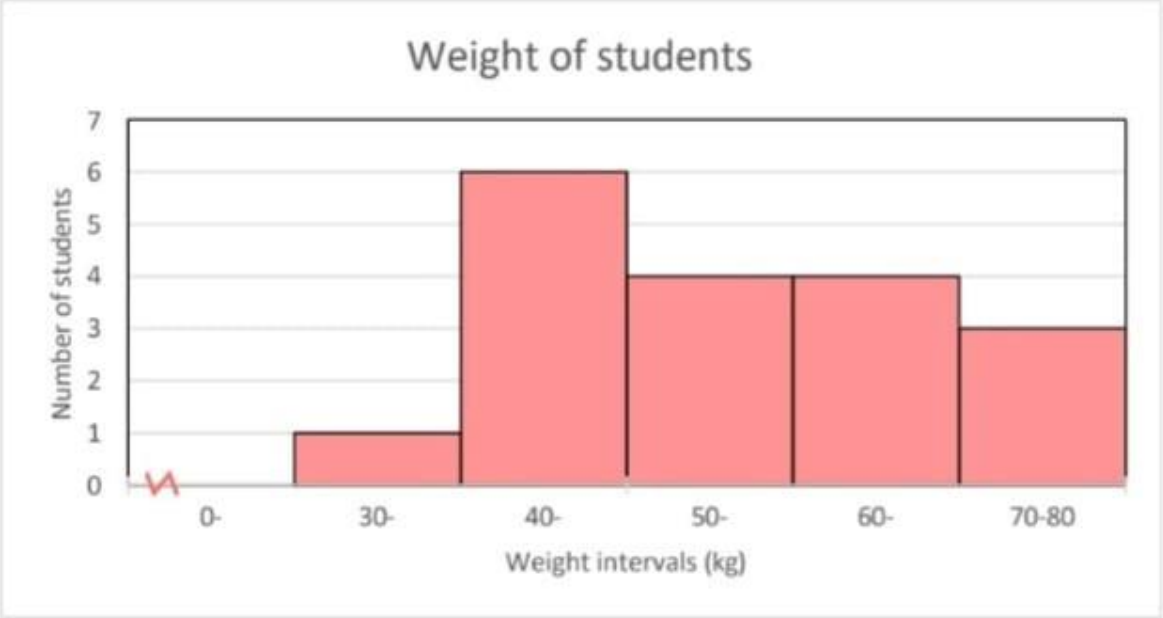
a) Label each interval on the x-axis.

b) Label the frequency on the y-axis.

c) Represent each interval on the x-axis with a bar according to its frequency.

d) Keep the bars next to each other with no space between them.

e) Represent the bars with the same order of the frequency table without changing the order of the intervals.



Example 2:

Some students recorded the hours spent in studying during last month.

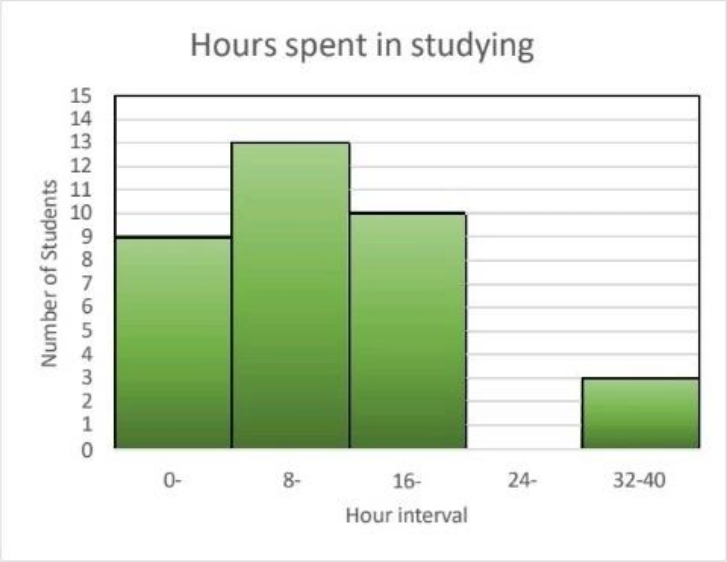
Use this data to make a frequency table & draw a histogram.

5	22	6	6	15	22	8
3	10	11	14	19	21	21
21	19	15	7	7	35	5
17	20	39	8	13	21	9
12	13	9	11	3	36	5

1st: find the range = biggest number - smallest number $= 39 - 336 = 40$.

2nd: I'll group my data into 5 intervals, length of each interval is 8.

Hour Intervals	Frequency
0 -	9
8 -	13
16 -	10
24 -	0
32 - 40	3
Total	**35**

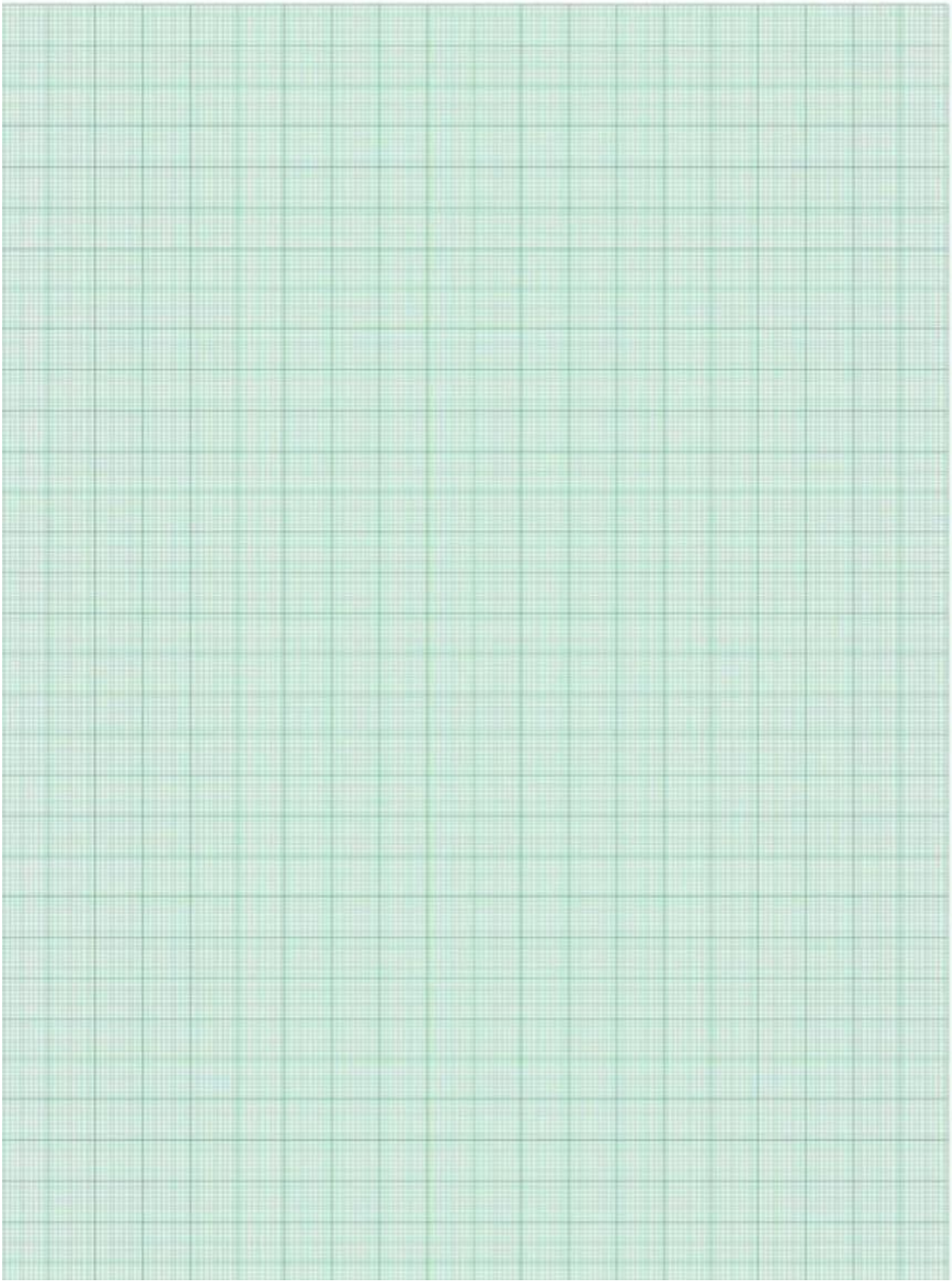


Let's Try

The following table shows the length of some iron rods in cm. represent this data by a histogram.

Lenth of rod	111	118	124	127	138	139	140	142	147	155	158	159
Frequency	5	8	3	6	5	2	7	3	8	4	6	1

Interval	Frequency

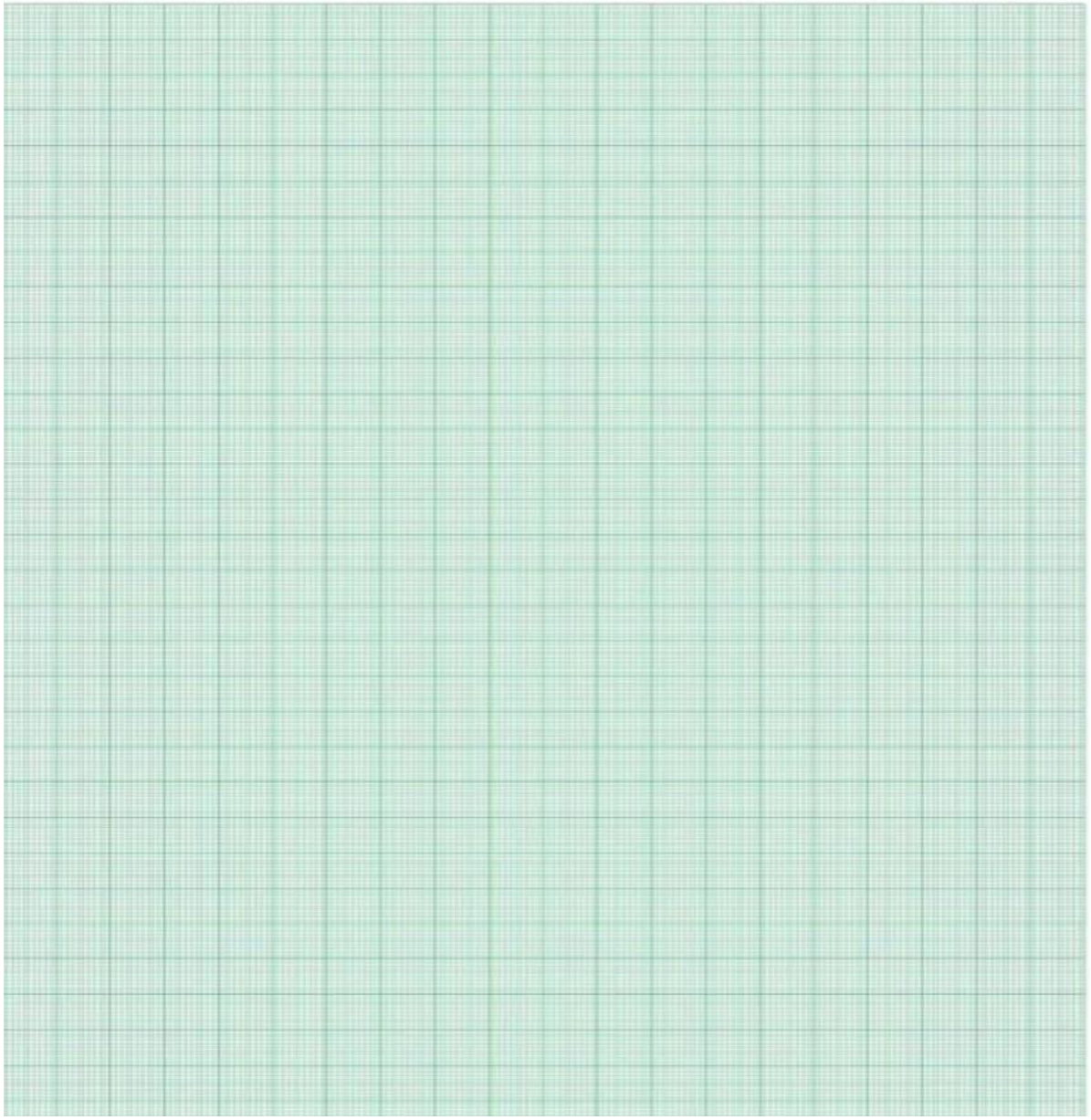


Evaluation

1) Use the following data to draw a histogram (make a frequency table with intervals if not given):

A) This table shows the frequency distribution of 200 students' marks in a certain examination. Draw a histogram to represent this data.

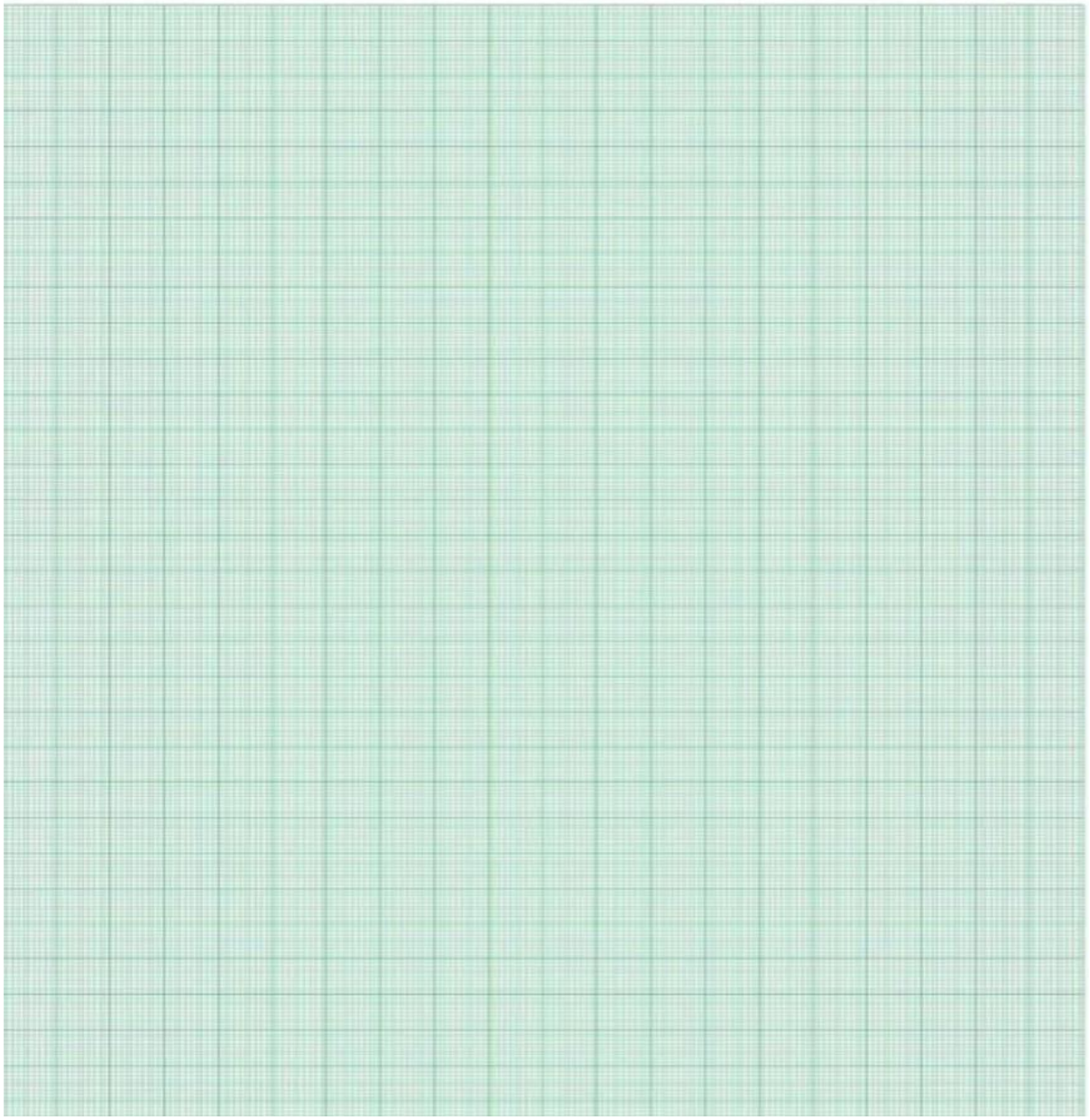
Set of marks	10_	20_	30_	40_	50_	60 _	70_	80-	90 – 100	Total
Number of students	10	10	25	20	45	15	20	35	20	200



B) This table shows the time (in minutes) spent in running 3 km. for a racing team. Draw a frequency table with intervals, then draw a histogram to represent this data.

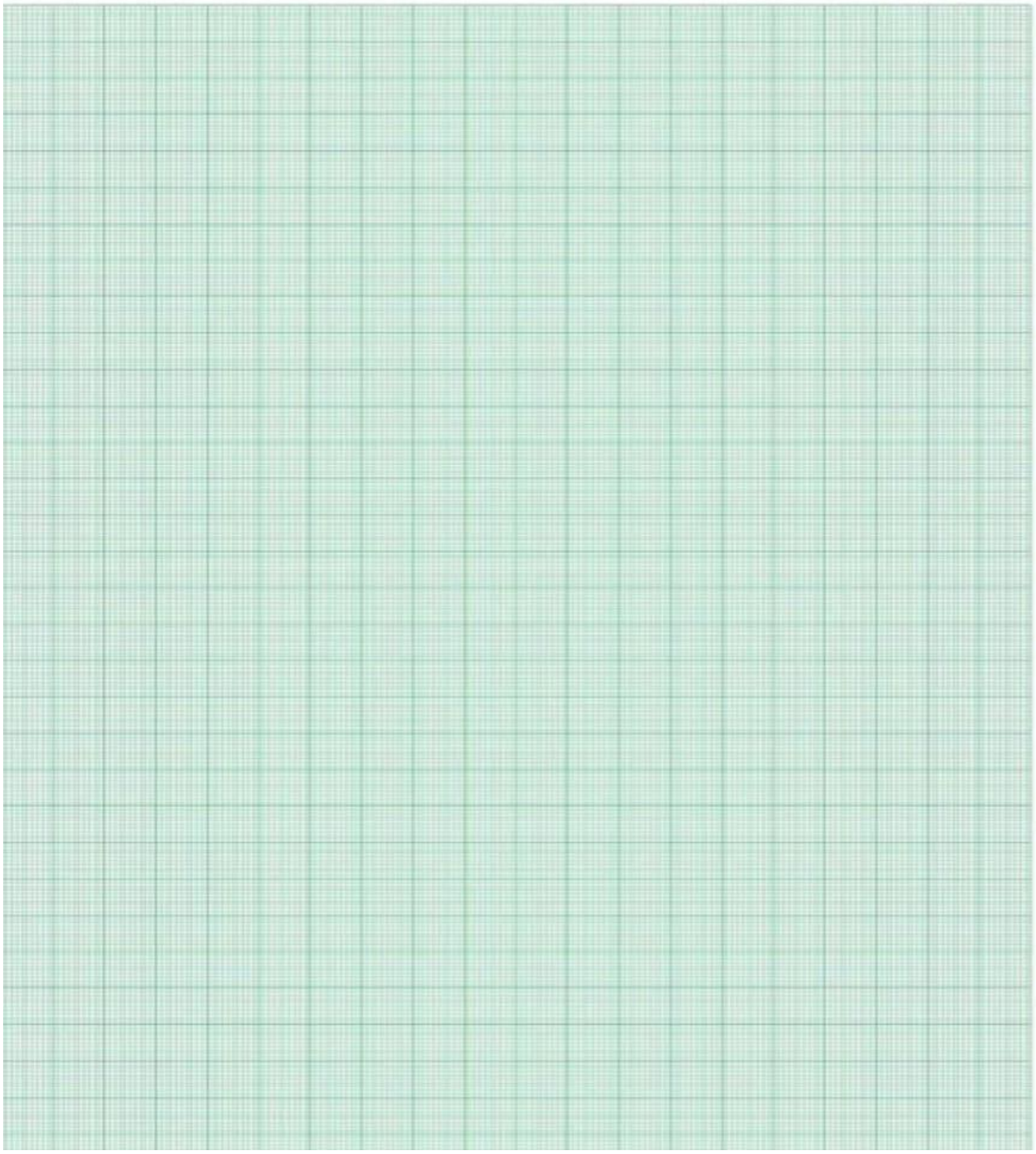
34	60	67	45	59	115	57	65	110	63
42	81	37	100	109	50	60	70	72	48
98	76	39	84	93	42	105	60	80	50

Interval		-					**Total**
Frequency							**30**



C) This table shows the frequency distribution of temperature recorded in 50 countries. Draw a histogram to represent this data.

Set of temperature	0 −	15_	30_	45_	60 − 75	**Total**
Number of countries	10	5	11	15	9	**50**



Date:……..

Lesson 4

Exploring Box plot

Box plot: is a way of representing data by showing a set of 5 numbers that are considered a summary for this data. This 5-numbers summary must include the minimum, lower quartile, median, upper quartile & maximum. To find the 5-numbers summary we 1st arrange the observations ascendingly then get each number as follows:

i) **minimum:** the smallest number observed.

ii) **lower quartile (Q1):** the middle value for the lower half of observations.

iii) **median:** the middle value for the set of observations:

➤ if the number of observations is odd, then the median is exactly the middle number.

➤ if the number of observations is even, then the median is half the sum of the 2 middle numbers = (1st middle + 2nd middle) /2

iv) **upper quartile (Q3):** the middle value for the upper half of observations.

v) **maximum**: the highest number observed.

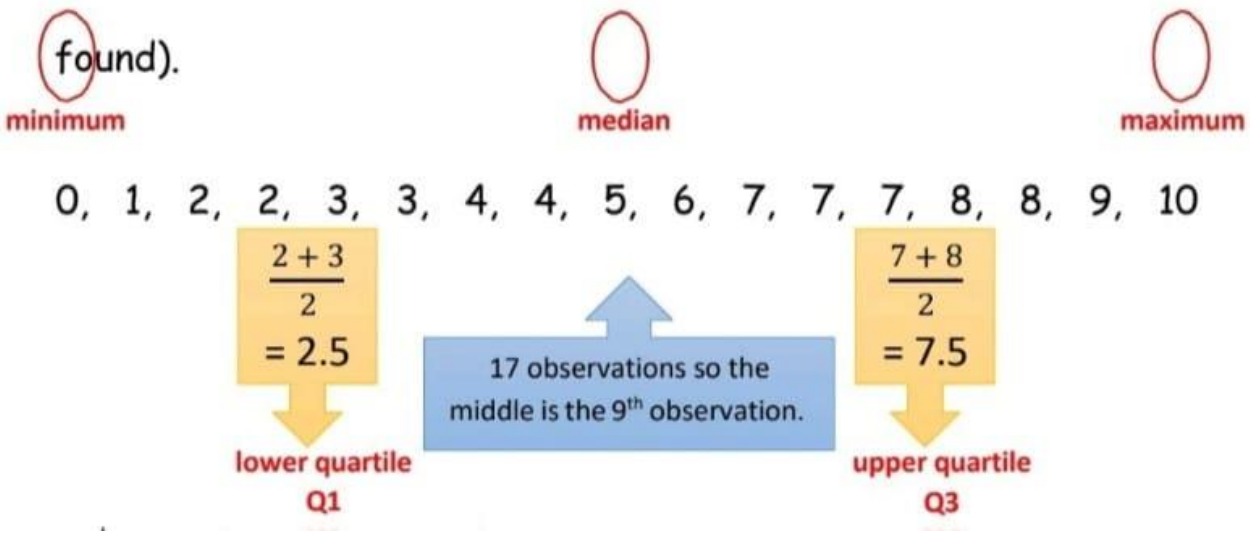
Example 1:

The following are the scores of some students in a math exam & we need to represent them using a box plot:

7, 3, 8, 7, 9, 8, 2, 4, 10, 5, 7, 0, 2, 1, 6, 4, 3

Steps for making the box plot:

➤ 1st: arrange the data ascendingly (show repetitions of data if



➤ 2nd: get the minimum = 0

➤ 3rd: get the maximum = 10

➤ 4th: get the median = 5

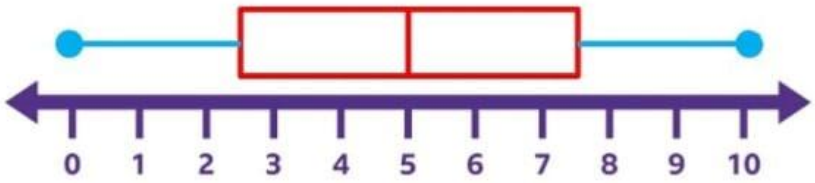
➤ 5th: get the lower quartile = 2.5

➤ 6th: get the upper quartile = 7.5

➤ 7th: draw a number line including the minimum & maximum.

➤ 8th: draw a rectangle for the lower quartile, median & upper quartile.

➤ 9th: extend a line from the lower quartile to the minimum & from the upper quartile to the maximum.



Let's Try

1) Find the median for each set of data:

a) 1, 2, 2, 3, 5, 7

The median =……..

b) 7, 7, 9, 11, 11, 12, 15, 16

The median =……..

c) 5, 7, 2, 9, 4, 5, 2, 10, 8

The median =……..

d) 7, 0, 11, 5, 7, 3,6,9,7,9

The median =……..

2) Write the 5-number summary for each set of data:

a) 1, 12, 6, 6, 5, 2, 0, 10, 7, 2

i) minimum =……..

ii) maximum =……..

iii) median =……..

iv) lower quartile =……..

v) upper quartile =……..

b) 0 , 1 , 7 , 4 , 18 , 12 , 0 , 9 , 12 , 11 , 13 , 17 , 15

i) minimum =……..

ii) maximum =……..

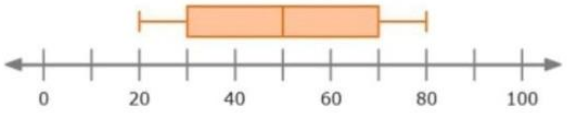
iii) median =……..

iv) Q 1 =……..

v) Q 3 =……..

3) Look at the box then answer the corresponding questions:

a) For this dot plot:



i) minimum =……

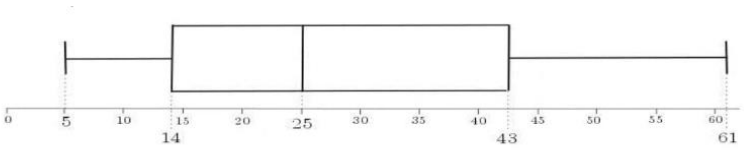
ii) maximum =……

iii) median =……..

iv) Q1 =……

v) Q3 =……..

b) For this dot plot:



i) minimum =……

ii) maximum =……

iii) median =……..

iv) Q1 =……

v) Q3 =……..

4) Draw a box plot for each set of data (show steps):

a) Following are the number of books read by a group in a reading club. Represent this data using box plot:

2, 7, 3, 5, 2, 8, 5, 7, 10, 3, 4, 9, 5, 2, 7

b) Ahmed scored below the time spent in front of T.V. (in hours) for his class students during last week. Represent this data using box plot:

14, 5, 3, 12, 8, 5, 7, 14, 20, 4, 6, 8, 13, 11

Evaluation

1) Find the median for each set of data:

a) 2, 4, 5, 6, 9, 10, 13

The median =……..

b) 12, 11, 15, 12, 14, 22, 17

The median =……..

c) 30, 10, 50, 60, 20, 30

The median =……..

d) 6, 12, 15, 13, 5, 8

The median =……..

2) Write the 5-number summary for each set of data:

a) 5, 8, 6, 9, 2, 4, 3, 1, 6, 8, 3, 5, 2, 1

i) minimum =……..

ii) maximum =……..

iii) median =……..

iv) Q1 =……..

v) Q3 =……..

b) 12, 4, 9, 15, 17, 6, 8, 4, 12, 7, 3

i) minimum =…….

ii) maximum =…….

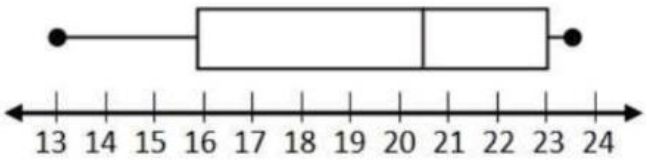
iii) median =…….

iv) Q 1 =…….

v) Q 3 =…….

3) Look at the box then answer the corresponding questions:

a) For this dot plot:



i) minimum =…….

ii) maximum =…….

iii) median =…….

iv) Q 1 =…….

v) Q 3 =…….

a) For this dot plot:



i) minimum =…….

ii) maximum =…….

iii) median =…….

iv) Q 1 =…….

v) Q 3 =…….

4) Draw a box plot for each set of data:

1) Dina recorded below the number of siblings each student has in her class. Represent this data using box plot:

0,5, 3, 2, 4, 1, 2, 3, 2, 4, 3, 2, 3, 1

Date:……

Lesson 5

Applications on Data Representations

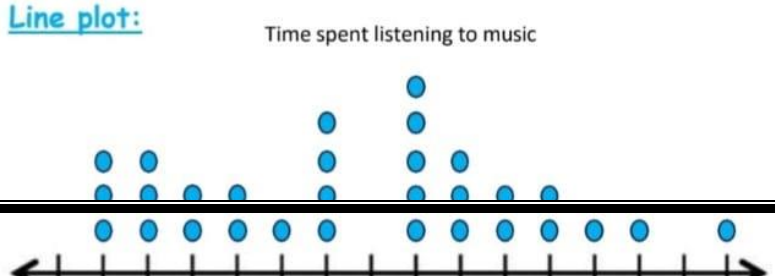
Throughout this unit we studied dot plot, box plot, bar graph & histogram. Notice that a certain set of data can be represented by more than 1 graph BUT not every graph will provide the same information.

Example 1:

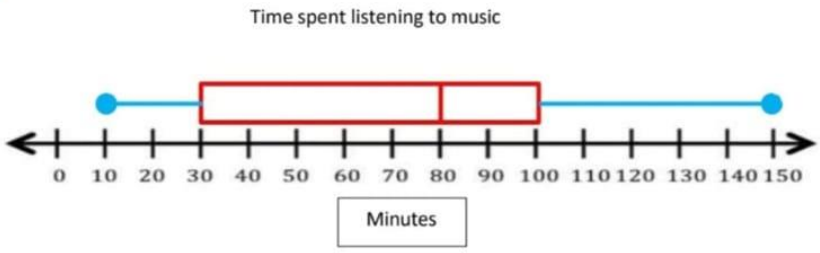
Time (in minutes) students spent listening to music last week					
60	30	50	130	10	110
100	60	40	80	120	90
80	10	60	90	90	10
40	20	30	80	20	20
80	110	60	80	150	100

Total Observations=30
Minimum Observations=10
Maximum Observations=150

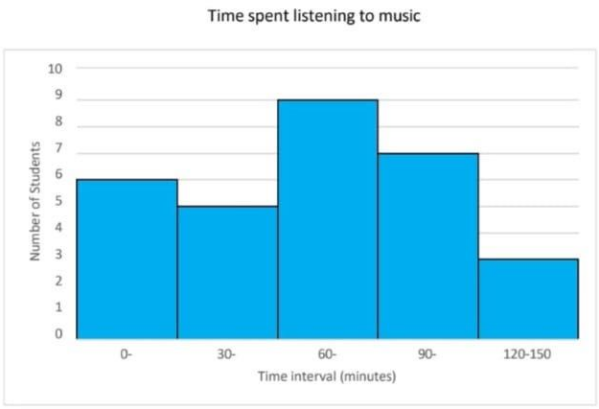
Line plot:



Dot plot:



Histogram:



I can tell from the dot plot that no student spent 70 minutes listening to music, but dot plot or histogram can't provide this information.

Dot plot shows individual data.

Histogram help to show information about grouped data but not individual observations.

Box plot shows summary of data, it doesn't show individual data.

Both the box plot & line plot show the median, minimum & maximum but not the histogram.

Both histogram & line plot show total number of observations, but box plot doesn't.

Let's Try

1) Write the type of graph that provides the best answer for each question (you may choose more than 1 answer):

a) How many people were surveyed?

……

b) What is the most frequent height?

……

c) How many scored 15 goals in the team?

……

d) Which graph shows the spread of data in each quartile?

……

e) What is the age group of most of the members in the club?

……

f) Which graph shows the minimum value?

.....

Evaluation

1) Write the type of graph that provides the best answer for each question (you may choose more than 1 answer):

a) Which graph shows individual data?

.....

b) What is the largest amount of money save?

.....

c) Which graph shows group intervals?

.....

d) How many countries registered temperature below 5°c?

.....

e) Which graph shows data summary?

.....

f) Which graph shows the longest time spent in reading?

.....

g) How many students saved between 50 & 100 L.E.?

.....

Date:……..

Unit 6 Assessment

1) Choose the correct answer:

a) What is the cost of each book in your bag? This question is

i) statistical question ii) non statistical question

iii) open ended question iv) essay question

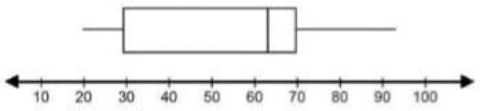
b) What is the minimum for this set of data? 3, 5, 9, 4, 1, 6, 2, 1

i) 1 ii) 3 iii) 2 iv) 4

c) The lower quartile is-------

i) 20 ii) 60

iii) 30 iv) 70



d) What is the favorite color for each of your classmates? This question provides data.

i) categorial ii) not accurate

iii) numerical iv) non statistical

e) The median for 7, 3, 5, 7, 8, 7, 2, 3 & 9 is

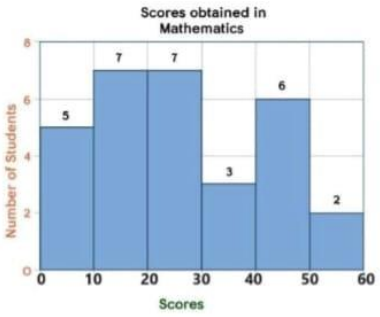
i) 2 ii) 7

iii) 5 iv) 9

f) How many students scored less than 30?

i) 7 ii) 5

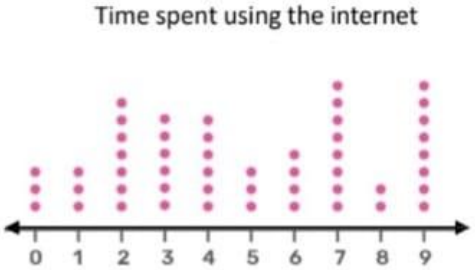
iii) 14 iv) 19



g) How many people spent 6 hours using the internet?

i) 4 iii) 8

ii) 0 iv) 2



h) Which graph represents individual data?

i) line plot iii) box plot

ii) histogram

iv) all of them

i) The median for 1, 3, 3, 4, 6, 8, 8, 9, 11 & 12 is

i) 4

iii) 6

ii) 7

iv) 8

j) How many families were surveyed?

i) 20

ii) 18

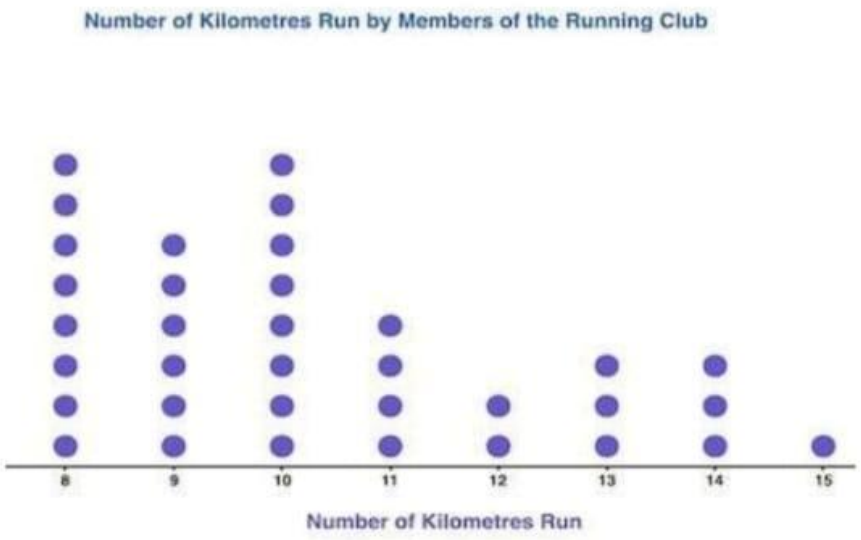
ii) 17

iv) 19



2) Look at each graph then answer its related questions:

i)



a)A suitable statistical question for this graph is

……

b) What is the type of data represented in this graph?

……

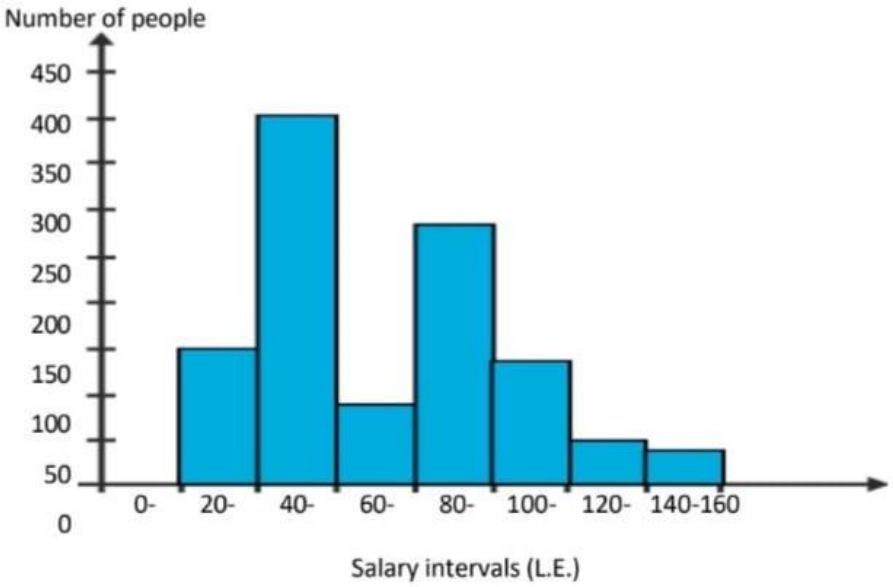
c) How many km did the majority of people run?

……

d) How many people were surveyed?

……

ii)



a) A suitable statistical question for this graph is

……

b) What is the type of data represented in this graph?

……

c) How many people gain salary less than 60 L.E.?

……

d) How many people gain salary between 100 & 160 L.E.?

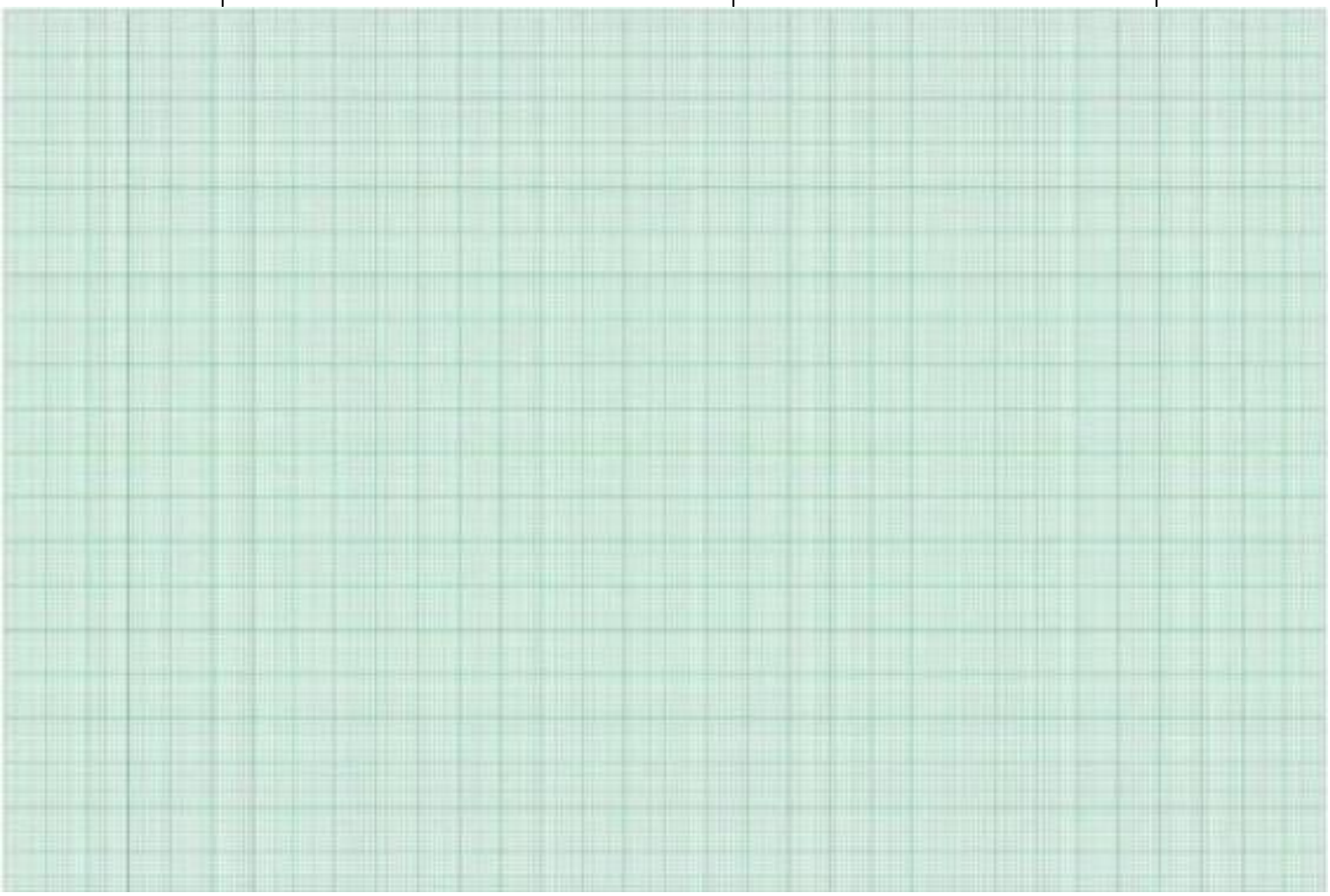
……

2) Use the following data to draw a histogram:

The following table shows the height of some students in cm. represent this data by a histogram.

Height (cm)	110	114	119	120	128	135	140	142	164	155	158	159
Number of students	3	4	1	7	2	1	6	3	2	4	4	2

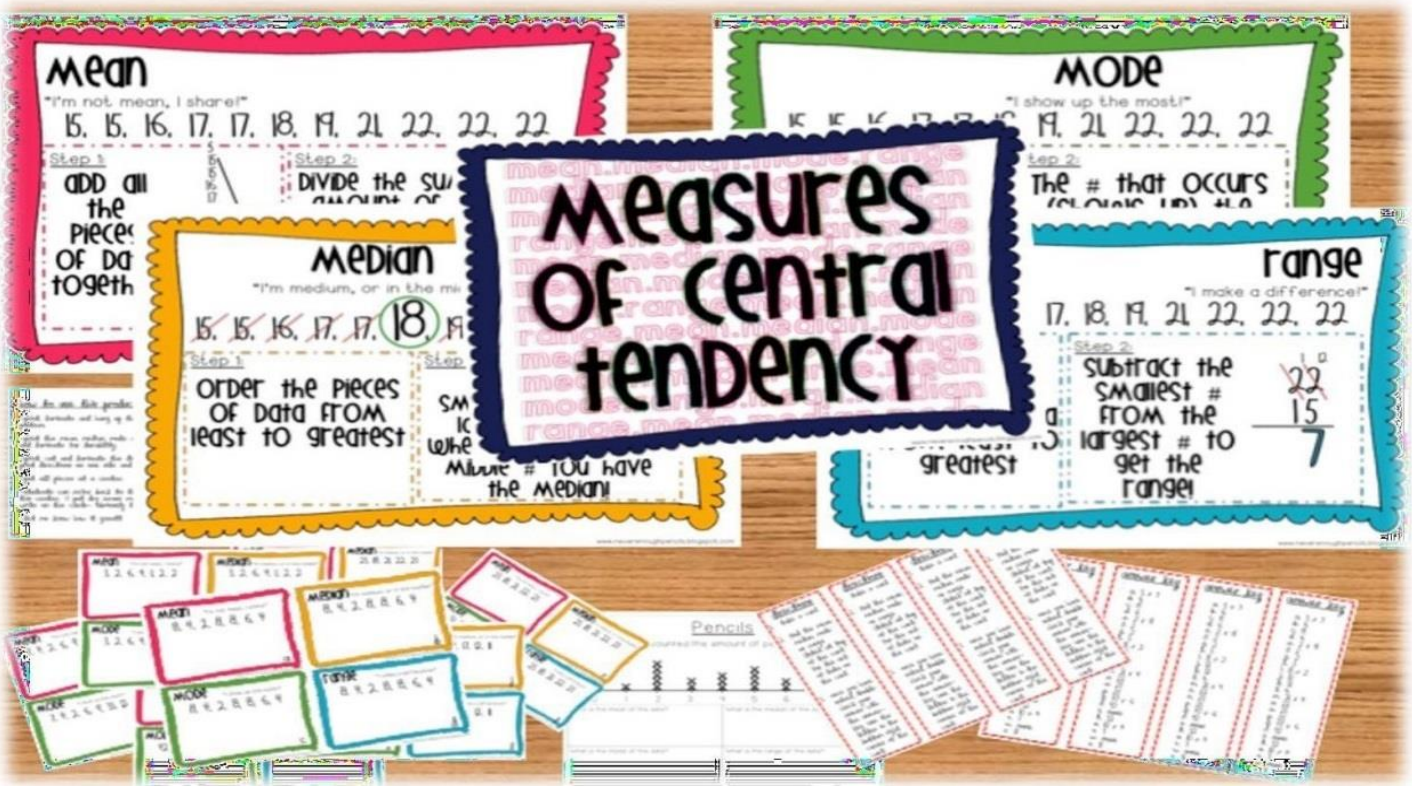
Interval	Frequency



3) Draw a box plot for each set of data (show steps):

a) Sally recorded the saving of her friends last month. Represent this data using box plot:

100, 120, 110, 100, 80, 120, 90, 120, 170, 110, 70, 90, 130



Unit 7, Concept 1

Concept 1: Exploring Measures of Central Tendency and Spread.

Lesson 1: Exploring the Balance of Data Sets.

Lesson 2: Interpreting Mean.

Lesson 3: Exploring Median, Mode and Outlier.

Lesson 4: Exploring Range.

Date:……

Lessons 1&2

Exploring the Balance of Data Sets- Interpreting Mean

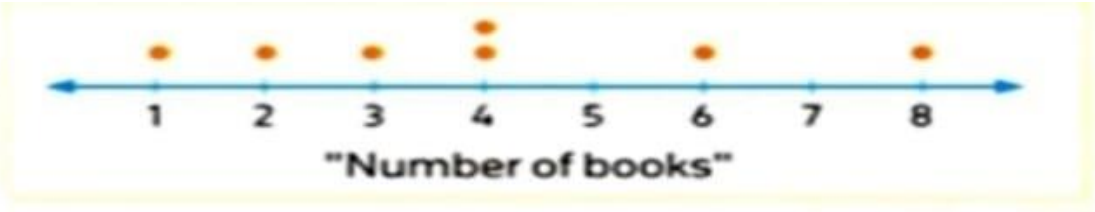
One way to find an average is to find a balance point.

For Example:

Ashraf asks 7 of his friends "How many books did they own?". Ashraf reports his findings as a list or a display such as a dot plot where each dot represents one response.

List: 1 . 2 . 3 . 4 . 4 . 6 and 8

• **Dot plot :**



Now: How can Ashraf find the balance point of this data?

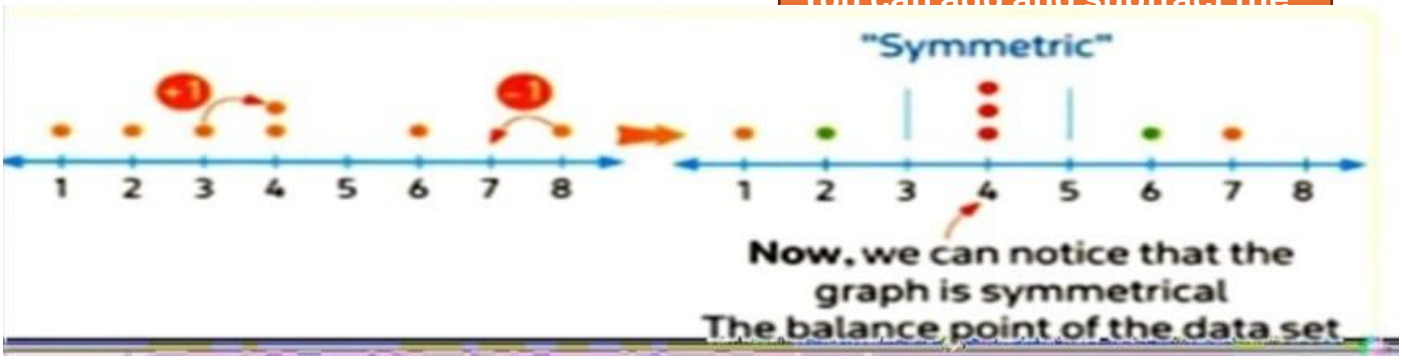
: $\mathbf{1+2+3+4+4+6+8=28}$

The sum of numbers in the list is So, if we add one to any number of the list and subtract one from another number of the list. the sum is still 28

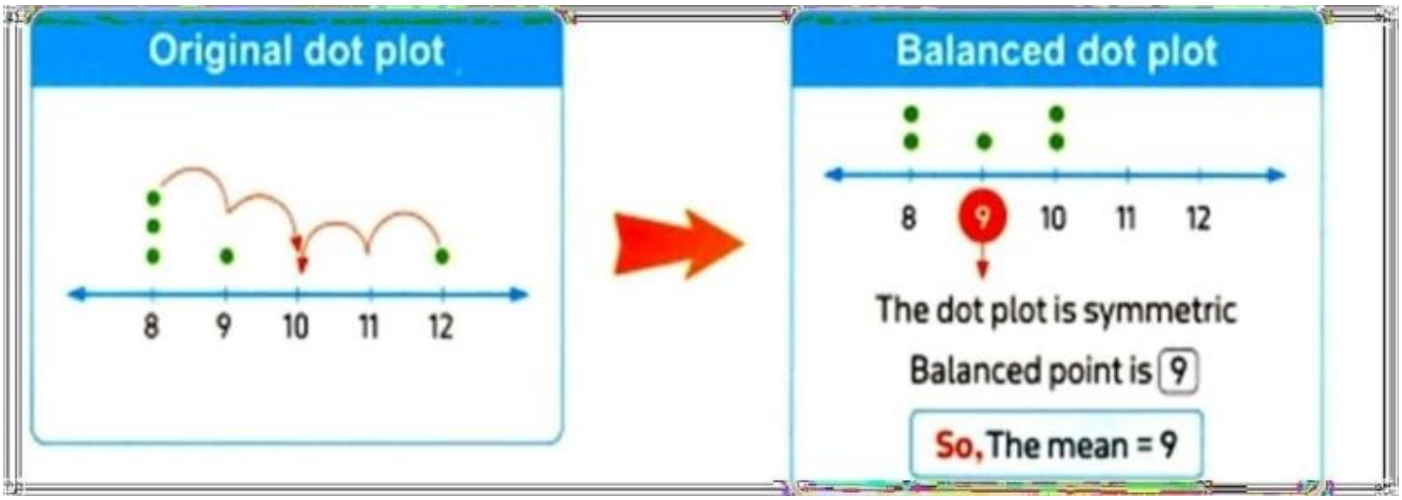
as : $\boxed{1+1}+2+3+4+4+6+\boxed{8-1}=28$

Note

You can add and subtract the



Finally, he can creat a new dot plot by moving the dots as a follow.



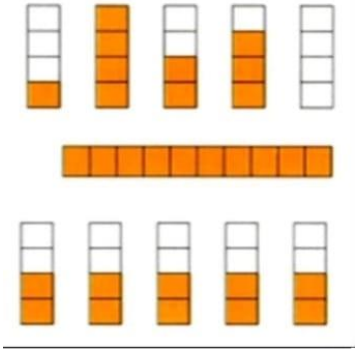
a) The balance point represents the mean

b) Mean as fair share

There are another way to find the mean as a fair share

for Example:

Suppose there are 5 bottles which have the following amounts of water: liter, 4 liters, 2 liters. 3 liters and 0 liters.



To find the mean, first we add up all of the values. We can think of as putting all of the water together: 1+4+2+3+0=10

To find the "fair share" we divide the 10 liters equally into the 5 bottles: 10÷5=2 Then, the mean = 2 liters.

b) Calculating the mean of a set of number



Example: find the mean of 2, 3,1, 5 and 4

Mean [Average] $= \frac{2+3+1+5+4}{5} = \frac{15}{5} = 3$

Let's Try

i. Find the balance point of each of the following data.

a. $\mathbf{3, 9, 6}$ ……

b. $\mathbf{5, 2, 0, 1}$ ……

c. $\mathbf{2, 3, 5, 6, 7, 7}$ ……

ii. Calculate the mean (average) of each of the following data set.

1. 5 , 2 , 0 , 1

.....

2. 1 , 2 , 4 , 5 , 6 , 6

.....

3. 8 , 8 , 10 , 10 , 11 , 2 , 2

.....

Evaluation

1) Complete the following:

a. The balance point the set of data $10 , 4 , 6 , 2 , 4$ is.....

b. The average of the values $7, 13 , 6$ and 4 is.....

c. The mean of the set of numbers $12 , 5 , 10 , 13$ and 20 is.....

d. The balance point of the data set $[9, 11, 20, 15, 5]$ is.....

2) Answer the following:

a. Clara's Can sold the following number of candles over the last 6 days: 5, 8, 12, 3, 7, 5. What was the average number of candles she sold each day?

.....

.....

b. Sam scored 8, 12, 15, 16 and 4 run in 5 matches. What was the mean score per match?

.....

……

c. The temperature of a city during a week was $10℃$, $16℃$, $14℃, 18℃, 13℃, 17℃, and\ 12℃$ What was the average daily temperature of the town for the week?

……

……

d. Jack earns L.E. 100 on the first day, L.E. 60 on the second day, L.E. 150 on the third day, L.E. 80 on the fourth day and L.E. 90 on the fifth day. What was his mean earnings?

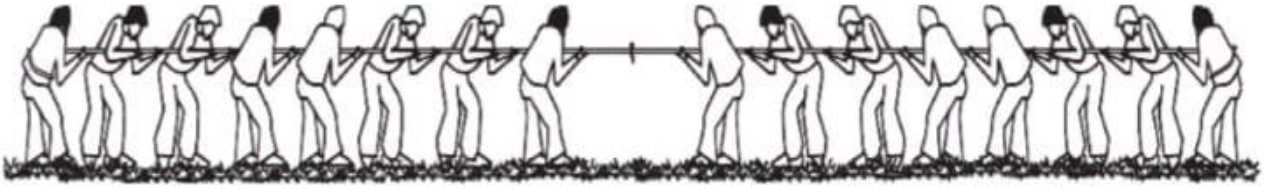
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e. The weights, in kilograms, of the 8 students of six primary tug of challenge team at a school sports are:

35, 35, 24, 26, 24, 26, 30,

28. Calculate the mean weight of the team.



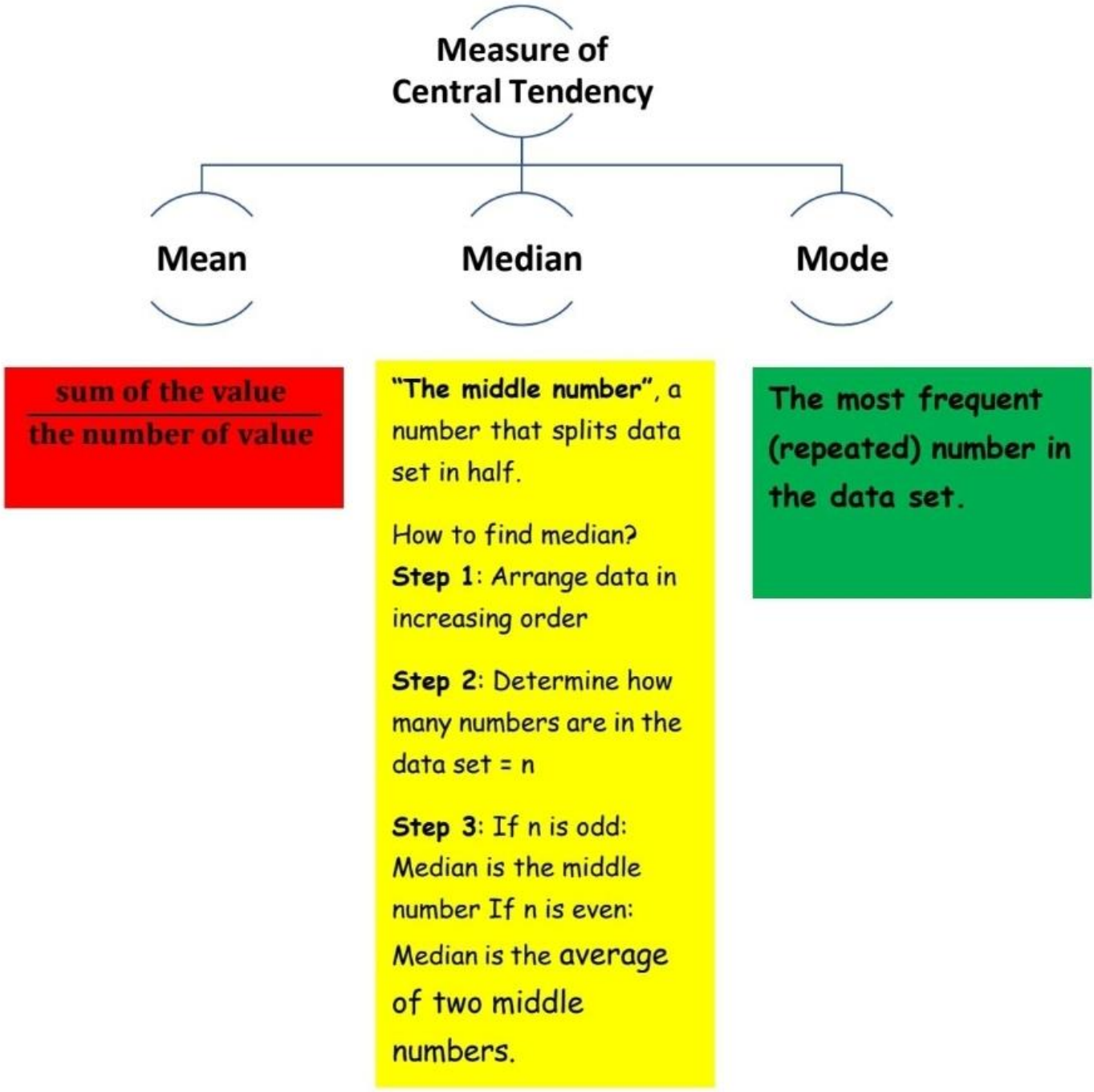
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Date:……

Lesson 3

[A] Exploring Measure of Central Tendency



Ex.1: Find the mean, mode and median for each of the set of data.

Solution:

$$5, 6, 2, 4, 7, 8, 3, 5, 6, 6$$

The mean is $\frac{5+6+2+4+7+8+3+5+6+6}{10} = \frac{52}{10} = 5.2$

To find the median, place all the numbers in order.

$$2, 3, 4, 5, 5, 6, 6, 6, 7, 8$$

As there are two middle numbers in this example, 5 and 6,

$$\text{median} = \frac{5+6}{2} = \frac{11}{2} = 5.5$$

From the list above it is easy to see that 6 appears more than any other number, so mode = 6.

Ex.2: Five people play gulf and their scores were:

$$3, 4, 4, 5, 7$$

For these scores find:

a) The mean b) The median c) The mode

Solution

(a) The mean is $\frac{3+4+4+5+7}{5} = \frac{23}{5} = 4.6$

(b) The numbers are already in order and the middle number is 4. So

median 4.

(c) The score 4 occurs most often, so,

mode 4.

Let's Try

1) Find the mean, median, mode of each set of number blow.

a) $\mathbf{2, 5, 8, 10, 5}$

b) $\mathbf{12, 8, 10, 2, 8}$

c) $\mathbf{0, 7, 7, 3, 0, 7}$

d) $\mathbf{100, 120, 200, 80, 100}$

f) $\mathbf{60, 55, 15, 30, 15}$

e) $\mathbf{90, 80, 70, 30, 10, 60}$

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Evaluation

1. Eight people work in a shop. They are paid hourly rates of

L.E.20, L.E.15, L.E.50, L.E.40, L.E.30, L.E.40, L.E.30, L.E.30.

(a) Find:

(i) the mean (ii) the median (iii) the mode.

(b) Which average would you use if you wanted to claim that the staff were:

(i) Well paid (ii) Badly paid?

Show you work out here:

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2. Two people work in a factory making parts for cars. The table shows how many complete parts they make in one week.

Worker	Mon	Tue	Wed	Thr	Fri
Fred	20	21	22	21	21
Harry	30	15	12	36	28

(a) Find the mean for Fred.

……

(b) What is the median for Harry?

……

(c) What is the mode for Fred?

……

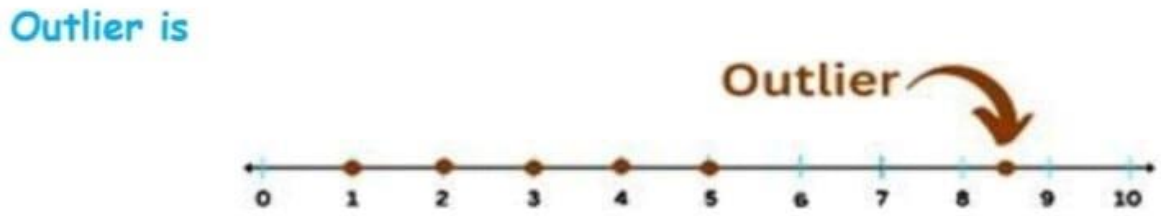
(d) What is the median for Fred?

……

Date:……

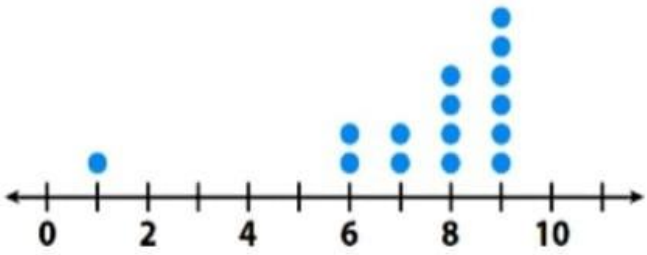
Lesson 3

[B] Exploring Mean, Median and Outlier



An outlier is a value in a data set that is very different from the other values. That is, outliers are values unusually far from the middle. In most cases, outliers have influence on mean, but not on the median, or mode.

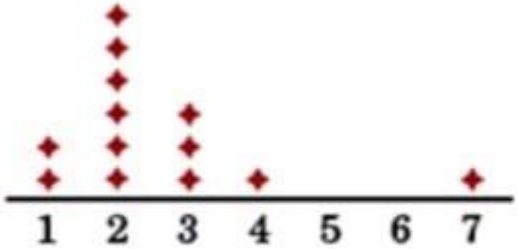




Ex.1: Find the mean, median, and the outlier of each of the below.

a. 3, 4, 7, 3, 5, 2, 6, 20

b.



c. 20, 18, 150, 25, 10

Which is better to use (Median or Mean) for dot plot graph?

a. If the dot plots are distributed in one side of the graph



The median is better

b. If the dot plots are distributed symmetrically on the graph



Either mean or median

c. If the dot plots are distributed evenly on the two sides of the graph without symmetry



The mean is better

How would the outliers affect the mean?

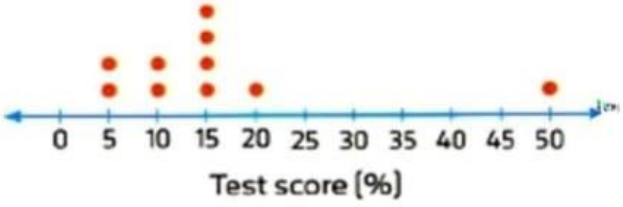
Use reasoning to assign each of the given description with the related graph.

Mean decreases Mean stays the same Mean increases

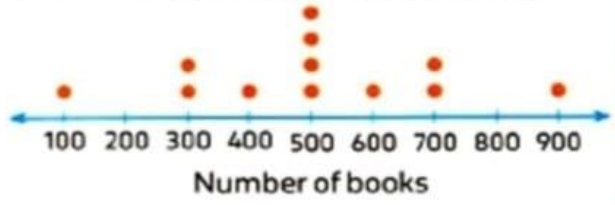
a. Saved money in 8 months



b. First test score



c. N Nmber of books in the library



d. Number of reading books in one week



Let's Try

Select the outlier in the data set.

22, 27, 25, 24, 26, 23, 99, 22, 25

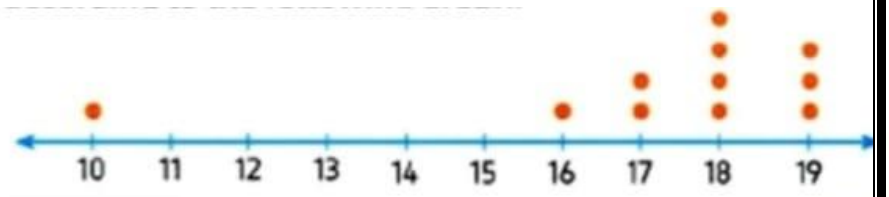
…….

2. Choose the best description of the mean if the outliers has been removed according to the following graph.

a. mean decreases.

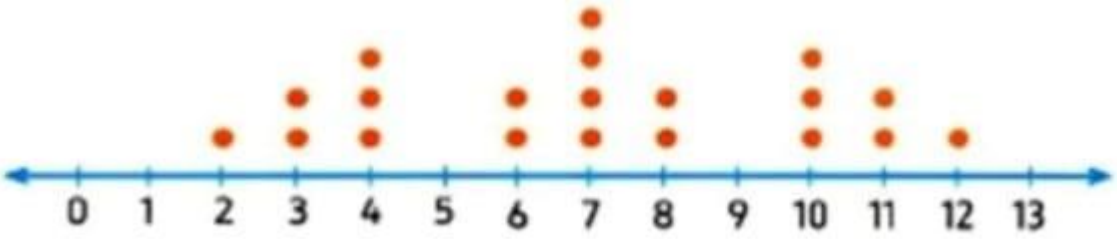
b. mean increases.

c. mean stay the same



3. Choose the better measure of central tendency you think to use.

(Mean , Median , Either)



4, Find the mode of the given data set.

$$2, 4, 5, 2, 4, 6, 2, 3, 5, 2, 4, 7$$

5. Complete:

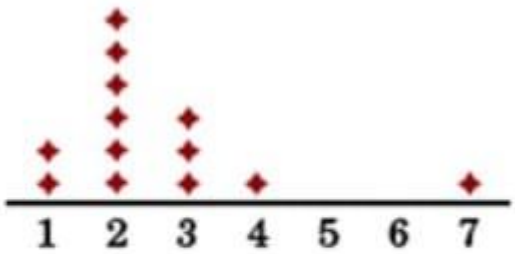
a) The outlier of the values [23, 10,27, 7,34, 35) is……..

b) …….. with outlier. is the measure of central tendency is changed more

c) The two outlier in the data set [123, 130,143,250, 50, 127,150] are ……..and……..

d) The median date set 12,5,13,16,3 and 6 is……..

e) The mode of the set of data is……..



Evaluation

1) Select the outlier for each of the following set of data.

a. 23, 12, 9015 19,25

…….

b. 103, 100, 130, 20, 143

…….

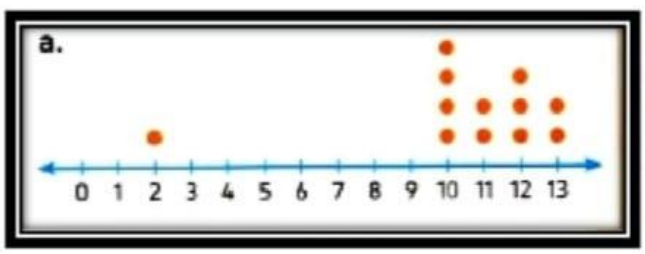
c. 54, 67, 59, 15,70, 43

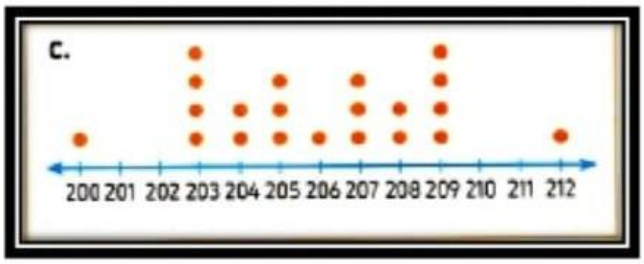
…….

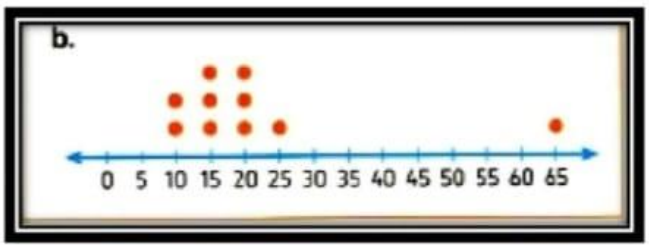
d. 76, 75, 43, 450,5, 66

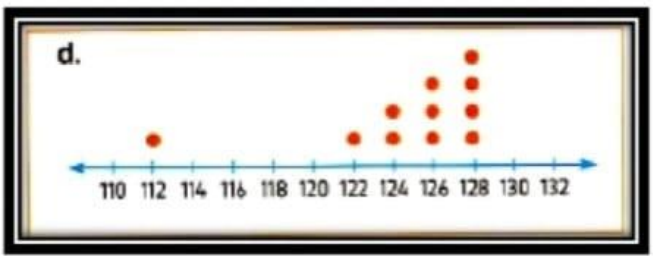
…….

2) Which of the following data represented on dote plots below affects the mean by decreasing?

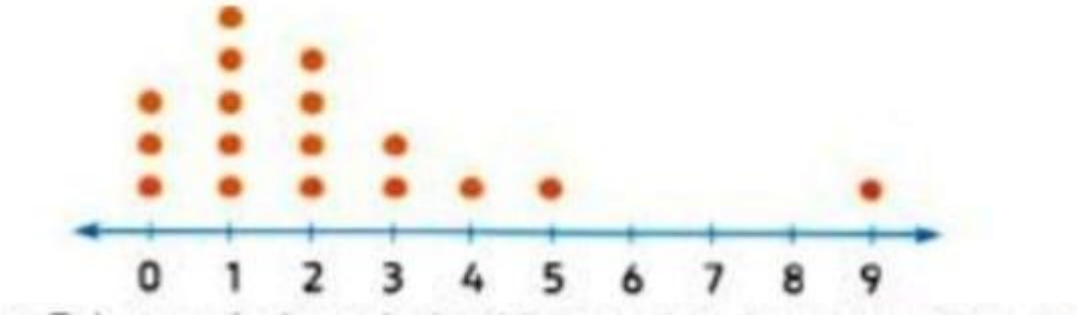








3) Number of sibling of primary 6 students



Here is the data from a Primary 6 class. In looking at the dot plot of the data about students' siblings, how does the outlier affect on the mean?

A. The outlier would not have an effect on the mean.

B. The outlier would cause the mean to be lower.

C. The outlier would cause the mean to be much higher.

D. The outlier would cause the mean to be higher, but the amount would be negligible.

Date:……

Lesson 4

Exploring Range



We can calculate the range using:

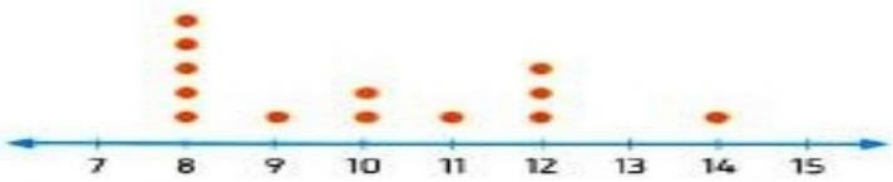
- Data set
- Data plot
- Box plot
- Histogram
- Tables

Example

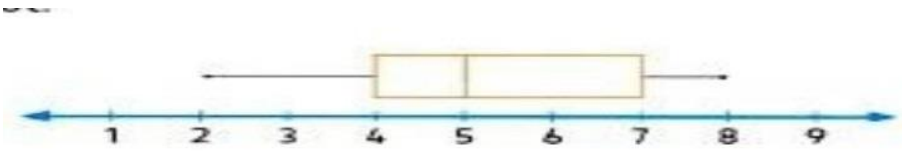
Find the range of each of the following:

a. The data set of values 6 , 7 , 5 , 1 , 9 , 12 , 3 , 4

b. The dot plot



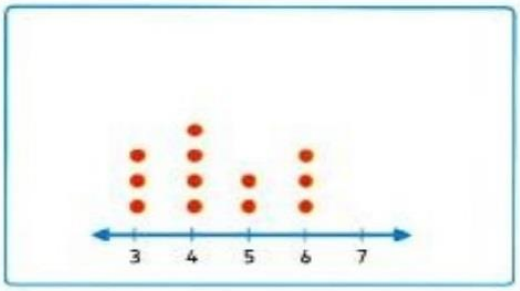
c. The Box plot



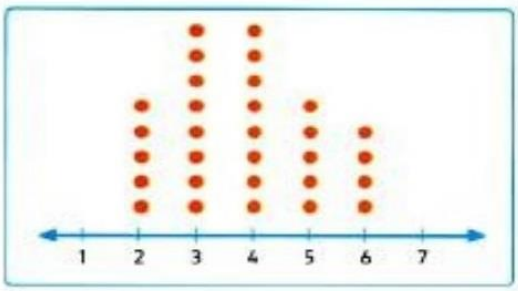
Let's Try

1) Find the range fot each of the following:

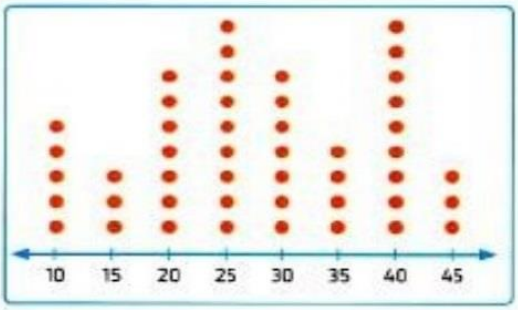
a.



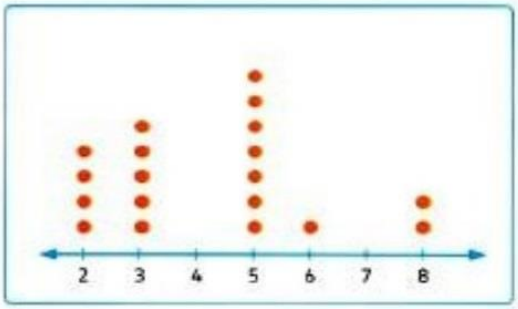
b.



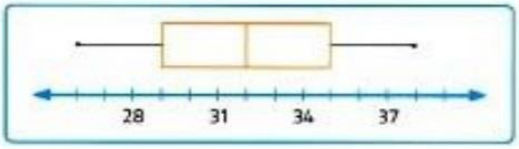
c.



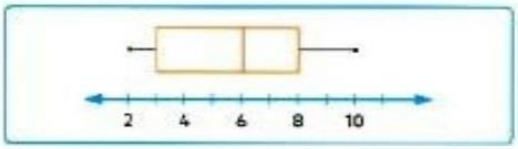
d.



e.



f.



a. ----------

b. ----------

c. ----------

d. ----------

e. ----------

f. ----------

2) Five people play golf at one hole their scores are:

3 , 4 , 4 , 5 , 7

For these scores , find:

a. The mean ----------

----

b. The median ----------

----

c. The mode ----------

----

d. The range ----------

Evaluation

1) Choose the correct answer:

1) The range = greatest value …….. smalest value

a) + **b) −** **c) ×** **d) ÷**

2) The range of the set of values 7 , 3 , 6 , 9 and 5 is ……..

a) 3 **b) 4** **c) 6** **d) 12**

3) If the marks of 6 pupils in one of the tests are 29 , 33 , 57 , 40 , 36 and 49. Then the range for these marks is equal to ……

a) 4 **b) 13** **c) 28** **d) 20**

4) If the values of data set start from 30 *to* 60 , then the range of this data = …….

a) 30 **b) 20** **c) 60** **d) 90**

5) The range of data set represented of the following graph is …….



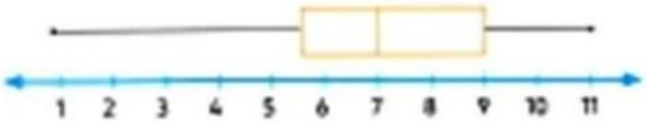
a) 14 **b) 28** **c) 6** **d) 3**

6) By using the following box plot , the range = ……..



a) 5 **b) 6** **c) 3** **d) 7**

7) The range of data set in the opposite box plot =…….

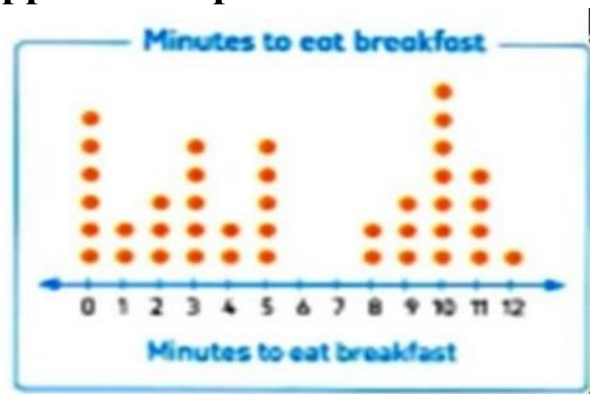


a) 2 **b) 6** **c) 8** **d) 10**

8) The range of data set by using the opposite dot plot=…….

a) 0 **b) 12**

c) 5 **d) 6**



9) What si the range of the quiz scores?

a) 80 **b) 54**

c) 7 **d) 26**

Quiz Number	Score
1	78
2	80
3	69
4	54
5	63
6	65
7	71

2) Complete the following:

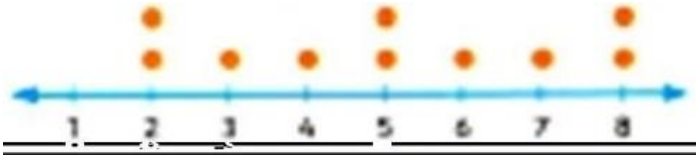
1. The ……= greatest value – smallest value

2. …… is the middle value of the data set

3. The value that lie outside most of the other values in a set of data called ……

4. The mode of 6 , 4 , 2 , 9 , 4 , 3 , 7 , 6 , 4 , 5 , 4 is .....

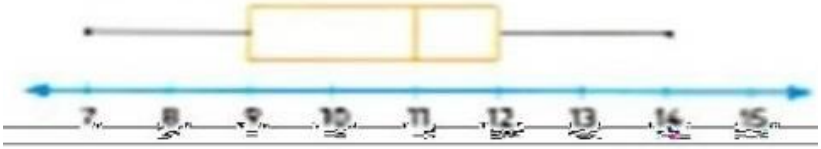
5. The average of 3 , 4 , 6 , 6 , 7 , 8 , 8 is ……

6. The median of the following data is which is represented by the dot plot  **is ……**

7. The outlier value of the following data set is ……

203	204	205	23	206	207	202	201

8. The range of the following data set by using the box plot is…….



Date : ……

Unit Seven , Concept One Assessment

1) Choose the correct answer:

1) The mean of data set [9 , 19 , 12 , 10] is …….

a) 10 **b) 12** **c) 12.5** **d) 15.5**

2) The balance of the following data set
17 , 18 , 20 , 20 , 20 , 21 , 21 , 21 , 22 is ……..

a) 18 **b) 20** **c) 20.5** **d) 21**

3) The mode of 7 , 9 , 7 , 8 , 7 , 6 , 7 , 10 is …….

a) 7 **b) 8** **c) 9** **d) 10**

4) The better measure of central tendencay for the following data set is …….



a) mean **b) median**

c) either

5) The outlier value of teh following data set is …….

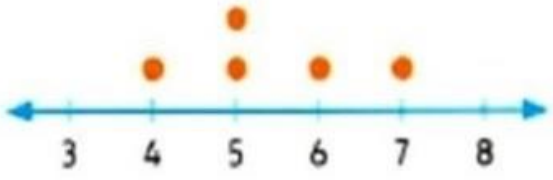
23	25	27	24	94	21	22	26

a) 21 **b) 27** **c) 49** **d) 94**

6) What is the range of the following data set?

a) 4 **b) 3**

c) 5 **d) 7**



7) The range of the set values 9 , 4 , 6 , 1, 7 is ……

a) 9 b) 8 c) 7 d) 6

2) <u>Complete the following:</u>

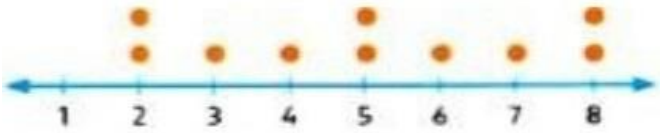
1. The ……= greatest value – smallest value

2. …… is the middle value of the data set

3. The outlier of the values { 13 , 10 , 27 , 70 , 34 , 35} is ……

4. The mean of data set {3 , 2 , 0 , 1 and 8 } is …….

5. The median of the folllowing data which represented by

The data plot 

is ……

6. The outlier of the following data {8 , 9 , 10 , 8 , 55 , 6} is ……

7. ……. Is the measure of central tedenecy is changed more with outlier

3) Eight people work in a shop . They are paid hourly rates of

L.E. 2 , L.E. 25 , L.E. 50 , L.E. 40 , L.E. 35 , L.E. 45 , L.E. 30 ,

L.E. 35 , a. Find:

i) The mean

ii) The median

iii) The mode

iv) The range

b. Find the outlier of this data set: is it effect the mean by increasing or decreasing ?

Show you workout here:

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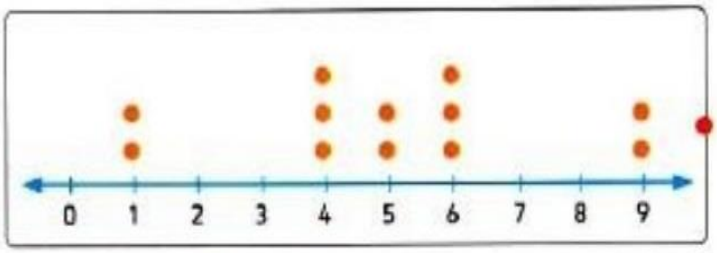
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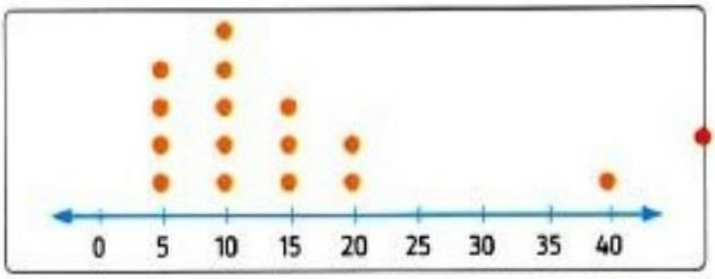
4. Choose the suitable descriptions with the related graph about the effect of outliers on the mean.

a.



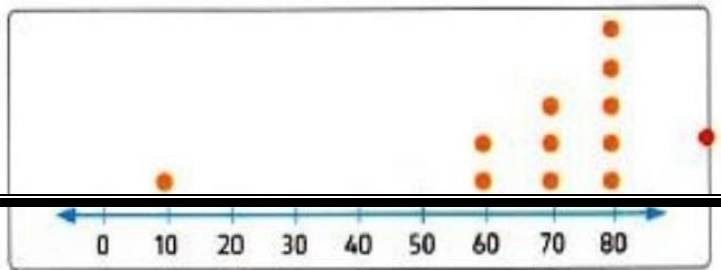
Mean increases

b.



Mean decreases

c.



Mean stay the same